

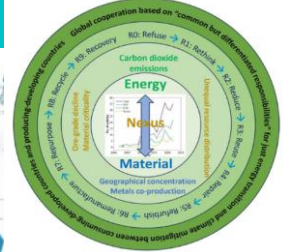


FINAL PROJECT

INTEGRATED PROJECT

INDUSTRIAL TRACK

ANALYSIS OF POLYCYCLE PORT IN BELGIUM



Group of PolyCycle Port

Rasoul Babaei Dargani
Ju Ting Hsu
Chang Ann Chen
Salih Ali
Mona Mirzae

Instructors:

Paola Ibarra González
Gijsbert Korevaar

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Chapter 1. Background and Context

1.1 Background

1.1.1 Group Members Description / Industry Track Selection

My name is Rasoul Babaei Dargani, with the student numbers S4735943 (Leiden) and 6466753 (TU Delft). My bachelor's degree (background) was in Industrial Engineering, and I've been working in an industrial zone, a polymer company, for more than 5 years. I was a quality assurance manager, and it is obvious why I chose the Industrial track! I have valuable experience and expertise in Energy-related subjects, Industrial design, manufacturing lines, and Analysis and economic assessment. Needless to say, my role in this project will be updated at the end of the course, along with the submission date. (Introduced Intervention for: CHP plant)

My name is Mona Mirzaei, and my TU Delft ID is: 6601812. I hold a bachelor's degree in animal science. Since high school, I have had a strong interest in biology and chemistry, which motivated me to pursue undergraduate studies in this field. For my master's degree, I decided to broaden my perspective and explore new academic environments. This led me to Industrial Ecology, a field that I believe offers a valuable opportunity to connect ecological and biological knowledge with industrial systems. My goal is to contribute to approaches that integrate industry with ecological principles in ways that benefit ecosystems, communities, and the planet as a whole. As part of this project involves chemical aspects, I believe my background and experience in chemistry will allow me to contribute effectively, particularly in tasks related to chemical analysis and interpretation. (Introduced Intervention for: PET plant)

My name is Ju Ting Hsu, and my student number for TU Delft is 6457193, and for Leiden is 4542398. My background is in economics. During my studies, I realized that economics often approaches sustainability from a rather theoretical and single-dimensional perspective, and does not sufficiently consider environmental constraints, which does not correspond to how I view the world. While this training provided me with strong analytical and quantitative skills, I felt the need to engage more directly with environmental problems. This motivated me to pursue Industrial Ecology, where sustainability is examined through material and energy systems. This is also why I chose the industrial track. I would like to delve deeper into the material and energy part of the industrial sector. In this project, I am particularly interested in contributing to the material analysis section, especially mass balance and material-related work, even though I am still developing my background in this area. (Introduced Intervention for: Olefines)

My name is Andrew (Chang Ann) Chen, and my student ID for TU Delft is 6599710 and for Leiden is 4696727. I have a bachelor's degree in financial accounting. After my studies at the university, I worked for 4 years in sustainable consultancy at accounting firms. During the years I worked as a consultant, I found that some companies did not really focus on sustainability; instead, they were more focused on the reputation and public image they could gain from the sustainability project. This motivates me to choose Industrial Ecology, which explores other perspectives on environmental sustainability that quantify environmental impacts and provide a systemic view of the world. Also, I realized the potential environmental impact of industries like manufacturing and electricity is much greater than that of other sectors. It makes me wonder whether there is any way to fundamentally reduce the impact of those highly polluting sectors. In this course, I am interested in the development and design of the circular economy within the industrial cluster, enabling resources to be used efficiently in the process and to stay within the loop. (Introduced Intervention for: Aromatics)

My name is Salih Ali, and my student ID is (Delft:6345409 and Leiden:4908155). I'm a mechanical engineer with a bachelor's degree and a strong academic and research background. My background includes research-driven engineering analysis and experimental/simulation-based evaluation of mechanical systems. Over time, I realized that traditional engineering often ignored wider environmental and systemic impacts. That's why I moved into Industrial Ecology. I aim to combine solid engineering skills with systems thinking, focusing on material flows, energy use, and circular economy strategies. In this group project, I will focus on material flow analysis, the mechanical and operational design of key process units, and energy systems optimization. I can work at both detailed technical and system levels, and my role will adapt as the project scope and team evolve. (Introduced Intervention for: EO/EG plant)



1.2. Context Overview

The industrial sector plays a central role in modern economies by producing goods, materials, and energy services, and is a major contributor to global emissions. Within this broader industrial sector, process and energy industries such as chemicals, iron and steel, cement and lime, food, and refining, as well as energy generation, underpin infrastructure, transport, housing, healthcare, and digital systems. They are deeply embedded in global value chains, provide substantial employment, and contribute largely to economic growth and technological development. Except for their economic role, these sectors are also important for innovation. New digital and clean-energy technologies are being introduced into factories and plants, including IoT, AI, cyber-physical systems, digital twins, blockchain, and smart energy management systems to improve efficiency, reliability, and carbon accounting. However, these industries also represent some of the most energy-intensive activities. In 2022, the industrial sector accounted for approximately 38% of global final energy and 25% of direct CO₂ emissions (Gailani et al., 2024). Due to their high energy consumption and significant greenhouse gas emissions, these industries are identified as priority sectors and represent major opportunities for industrial transition and decarbonization planning (Gailani et al., 2024, Challenge Validation, n.d.w). Globally, industries face a dual challenge: decarbonizing while remaining competitive. The potential or ongoing transition pathway includes increased electrification, hydrogen, CCUS, circularity, and energy efficiency, alongside improved digital control and flexibility; this same technology mix is highlighted both in the EU Commission's mapping of industrial demonstrator projects and in the academic review on Industry 4.0 and clean energy integration. However, the transition still faces difficulties because many solutions require high upfront costs, higher operating costs in some cases, limited market maturity, infrastructure gaps, interoperability problems, cybersecurity risks, fragmented regulation, and shortages of workers who understand both industrial processes and digital energy systems. These industries are especially important for climate change and energy security because they consume large amounts of energy and create a large share of emissions, so decarbonising them has a big effect on overall climate targets, but they also need stable and affordable energy to avoid supply chain disruption and loss of competitiveness.

The Energy Strategy Reviews paper shows this clearly for Europe by noting that in 2022 EU industry accounted for 25% of final energy demand and 20% of EU greenhouse gas emissions, and by showing that future industrial decarbonization depends heavily on both electricity and hydrogen, with outcomes strongly affected by renewable deployment and cross-border electricity and hydrogen infrastructure. At the EU level, the direction is to push industry toward climate neutrality by 2050 and to accelerate market uptake of clean technologies before 2030 through policy, funding, and demonstration support; the Commission's 2023 report on 184 demonstrator projects (over €3 billion investment) shows this strategy in practice, with support mainly from Horizon Europe, Horizon 2020, the Innovation Fund, and IPCEI related efforts, while stressing the need for follow up investment to commercialize technologies by 2030.

For Belgium and the regional Antwerp context, the policy frame is the NECP process (2021–2030), which the European Commission describes as the main 10 year planning tool for climate and energy transition and investment, while Antwerp itself is a strong regional example because the R-ACES eco-region shows high industrial concentration, two focus areas (Terbekehof and Kanaalkant), a city heat demand of 31 PJ, and major nearby excess heat potential, which makes industrial symbiosis and district heating expansion realistic transition strategies. The Eurelectric Antwerp Dialogue adds the competitiveness perspective by arguing that electrification should be accelerated, but only with policies that also reduce energy cost pressure, improve infrastructure, and support investment, so decarbonization and industrial competitiveness can move together rather than conflict.

These global and European developments converge in the Port of Antwerp-Bruges, which serves as the central case study for this project. As Belgium's largest industrial cluster and a pivotal hub in Europe's chemical supply chains, it stands at the forefront of the decarbonization transition, reflecting both the immense challenge and the significant potential for systemic transformation. The Antwerp cluster offers a compelling setting for exploring eco-industrial park transitions due to its concentration of energy-intensive industries, abundant excess heat, and emerging hydrogen infrastructure. The subsequent chapters use this context to assess the practical application of industrial symbiosis, circularity, and low-carbon energy integration within the Antwerp industrial cluster.



Chapter 2. Eco-Industrial Parks

2.1 Industrial Parks and the Transition Toward Eco-Industrial Parks

Historically, industrial parks emerged as spatial instruments for economic optimization. Their primary function was to generate agglomeration economies by clustering manufacturing activities within a defined geographic area. Through co-location, firms benefited from shared infrastructure, including transport corridors, port access, electricity grids, and utility systems. This configuration reduced transaction and transportation costs while providing access to a pooled labor market and specialized suppliers. In this early model, efficiency was understood primarily in economic terms. However, the contemporary context has fundamentally altered the role of industrial clusters. The industrial sector remains a major contributor to global environmental pressures. According to the International Energy Agency and data compiled by Our World in Data, industry accounts for roughly one quarter of global greenhouse gas emissions, with energy-intensive sectors such as petrochemicals, cement, and steel production contributing substantially. From a systems perspective, traditional industrial parks operate predominantly as linear metabolic structures: resources enter the system, are transformed into products, and exit alongside residual waste and emissions. This Take–Make–Waste configuration externalizes environmental costs and generates cumulative ecological degradation.

Given societal demands, the industrial park can provide great opportunities for the development of service industries of the local area where it is built and put into operation, boosting the income level of local communities. Furthermore, industrial parks could create working opportunities for domestic workers and also increase the attractiveness and competitiveness of the investment environment for foreign investors. At the same time, institutional and market pressures are reshaping industrial governance. Policy frameworks such as the European Green Deal introduce stricter climate targets and regulatory instruments, including carbon pricing mechanisms and emissions trading systems. Markets increasingly demand low-carbon and resource-efficient products. As a result, industrial clusters can no longer function solely as co-location zones. They must evolve into integrated systems that simultaneously address economic performance, environmental responsibility, and social legitimacy.

2.2 Measures and Approaches to Enhance the Sustainability of Industrial Parks

To improve the sustainability and circularity of industrial parks, new concepts and methods are being developed and implemented worldwide. These measures include industrial symbiosis, circularity, renewable energy integration, and other sustainable development practices. Industrial Symbiosis forms the foundation of EIPs, emphasizing resource exchanges between nearby companies to achieve a combined benefit that exceeds individual efforts. Possible applications include By-product Exchange, where waste from one facility becomes raw material for another, and Utility Sharing, which involves centralized systems for steam, compressed air, or wastewater treatment to minimize duplicated infrastructure and energy waste. Transitioning from linear models to circular systems helps extend the use of resources as long as possible. Water Cascading involves reusing high-quality process water for secondary purposes like cooling or irrigation after minimal treatment. Material Recovery includes establishing on-site recycling practices to convert specialized industrial waste, such as electronic scrap or metal shavings, into reusable secondary raw materials. Energy transition is essential for reaching climate neutrality in industrial clusters. The Distributed Energy Resource method aims to deploy rooftop solar PV, wind turbines, and battery energy storage systems directly within the industrial park. Additionally, Waste Heat Recovery technologies capture low- to medium-grade heat from industrial processes for power generation or district heating. The Kalundborg Symbiosis in Denmark is a perfect example of industrial symbiosis. In 1972, they developed the world's first industrial symbiosis with a circular approach to production. The main principle is that a residual flow in one company becomes a resource in another, benefiting both the environment and the economy. The industry cluster has reduced carbon dioxide emissions by 80% in Symbiosis since 2015, and the local energy supply has reached the carbon-neutral goal. To gain a comprehensive understanding of the industrial park's condition, it's essential to analyze the main drivers, benefits, and barriers affecting the implementation of a sustainable approach. Key drivers include regulatory pressure, market competitiveness, and financial incentives. The primary benefits are notable positive environmental, economic, and social impacts. Nevertheless, there are significant challenges, such as mistrust among companies, which hinders resource sharing; timing and quality mismatches in resource flows; and



high upfront investments with long payback periods for shared infrastructure, such as pipelines or microgrids.

2.3 Problem Statement

The Port of Antwerp-Bruges is a critical part of the economic engine for Belgium, operating as the second-largest port in Europe by supplying to neighboring countries. While being an essential part of Belgium's economic engine, the PolyCycle Industrial site directly indicates the systemic lock-in of the Belgian cluster. For instance, the cluster's dependency on a 50 MW natural gas-fired CHP plant in order to supply 125,000 kg/h of steam for olefins and aromatics cracking, which develops an infrastructure inertia and makes rapid decarbonization both financially and technically un-feasible. The primary barrier for this site is due to a systemic fragmentation over three different levels:

1. **Energy efficiency and fossil fuel dependency:** In the current industrial park, the CHP plant is dependent on natural gas and functions as the primary supplier of electricity and steam to the industrial network. This creates a strong reliance on fossil-based utilities and exposes the industrial park to future carbon pricing mechanisms, energy price volatility, and decarbonization pressures. In addition, high-temperature exhaust gases from the CHP system are not fully utilized, resulting in avoidable energy losses. The current configuration also has limited integration of renewable energy sources, increasing exposure to future carbon pricing mechanisms, fuel price volatility, and decarbonization pressures. Within the current system, all firms heavily rely on fossil fuel, especially the CHP plant and the olefin plant. The former relies on natural gas for utility generation, while the latter depends on virgin naphtha as its primary feedstock and generates hydrocarbon by-products that are not fully utilized within the industrial network.
2. **Lack of recycling:** Currently, the site operates by using a linear "Naphta-to-PET" model. The site is not only fragile to shocks of fossil fuel prices, but also regarding the dependence on virgin materials - since the site is without a proper chemical recycling loop that can return polymer waste to the Steam Cracker.
3. **Wastewater generation and limited water reuse:** The current industrial park generates wastewater streams that require treatment before discharge, particularly from the Aromatics plant. The lack of wastewater recovery and reuse increases freshwater demand and reduces the overall resource efficiency and circularity of the industrial system.

2.4 Main Objective

The primary objective of this project is to establish a detailed Coordinated Transition Plan, which can enable PolyCycle's Industrial Site (Port of Antwerp), to transition its site by 2050, into a high-quality-performance Eco-Industrial park (EIP). To make sure this ambitious goal is possible, this project will follow a narrow scope to focus on three distinct highly influential Symbiosis Pathways to be implemented by 2030.

1. **Heat recovery & Electrification:** Fossil-fuel-depending steam will be reduced through CHP Utility-Hub Retrofit measures, including waste heat recovery, renewable electricity integration, and electric boiler deployment. This facility can improve energy efficiency, reduce CO₂ emissions, and increase steam flexibility within the industrial park.
2. **Closed-loop recycling:** Create a closed-loop chemical recycling system (e.g. Pure Cycle technology) which can reduce the site's consumption of virgin fossil fuels, as the closed loop is able to transfer the secondary PET waste back into the Circular Naphta.
3. **Wastewater treatment and reuse:** Wastewater generated within the Aromatics plant will be treated through advanced treatment technologies and reused within the industrial park. This intervention aims to reduce freshwater demand, lower wastewater discharge, and improve water circularity.
4. **Main Research Question:** How can the PolyCycle Industrial Site utilize integrated energy, material, and water interventions to reduce fossil-fuel dependence, improve resource and energy efficiency, and strengthen industrial symbiosis within the industrial park?



CHAPTER 3. PROJECT SCOPE AND RESEARCH APPROACH

3.1. Project scope- Case Study: Polycycle Port Antwerp (Belgium)

Location: NextGen District, Port of Antwerp-Bruges, Belgium.

The Port of Antwerp is the second largest integrated petrochemical cluster on a global scale. Locating the hub within the NextGen District, which is an area specifically dedicated to circular economy initiatives, provides immediate access to existing pipeline infrastructure, with terminals consisting of high-volume crude and naphtha, and a strategic geographic position to manage and reuse the industrial waste heat. The industrial park consists of a complete connection of five facilities and their activities to establish an integrated polymer supply chain, which starts from raw fossil feedstocks and ends with final plastic resin production. The system aims to improve resource utilization through the integration of material, energy, and utility flows across the industrial park. The following five facilities comprise the industrial park:

1. Natural Gas CHP (Combined Heat and Power) Plant

The Combined Heat and Power (CHP) facility functions as the central energy hub of the industrial park. Its primary role is to simultaneously generate electricity and recover thermal energy for industrial use. In this configuration, natural gas is combusted in a gas turbine to produce electricity. Rather than releasing the high-temperature exhaust gases into the atmosphere, the system directs them to a Heat Recovery Steam Generator (HRSG), where the residual heat is utilized to produce both High-Pressure Steam (HPS), Medium-Pressure Steam (MPS), and Low-Pressure Steam (LPS). This recovered steam is then distributed across the industrial network, particularly to the olefins, aromatics, and EO/EG production units. This integrated configuration significantly improves overall energy efficiency by converting what would otherwise be waste heat into a usable industrial resource. Production capacity is approximately 50 MW of electricity and around 120–150 tones per hour of steam. Approximate main inputs include natural gas at about 12,000 kg/h and boiler feed water at approximately 130,000 kg/h. Main outputs consist of electricity (around 50 MWh/h), steam streams totaling roughly 125,000 kg/h distributed to downstream chemical plants, and carbon dioxide emissions of about 33,000 kg/h. As the central utility provider, the CHP plant plays a critical role in determining the energy efficiency and carbon intensity of the industrial park. An example in the Antwerp cluster: The Zandvliet Power CHP plant located at the BASF Antwerp site and the INEOS CHP facility in Lillo.

2. Olefins Production Plant

The olefins production unit operates as the primary feedstock generator within the petrochemical complex. Its core process is steam cracking, a thermally driven reaction in which hydrocarbon feedstocks are broken into lighter and more reactive molecules. In this process, naphtha is mixed with steam and heated to temperatures of approximately 850°C in cracking furnaces. Under these extreme conditions, long-chain hydrocarbons undergo thermal decomposition, producing a mixture of smaller molecules, primarily ethylene and propylene. These olefins represent the fundamental building blocks for a wide range of downstream petrochemical products. The facility operates at a production capacity of roughly 876 kilotons per year of ethylene. The main material inputs include approximately 330,000 kg/h of naphtha and around 80,000 kg/h of high-pressure steam supplied by the CHP plant. Outputs include about 109,500 kg/h of ethylene, of which approximately 14,000 kg/h is directed to the on-site EO/EG production unit (matching EO/EG's stated ethylene input) and the remaining approximately 95,500 kg/h is exported via the regional ethylene pipeline network to external petrochemical users, roughly 50,000 kg/h of propylene typically sold to external markets, and approximately 55,000 kg/h of pyrolysis gasoline (pygas), which is transferred to the aromatics production unit for further processing. An example in the Antwerp petrochemical cluster: The BASF Antwerp steam cracker and the INEOS Olefins Belgium facility.

3. Aromatics Production Plant

The aromatics plant processes complex hydrocarbon streams to isolate high-value aromatic compounds. Its main function within the industrial network is to convert pyrolysis gasoline and related hydrocarbon mixtures into benzene, toluene, and xylene derivatives, collectively known as BTX aromatics. The process involves several sequential



operations, including catalytic reforming, liquid–liquid extraction, and fractional distillation. A key operational objective is the separation and purification of paraxylene (p-xylene) from other xylene isomers through isomerization and selective separation processes. The facility has an approximate production capacity of 754 kilotons per year of paraxylene. Primary inputs include pyrolysis gasoline from the olefins plant at roughly 55,000 kg/h, additional reformat or naphtha streams of around 150,000 kg/h to maintain full operational capacity, and about 40,000 kg/h of steam supplied by the CHP plant. The resulting outputs include approximately 94,250 kg/h of paraxylene, of which approximately 8,600 kg/h is directed to the on-site PET production facility (matching PET's scaled paraxylene requirement) and the remaining approximately 85,650 kg/h is sold to external markets, around 30,000 kg/h of benzene typically sold to external markets, and roughly 10,000 kg/h of raffinate composed of lower-value hydrocarbons. An example in the Antwerp region is the ExxonMobil Antwerp refinery aromatics unit, which represents one of the largest BTX production facilities in Europe.

4. Ethylene Oxide and Ethylene Glycol (EO/EG) Plant

The EO/EG facility converts ethylene into intermediate chemicals widely used in polymer production. The process consists of two primary reaction stages. First, ethylene undergoes catalytic oxidation in the presence of oxygen to form ethylene oxide (EO). In the second stage, ethylene oxide reacts with water through thermal hydration to produce monoethylene glycol (MEG). Both reactions are strongly exothermic, releasing significant amounts of thermal energy. As a result, the plant requires extensive cooling systems. This cooling demand creates a large stream of low-grade heat, typically carried by hot water, which can be recovered and utilized elsewhere in the industrial system. The plant has a production capacity of approximately 135 kilotons per year of monoethylene glycol. Main inputs include around 14,000 kg/h of ethylene supplied by the olefins plant, approximately 16,000 kg/h of oxygen, and about 10,000 kg/h of water used in the hydration reaction. Outputs include roughly 16,800 kg/h of monoethylene glycol, of which approximately 4,200 kg/h is supplied to the PET production facility (matching PET's scaled MEG requirement) and the remaining approximately 12,600 kg/h is sold to external markets, about 1,500 kg/h of diethylene glycol as a secondary by-product, and an estimated 30 to 50 MW of recoverable thermal energy that can potentially be recovered and utilized elsewhere within the industrial park. At 8,000 operating hours per year, the 16,800 kg/h MEG output corresponds to 134.4 kt/year, consistent with the approximately 135 kt/year capacity stated above and forming the production basis for the Chapter 5 EO/EG utility import (396 kJ/kg MEG) and waste heat (732 kJ/kg MEG) figures. This is consistent with the Scope 1 GHG figure of 305,088 tCO₂e/year (2.27 tCO₂e/t MEG × 134,400 t/year) reported in Chapters 4 and 5, corrected from an earlier draft value of 2,661.63 tCO₂e/year that did not match this production basis. An example is the Antwerp cluster: INEOS Oxide in Zwijndrecht and the BASF EO/EG production unit.

5. PET Production Plant

The PET production facility synthesizes polyethylene terephthalate, one of the most widely used polymer materials in packaging and textile industries. The process involves a polycondensation reaction between purified terephthalic acid (PTA) and monoethylene glycol (MEG). PTA, derived from paraxylene in the aromatics unit, reacts with MEG under elevated temperatures and reduced pressure conditions. This reaction forms long polymer chains of PET while releasing water as a by-product. The molten polymer is subsequently cooled, solidified, and cut into pellets for commercial use. The base capacity listed in the reference data sheet is approximately 1 kiloton per year. However, in order to balance the upstream flows from the aromatics and EO/EG plants, the facility would need to be scaled up by roughly two orders of magnitude in the system mass balance analysis. Under a scaled configuration of approximately 100 kilotons per year, the plant would require about 8,600 kg/h of paraxylene (converted to PTA) and around 4,200 kg/h of MEG supplied from the EO/EG unit. Outputs would include approximately 12,500 kg/h of PET resin in pellet form and about 1,500 kg/h of wastewater generated from the esterification reaction. All energy figures and intervention calculations presented for the PET plant in Chapter 5 correspond to this scaled 100 kt/year configuration, not the 1 kt/year reference case. With a production rate of 12,500 kg/h and 8000 operating hours per year, using the same operating basis applied to the CHP baseline in Chapter 5, the output stream corresponds to an annual production of exactly 100 kt. This confirms the scaled mass balance used in the Chapter 5 PET energy assessment. The reported recoverable heat potential of 5.684 TJ/year and heat demand of 6.049 TJ/year equate to approximately 56.8 KJ/Kg and 60.5 KJ/kg PET respectively. These values are consistent with the thermal energy requirement reported in the literature for esterification and polymerization processes, supporting the validity of the underlying assumption. An example of large-scale PET manufacturing in the region is the JBF Global Europe facility located in Geel along the Albert Canal, which is directly integrated with the Antwerp petrochemical cluster.



In addition to the five core facilities, the system boundary includes links to several external symbiosis partners that are not located within the industrial park but are relevant to the proposed interventions discussed in later chapters. A Urea Plant situated outside the park is identified as a potential recipient of captured CO₂ from the EO/EG plant, creating an opportunity for carbon capture and utilization (CCU). Similarly, nearby greenhouses with a demand for low-grade heat (70–85°C) are considered potential off takers of upgraded waste heat recovered from the EO/EG plant through a high-temperature heat pump. As these collaborations have not yet been established, the external entities are included in the stakeholder analysis as prospective rather than confirmed symbiosis partners.

3.1.1 Stakeholder Analysis

The implementation of the PolyCycle Hub Antwerp requires coordination among multiple actors that influence the functioning of the industrial ecosystem. A stakeholder analysis was conducted to identify the key actors involved in the system and understand their roles. Internal stakeholders include the companies operating inside the industrial park. Chemical plant managers responsible for facilities such as Olefins, Aromatics, EO/EG, and PET plants are central to the system. These plants are typically operated by companies active in the Antwerp cluster such as BASF, INEOS, Borealis, and TotalEnergies. Their role is to manage the production of chemical intermediates and polymers while maintaining safety, efficiency, and profitability. These facilities exchange material streams such as ethylene and pygas and rely on shared utilities like steam and heat networks, making them key actors in maintaining the industrial symbiosis polymer loop. Another important internal stakeholder is the natural gas combined heat and power (CHP) operator, often operated by energy providers such as ENGIE or Electabel in the Antwerp region. The CHP plant produces electricity and high-pressure steam that are distributed throughout the industrial park, while also generating significant CO₂ emissions associated with natural gas combustion. Because most facilities depend on this energy supply, the CHP operator acts as the central energy provider within the system. External stakeholders also play a significant role. The Port of Antwerp–Bruges Authority is a key actor because it manages the NextGen District where the PolyCycle Hub is located. The authority coordinates land leases, logistics infrastructure, and collaboration among industrial tenants. Government and regulatory bodies influence the system through environmental and economic policy. Important institutions include the European Commission and the Flemish Environmental Agency. These organizations regulate emissions, enforce environmental permits, and implement policies such as the EU Emissions Trading System (ETS), which directly affects the economic viability of shared heat and energy networks. Universities and research institutions contribute to technological innovation and sustainability assessments. Key partners may include the University of Antwerp (UAntwerp) and VITO (Flemish Institute for Technological Research). These institutions support the project through research on process optimization, industrial symbiosis design, and environmental performance evaluation. Investors and financial institutions such as commercial banks, infrastructure funds, and corporate shareholders provide the capital required to build shared infrastructure, including shared utility infrastructure, wastewater treatment facilities, and energy-integration projects. Finally, the local community and environmental NGOs represent societal stakeholders who are affected by emissions, noise, and economic development in the port area and can influence the project through public opinion and regulatory processes.

3.1.2 - Long-Term Goals & Vision Development

This section consists of three parts; first the climate targets and introduction to the project's vision development will be discussed, followed by the alignment of policies and strategically mapping, including three specific research questions to guide companies in the system boundary to reach their sustainability objectives. The last section describes how the project can be translated from vision to tangible objectives, by following a three-step timeline, by using the Back Casting methodology.

3.1.2.1 Climate targets & Vision development introduction

The PolyCycle Port aims to improve resource efficiency, reduce fossil-fuel dependence, and strengthen industrial symbiosis within the industrial park. Achieving these objectives requires the integration of energy, material, and water flows across the industrial network. As a major industrial cluster, the Port of Antwerp–Bruges faces increasing pressure to reduce greenhouse gas emissions while maintaining industrial competitiveness and resource security. Within the defined system boundary, the CHP plant remains a major source of direct CO₂ emissions due to its dependence on



natural gas. Therefore, the project's vision focuses on reducing the carbon intensity of the industrial park through energy-efficiency improvements, renewable-energy integration, material recovery, and water reuse strategies. The PolyCycle Port's vision prioritizes "Water Autarky" as a response to the severe risk of water scarcity in Flanders. In particular, the vision replaces general regional targets with an "Internal Water Backbone", designed to recover and treat wastewater streams generated by the Aromatics plant for reuse within the industrial park. This vision supports the principles of both "Waterkracht" and "CHERISH20" by improving water reuse, reducing freshwater consumption, and strengthening water circularity within the project boundary. The final pillar of the vision is to reduce dependence on virgin fossil-based naphtha by increasing the recovery and utilization of post-consumer PET waste and other secondary material streams as alternative feedstocks within the industrial park. By valorizing hydrocarbon by-products and improving material circulation across the petrochemical network, the system can improve resource efficiency and strengthen material circularity. This transition may be further supported by digital monitoring and resource-tracking tools to improve transparency and decision-making across the industrial ecosystem.

3.1.2.2 Aligning policies & mapping strategically

The Port's central position in Europe's strategy for achieving "Open Strategic Autonomy" aligns with the objectives of the Net-Zero Industry Act (NZIA). Antwerp also plays a critical role in the Flemish Energy and Climate Plan (VEKP), serving as a strategic location for industrial innovation and sustainability transitions. To reach these objectives, the project vision is aligned with the Warmtenet Antwerpen framework, which promotes the utilization of industrial waste heat to reduce fossil-fuel consumption. Furthermore, the project supports regional ambitions for water reuse and resource efficiency through improved water management and industrial symbiosis practices.

The vision is in line with the UN 2030 Agenda, as it promotes SDG 6 (Clean water), SDG 7 (Affordable energy) and SDG 9 (Infrastructure). By implementing energy-efficiency measures, renewable-energy integration, material recovery, and water reuse strategies, the park is able to separate industrial growth from carbon emissions (Atlas Copco, 2022). This project will use three specific research questions to guide the analysis of how the 5 companies in the system boundary can reach their sustainability objectives.

1. How can CHP Utility-Hub Retrofit measures, including waste heat recovery and renewable-energy integration, reduce the fossil-fuel dependence and carbon intensity of the industrial park?
2. How can the recovery and utilization of hydrocarbon by-products and reintroduce post-consumer PET waste reduce dependence on virgin fossil-based feedstocks and improve material circularity?
3. How can wastewater treatment and reuse reduce freshwater demand and improve water circularity within the industrial network?

3.1.2.3 - Transitioning from Vision to Tangible Objectives

To achieve translating PolyCycle's vision into tangible objectives, this project combines the back Casting methodology, with the SMART criteria. To be able to make the substitution of fossil fuel-based steam a reality, the energy objective focuses on reducing the industrial park's dependence on fossil-fuel-based utilities through CHP Utility-Hub Retrofit measures. These include waste heat recovery, renewable-energy integration, and electrification strategies that improve energy efficiency while reducing the carbon intensity of steam and electricity generation. The material objective aims to reduce dependence on virgin fossil-based feedstocks by increasing the recovery and utilization of hydrocarbon by-products and secondary material streams.

By improving material circulation within the petrochemical network, the industrial park can strengthen resource efficiency and circularity. The water objective operationalizes the Flemish Blue Deal 2.0 by establishing a common water backbone. By treating and reusing BTX-containing wastewater streams, the project aims to reduce the demand for external freshwater within the industrial park, particularly for utility and process applications. This supports improved water circularity and resource efficiency across the system. The vision is translated into a three-step timeline, defined below.

- I) By 2027 - Implement CHP Utility-Hub Retrofit measures, including waste heat recovery systems and renewable-



energy integration.

II) By 2028 - Establish hydrocarbon recovery and authorize the PET-to-Naphta chemical recycling pilot (PureCycle) to begin decoupling from virgin feedstock.

III) By 2030 - Achieve significant wastewater reuse within the industrial park and reduce freshwater consumption through the Internal Water Backbone.

3.2 Stakeholder Analysis

A condensed overview of key stakeholder roles is presented below. Detailed descriptions of stakeholder functions and interactions are provided in Appendix A.

3.2.1 Roles and Operations

Stake Holder	Key Roles	Role in the Symbiosis Network	Motivation
Port of Antwerp-Bruges	Port Authority manages land concessions, shared infrastructure, and permitting	Approves symbiosis infrastructure: coordinates shared investments (pipelines, utilities)	Advance NextGen District's circular mandate and maintain Antwerp's competitive position
EO/EG Plant Operator (eg. INEOS / Borealis)	Asset owner manages operations, catalysts, emissions, and CAPEX	Central node for production, utilities, and process upgrades	Reduce EU ETS liability and avoid stranded asset risk under Fit-for-55
Olefins Plant (eg, BASF TotalEnergies)	Producer of ethylene via steam cracking	Supplies ethylene feedstock required for EO production through a dedicated pipeline	Secure long-term off-take and reduce shared fossil feedstock price exposure
PET Plant	Downstream polymer production facility using MEG as a co-monomer	Main offtake partner receiving MEG from the EO/EG plant	Ensure MEG supply continuity and meet downstream sustainability requirements
CHP Plant (eg. ENGIE / Electrabel)	Combined heat and power facility supplying steam and electricity to the industrial park	Provides process steam required for EO/EG operations through the shared steam grid	Transition fuel mix toward hydrogen / biomethane ahead of rising carbon costs
Urea Plant	Fertilizer production facility using CO ₂ as feedstock	Potential receiver of captured CO ₂ from EO/EG operations through CCU symbiosis	Reduce external CO ₂ procurement costs and improve emissions profile
Centralised WWTP	Shared wastewater treatment facility serving multiple park plants	Treats purge water streams from EO/EG operations and enables recovery or compliant discharge	Expand through pilot and demonstrate circular water management under Flemish

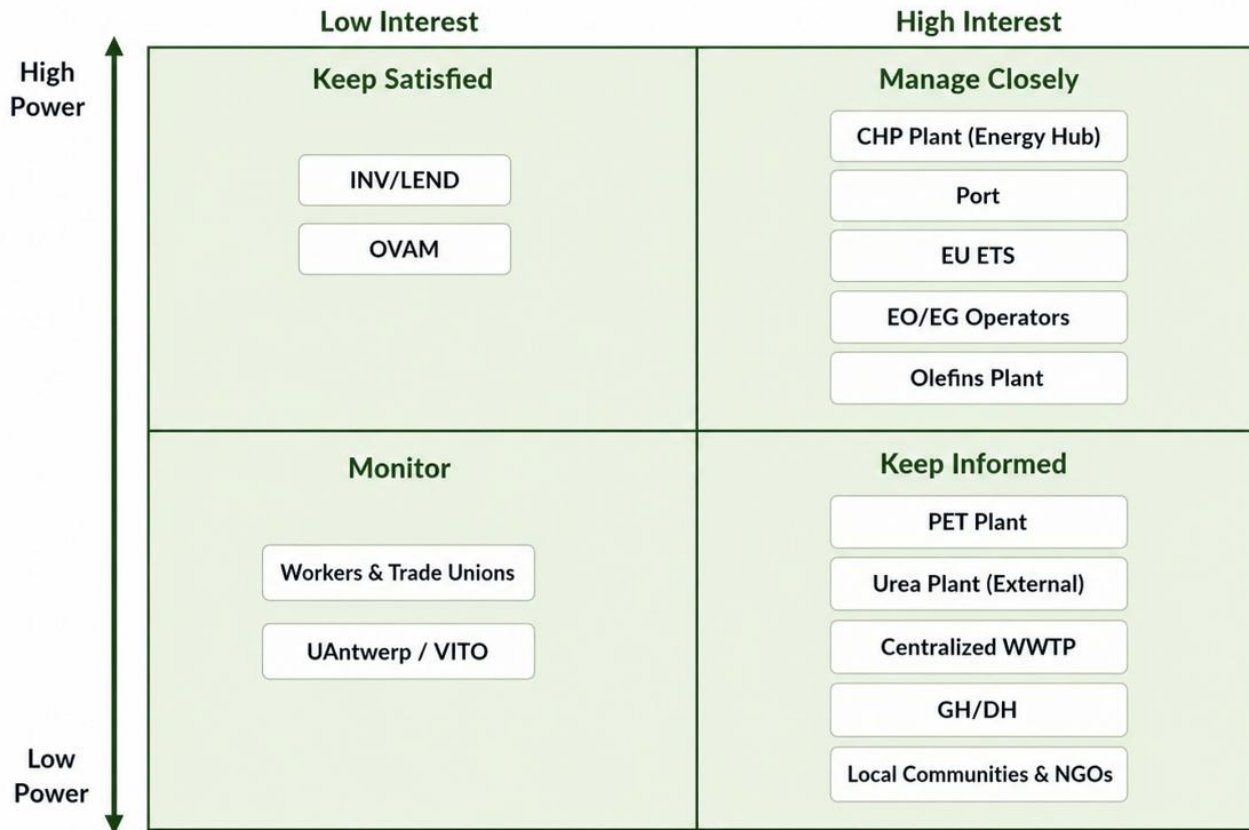


			Blue Deal 2.0
European Commission / EU ETS	Policy authority establishing carbon pricing and climate regulation	Indirect drivers influencing the economic feasibility of emissions reduction interventions	Accelerate industrial decarbonization toward Fit-for-55 targets
Flemish Government Agency (OVAM/VMM)	Regional environmental regulator enforcing VLAREM II standards	Issues environmental permits and monitors compliance with emissions and discharge limits	Demonstrate measurable industrial transition progress under VEKP 2021-2030
Local Communities & NGOs	Residents located near the industrial park	Affected by emissions, safety, and environmental performance of industrial operations	Achieve tangible air quality improvements and visible environmental progress
UAntwerp / VITO	Process optimization and industrial symbiosis assessment	Applied research, pilots, and LCA assessments	Secure research funding & contribution to regional transition policy
Investors / Lenders	Financial Institution providing capital for industrial projects	Evaluating investment viability based on ESG performance and EU Taxonomy alignment	Deploy capital into Taxonomy-aligned assets and meet institutional ESG mandates
Workers & Trade Unions	Workforce operating and maintaining the plant	Participate in operations, safety, and transition planning	Secure employment continuity and retraining opportunities during upgrade

Table 3.1 Stakeholder role, interaction, and motivation

3.2.2 Interests and Influence

Figure 3.1 maps all stakeholders using the Mendelow (1991) power-interest framework. Power reflects a stakeholder's capacity to influence operations, investment, or regulation. Interest reflects the extent to which they are directly affected by the industrial park's performance, with particular emphasis on the CHP plant given its role as the central energy hub for the integrated intervention strategy. Four management strategies follow from this mapping.



Note: CHP Plant is Shown as the central Energy Hub, consistent with the integrated intervention strategy presented in the final project deliverable. PET plant is placed in “Keep Informed” given its role as a downstream offtake receiver with comparatively limited direct influence over upstream intervention decisions.

Figure 3.1 Power-interest grid for PolyCycle Port stakeholders (Mendelow, 1991).

Quadrant	Stakeholder	Engagement Strategy
Manage Closely	CHP Plant (Energy Hub), Port of Antwerp-Bruges, EU Commission/ETS, EO/EG Plant Operator, Olefins Plant	Core Decision-makers- ongoing involvement in investments, design, and regulatory approvals
Keep informed	PET Plant, Urea Plant, Centralized WWTP, Greenhouses/DH Operator, Local communities	High Interest, limited unilateral power-Provide updates and symbiosis communications
Keep Satisfied	Investors, Lenders, Flemish Government (OVAM/VMM)	Maintain through KPI reporting and EU Taxonomy alignment
Monitor	Workers Trade Unions, UAntwerp/VITO	Low operational engagement-escalate involvement during turnarounds and capital projects

Table 3.2 Stakeholder management strategies by power-interest quadrant.



Potential Conflicts and Tensions

Despite the above-structured engagement, key risks remain. Misalignment between the investment priorities of the EO/EG operator and the port Authority's timeline may delay implementation, which is a common issue in industrial symbiosis (Tudor et al. 2007). Continued reliance on gas in the CHP plant risks reinforcing technological lock-in and complicating compliance with EU decarbonization policies (European Commission, 2024). Failure to meet EU Taxonomy criteria may also weaken investor confidence (European Commission, 2020; Veugelers et al., 2024). Limited evidence of emissions reductions could further undermine social license to operate (Jijelava & Vanclay, 2017).

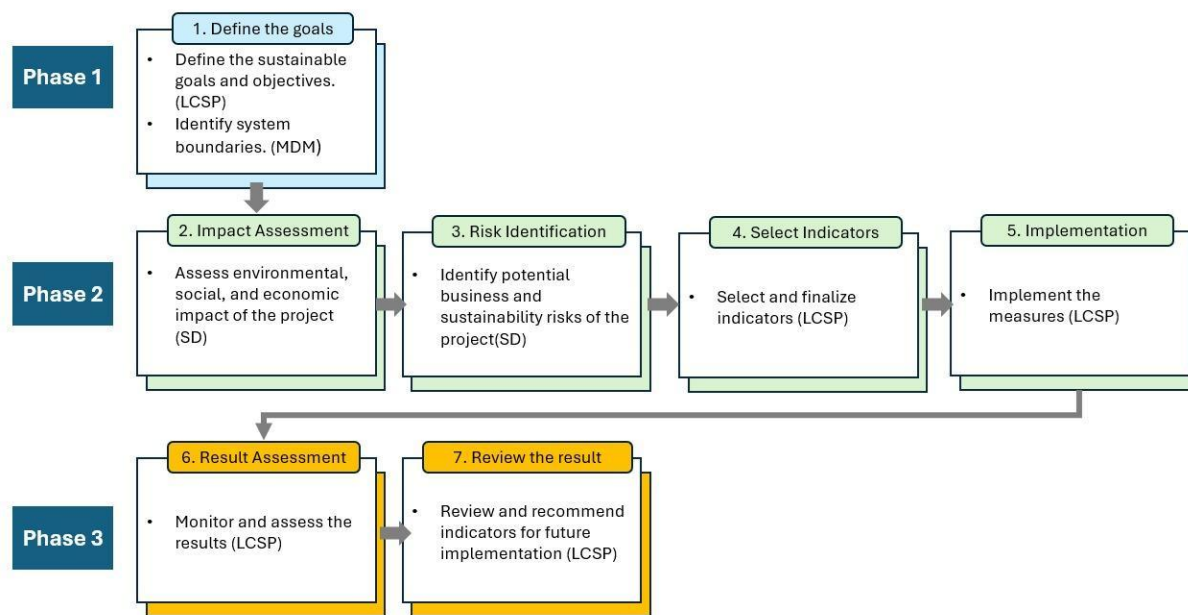
Integration with Methodology

These risks, as indicated above, inform the methodology, where key tensions are treated as systemic risks and guide indicator selection based on stakeholder priorities (Mendelow, 1991).

3.3. Research Approach

This report adopts the Lowell Center for Sustainable Production (LCSP) Framework as the primary research design, complemented by the multilevel design model (MDM) and the systemic double-flow scenario method (SD), which together integrate sustainability principles, system-level analysis, and transition-oriented scenario thinking. The strength of LCSP is that it clearly defines indicators in multiple perspectives while also providing solid frameworks and steps to follow. These combined frameworks broaden and deepen the analytical perspective when developing a comprehensive transition plan for the Polycycle Industrial Site at the Port of Antwerp.

The research approach is designed as three phases method:



Phase 1: Finding Current Problems

a. Define the goals and system boundaries

The project primarily focuses on environmental sustainability, resource efficiency, circularity, and industrial symbiosis, while acknowledging the broader organizational and social dimensions identified within the LCSP framework. Additionally, by adding the MDM, the relevant system boundaries are further defined across multiple levels, ranging from a very specific Product-Technology System to broader societal contexts (Product-Service System, Socio-Technical System, and Societal System). In this way, the scope of the research question is clarified by positioning the Polycycle Industrial Site within a broader industrial and societal transition towards circular economy practices.



Phase 2: System Analysis and Strategy Development

a. Impact Assessment

Based on the defined goals and system boundaries, the SD method is applied to analyze material and energy flows, mass and energy balances within the industrial ecosystem across the defined system levels, thereby enabling the identification of key leverage points to improve circularity and sustainability performance. In addition, stakeholder analysis, policy alignment, and preliminary economic considerations are assessed to understand the broader factors influencing the transition of the industrial park.

b. Risk identification

Identifying risks is a necessary step to uncover potential challenges that might be overlooked when pursuing a high-performance, circular industrial site. This concept stems from the SD, which is used to identify sustainability risks, business risks, and possible mitigation/adoption measures. In addition, the relationships between these risks and potential responses are analyzed to better understand how different factors may influence the transition process. These risks are examined across the defined system levels to understand how technological, organizational, and societal factors influence the transition process.

c. Select Indicators

After the previous steps, the system's objectives, system dynamics, and relevant risks have been identified. Based on this understanding, appropriate indicators are selected to measure the outcomes of the proposed transition strategies. These indicators, both qualitative and quantitative, enable performance quantification, support the evaluation of circularity improvements, and provide a basis for monitoring progress and guiding future adjustments.

d. Implementation

In this stage, potential intervention scenarios are developed and evaluated. Based on the identified system inefficiencies, risks, and performance indicators, targeted energy, material, and water interventions are designed and evaluated. This may involve collaboration between industrial actors, infrastructure adjustments, and the development of industrial symbiosis to facilitate material, water, and energy exchanges within the site.

Phase 3: Evaluation and Completion

a. Result Assessment:

After developing the proposed intervention strategies, a results assessment is conducted to evaluate their effectiveness and efficiency in achieving the defined objectives using the selected indicators. The outcomes of the transition measures are analyzed in relation to the project goals, system performance, and identified sustainability challenges. This assessment provides insights into whether the proposed strategies effectively contribute to the desired outcomes and support the transition towards a more sustainable industrial system.

b. Review the result

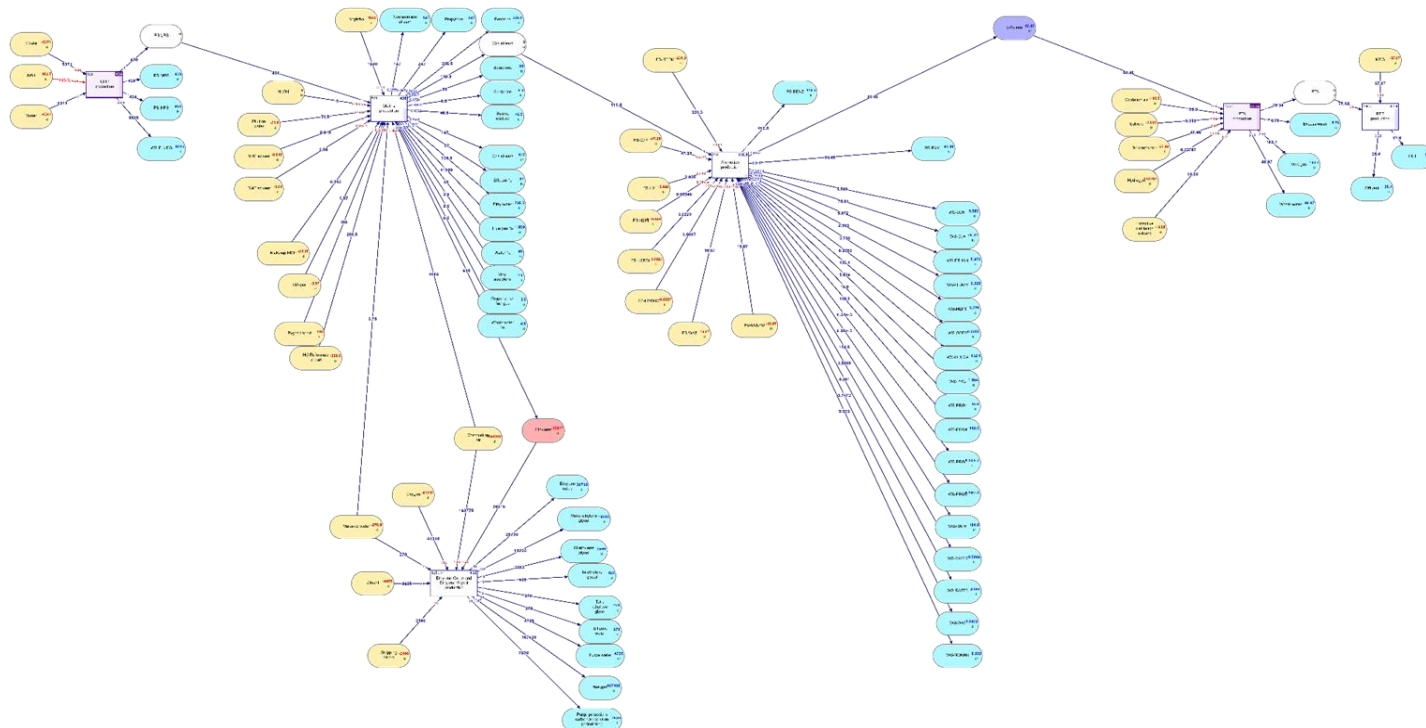
Following the previous step, the results are reviewed to determine whether the proposed measures are sufficient to address the identified sustainability challenges and risks. Based on this reflection, potential improvements and recommendations are proposed to further enhance the transition process and support the long-term achievement of the defined goals and objectives.

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Chapter 4. Material Analysis

4.1 Mass balance analysis



System Material Interaction Diagram of the PolyCycle Industrial Park

4.1.1 Material Analysis of PolyCycle Port in Belgium

In a petrochemical eco-industrial park, carbon serves as the core building block, a primary energy source, and the key indicator of environmental performance. The carbon cycle starts at the system boundary with a Combined Heat and Power (CHP) plant that transforms natural gas into steam and energy. Additionally, the Olefins plant uses all types of Naphtha, while the Aromatics plant depends on dense Reformate and Pygas. Once inside the system, carbon experiences significant structural changes while its total mass remains constant. In steam crackers, long aliphatic chains are broken down into valuable monomers like ethylene, along with heavy byproducts such as pygas. These byproducts are not wasted but directed to the Aromatics plant, where the carbon is reorganized into stable benzene rings, mainly producing p-Xylene and Benzene.

Carbon thus serves as the physical bridge connecting disparate facilities. Ethylene flows to the EO/EG plant to form MEG, while p-Xylene travels to the PET plant. Ultimately, the PET facility serves as the primary carbon sink, permanently locking the carbon in flowing streams into stable commercial polymers. Conversely, surplus carbon intermediates are exported to external markets, while carbon oxidized during operations—such as in the CHP and EO/EG plants—is released as Carbon Dioxide, representing the system's environmental footprint.

4.1.2 Methodology: Constructing the Sankey Diagram

The methodology began by defining the system boundaries. Because the five facilities reported data using different metrics, all mass flows were normalized to a functional baseline production capacity for the final PET product. Next, to prevent non-carbon elements (such as hydrogen and oxygen) from skewing the visual balance, chemical stoichiometry was applied to convert total stream weights to pure elemental carbon mass. For complex mixtures, detailed Gas Chromatography (GC) compositional data were used to calculate highly precise harmonic-mean molecular weights and carbon mass fractions (e.g., 88.86% for Pygas and 87.60% for Reformate). The key analytical step was mass balance reconciliation. Applying the Law of Conservation of Mass at each node revealed hidden

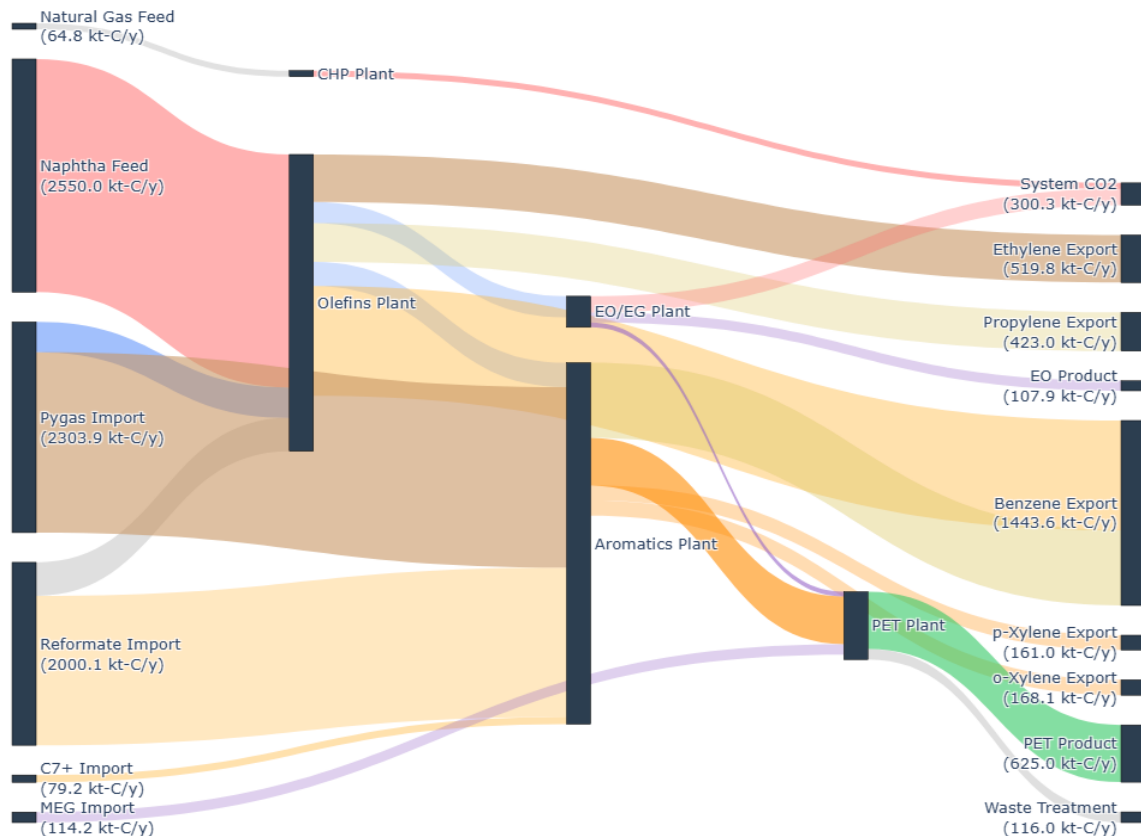


discrepancies in the system. For example, stoichiometric auditing exposed a Pygas production shortfall at the Olefins plant, mathematically demonstrating the need to import external Pygas to meet the Aromatics plant's requirements.

Finally, the open-source Python library `plotly.graph_objects` was used for mapping precise source nodes, target nodes, and reconciled carbon values. This dynamic model consolidates fragmented operational data into a unified macro-level overview, forming the critical foundation for computing the park's overall environmental performance indicators.

Eco-Industrial Park: Ultimate Elemental Carbon Flow (kt-C/y)

Validated with detailed compositions and mass balance



Elemental Carbon Flow Sankey Diagram of the PolyCycle Industrial Park

4.1.3. Water Flow Analysis

For the water flows within the industry cluster, we identify all potential inflows and outflows that contain water. In addition, the level of water composition and the status of the stream is considered to select the most relevant material flows to be analyzed. For example, the In the next steps, the flows are analyzed across categories, including sources, process use, water types, and the main impurities in the water stream. The sources indicate the previous production step in the current flow, and the water sources are assumed to be either municipal or surface water, depending on the required water purity. In addition, the process use of the stream is the major highlight of the analysis, as we identify its purpose in the water stream's production process. It enables us to fully understand the role of water in different processes and to determine whether there are any alternative water solutions beyond the current ones. After analyzing the stream's origin and function, the study focuses on treating water flows, as the primary goal of industry symbiosis is to increase the circularity of water use within the industrial park. The main impurities and their percentage in the aqueous solution determine the water treatment process and water types. The quality of wastewater is classified into three types: biodegradable, poorly biodegradable, and toxic organics. Furthermore, we select only two compounds: the top two and those over 1% in the stream to present in the table. For more detailed information on the water flow analysis, please refer to Appendix C- Water Flow Analysis.



4.2 Performance evaluation

To evaluate the current performance of the Polycycle industrial park and support the development of potential interventions. As discussed in chapter 2, the industrial park faces three main challenges: fossil-fuel dependence and carbon emissions, inefficient use of material resources, and limited water circularity. Therefore, the selected indicators aim to assess resource efficiency, waste generation, and carbon intensity across the industrial network. The results provide a baseline for identifying critical inefficiencies and support the selection and evaluation of interventions in subsequent chapters.

Firstly, to evaluate the dependence on virgin materials and identify opportunities for improving resource efficiency and circularity, material intensity (mass intensity) is selected as a key indicator and is included below to analyze the performance in current practice assess if the dependence on the virgin material is reduced and measure the effectiveness of the transition from a linear system to a circular material loop in later chapter.

Production	Formula	Material intensity
CHP	Energy provider	-
Olefins	(Naphtha + NaOH + NFM solvent + DMF solvent +make-up NMP + H2 gas + pygas+ HC reformat) / Ethylene	≈4.35
Aromatics	(REFM + C7+ pygas + H2 + H2IN + H2TOL + HYDRO + SULFO) / p-Xylene	≈4.36
Ethylene Oxide and Ethylene Glycol	Ethylene / Mono Ethylene Glycol	≈1.99
PET	(p-Xylene + solvent + hydrogen + Extractive distillation solvent + MEG) / PET	≈1.24

Material intensity Assessment of the PolyCycle Industrial Park

Based on the results above, the PET and the EO/EG production processes show relatively lower mass intensity values, indicating higher material efficiency compared to upstream processes. On the other hand, the olefins and aromatics processes demonstrate significantly higher material intensity, which means that the material losses are significant and chemical transformation could be inefficient. This highlights that upstream processes are the main contributors to resource inefficiency and therefore represent key targets for interventions aimed at reducing virgin material and improving circularity within the system. Secondly, the E-factor is included to evaluate process sustainability by measuring the amount of waste generated per unit of product, which helps determine the efficiency of material use and identify processes with high levels of waste generation. This indicator is particularly relevant for assessing waste generation and identifying streams with the highest potential for valorization and industrial symbiosis.



Production	Formula	E-factor
CHP	Energy provider	-
Olefins	(Flue gas + Propane rich fuel gas + Vinyl acetylene + C7+ stream) / Ethylene	≈27.74
Aromatics	(2C9+ + C9+ + ETHAN + HEAVY + HEPT + OFFG + OFFGA + PRG + PRG1 + PRG4 + PRG5 + PRG6 + RAFF + RAFF2 + RAFF3 + NONAR) / p-Xylene	≈5.05
Ethylene Oxide and Ethylene Glycol	(Flue gas + Purge potassium carbonate solution)/ Mono Ethylene Glycol	≈9.61
PET	Vent gas / PET	≈2.09

E-factor Assessment of the PolyCycle Industrial Park

The result indicates that olefins and EO/EG have relatively high E-factors that exceed the Dtolerance (typical bulk chemicals should be below 5), with olefins showing particularly high levels of waste generation. Aromatics also shows slightly higher than the tolerance. These findings suggest that significant quantities of potentially valuable material streams are currently leaving the system without being recovered or utilized. As a result, the Olefins and Aromatics plants represent key targets for material recovery and industrial symbiosis interventions. By valorizing by-products and redirecting waste streams into productive applications, the industrial park can reduce material losses, improve resource efficiency, and strengthen overall circularity performance. Thirdly, the GHG Emission Intensity Ratio (Scope 1) is selected to evaluate the degree of fossil-fuel dependence and support future decarbonization measures and monitor the intensity of GHG emissions relative to production output.

Production	Formula	GHG Emission Intensity Ratio (Scope1)
CHP	FLUEG * 0.059 / (LPS + MPS + HPS)	≈0.25
Olefins	Flue gas * 0.0774 / Ethylene	≈0.026
Aromatics	No direct CO2 emissions assigned	-
Ethylene Oxide and Ethylene Glycol	(0.0086 * Ethylene oxide + 0.2 * Flue gas + 0.98 * Purge potassium carbonate solution)/ Mono Ethylene Glycol	≈2.27
PET	0.0021 * Vent gas / PET	≈0.004

GHG Emission Intensity Ratio (Scope1) of the PolyCycle Industrial Park



The calculation shows that EO/EG is the main contributor to direct GHG emission among the processes, while PET is the least carbon-intensive activity. Therefore, emission reduction efforts should focus on energy-intensive processes, such as EO/EG and CHP production.

Overall, the selected indicators establish a quantitative baseline for the PolyCycle industrial park and support the evaluation of research objectives and assessing the implementation outcomes in later chapters. Material Intensity and E-factor identify opportunities for improving resource efficiency and material circularity, while GHG Emission Intensity highlights key sources of carbon emissions. These three indicators help understand the performance in each plant and guide the selection of energy, material, and water interventions presented in the following chapters and provide a basis for evaluating their effectiveness.

4.3 Identification of Industrial Symbiosis Opportunities

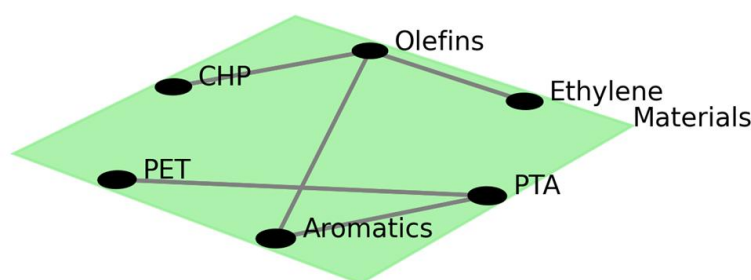
The industrial symbiosis network is first illustrated at the company level, as shown in the diagram and matrix below, to highlight the key exchanges between major production units.

The network is structured around two main material pathways. The first pathway is based on ethylene, where the olefins plant supplies ethylene to the EO/EG unit. CHP plant also provides steam to olefins plants. The second pathway is based on aromatics, where p-xylene is produced in the aromatics plant and supplied to the PTA plant, forming a critical precursor for PET production.

These two pathways are interconnected through the exchange between the olefins and aromatics units. Pygas produced in the olefins plant is transferred to the aromatics unit, where it is further processed to extract valuable BTX components, including p-xylene.

In addition to material exchanges, the CHP plant supports the system by supplying steam to multiple units. Although this flow is energy-based rather than material-based, it enables the key chemical transformations within the network.

Overall, the network highlights a highly interconnected system in which a limited number of core plants control the majority of material flows, forming the backbone of the industrial symbiosis structure.



Material Flow Network of the PolyCycle Industrial Park



	CHP	Olefins	Aromatic s	Ethylene	PTA	PET
CHP	0	1	0	0	0	0
Olefins	1	0	1	1	0	0
Aromatics	0	1	0	0	1	0
Ethylene	0	1	0	0	0	0
PTA	0	0	1	0	0	1
PET	0	0	0	0	1	0

Binary Connectivity Matrix of Material Exchanges

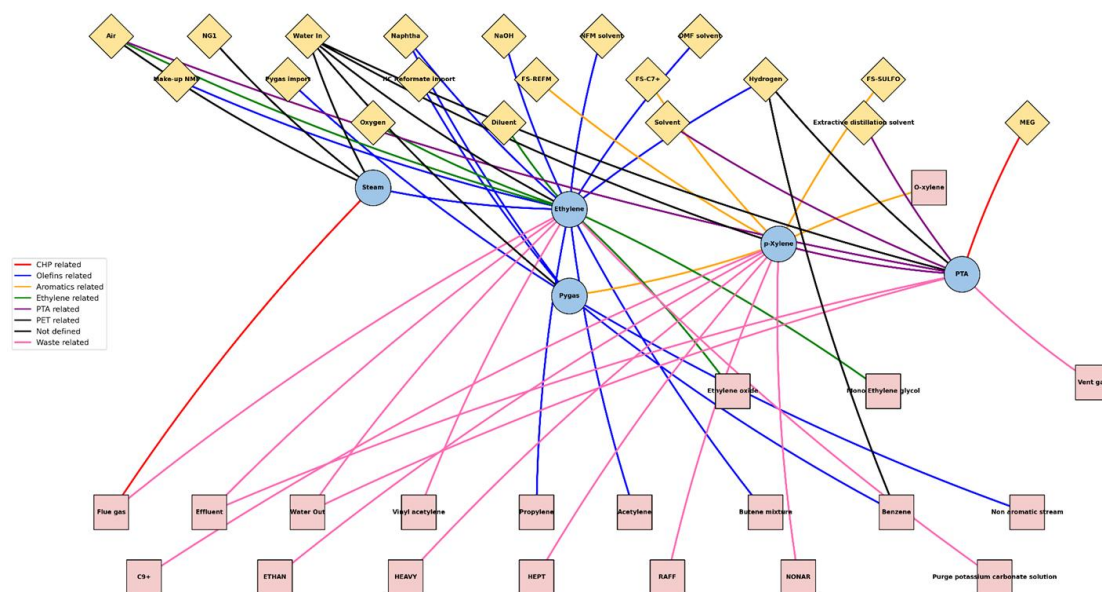
To further identify potential industrial symbiosis opportunities, a detailed material flow structure of the industrial park is presented. This figure illustrates the transformation of key raw materials, intermediate products, and waste streams across different production units, providing an overview of material flows within the system and a basis for evaluating the potential reuse of by-products and waste. At the center of the network, ethylene acts as the main hub that connects multiple upstream inputs and downstream outputs. A wide range of raw materials, including naphtha, steam, solvents, hydrogen, and imported streams, converge into the olefins process, where ethylene is produced. From this central node, ethylene is distributed to various pathways, including multiple by-product streams such as propylene, acetylene, and vinyl acetylene. A second important cluster is formed around p-xylene and PTA. Pygas from the olefins unit is transferred to the aromatics process, where it is converted into p-xylene. This p-xylene is then supplied to the PTA plant, which serves as a key intermediate step toward PET-related production. Compared to ethylene, this pathway is more sequential and represents the aromatics-based material route. The diagram also highlights the large amount of waste and by-product streams generated throughout the system. These include heavy fractions such as C9+, raffinate, non-aromatic streams, as well as emissions such as flue gas, effluent, and purge streams. Many of these flows are currently directed out of the system, indicating a predominantly linear material structure.

Overall, the network reveals a highly centralized and resource-intensive system, where ethylene and p-xylene act as key intermediates, while a significant portion of material exits the system as waste. This structure provides a clear basis for identifying opportunities to recover by-products and improve material circularity. Based on the identified material flow structure, several potential industrial symbiosis opportunities can be observed.

Firstly, several potentially valuable by-products are generated in the olefins unit, such as propylene, acetylene, butene mixture, benzene, and non-aromatic streams, could be utilized either internally or by nearby facilities instead of being treated as waste or low-value outputs. For example, propylene can be feedstock of nearby petrochemical facilities for further processing, such as polymer or chemical production. This is technically feasible in a larger scope of industrial park, where this manufacturing exists, and this could reduce material losses and improve overall resource efficiency. Acetylene can be hydrogenated into ethylene and reintroduced into the system, while can be further utilized as a feedstock in aromatic processing units, given its role as a key component in BTX chemistry. These are technically feasible as hydrogenation is a standard industrial process used to remove impurities. This allows recovery of valuable carbon, improves resource inefficiency and increases overall yield while reducing material losses. Second, heavy fractions from the aromatics process, including C9+, raffinate, and non-aromatic streams, can be utilized as low-grade fuel in the CHP unit or reintroduced into the olefins process as secondary feedstock. This is technically feasible due to their hydrocarbon composition and compatibility with combustion and cracking processes. Utilizing these streams would reduce waste generation, decrease reliance on virgin feedstock, and improve overall resource efficiency.

Finally, wastewater and purge streams generated across multiple units could be treated and reused within the system, which can improve overall water circularity within the park.

In general, a number of currently underutilized flows could be redirected within the system, enhancing circularity and reducing material losses.



Industrial Symbiosis Material Network of the PolyCycle Industrial Park

4.4 Selection of companies for further assessment and intervention design

ExxonMobil was selected for this study as it is the largest operator within the Antwerp port industrial cluster. According to its 2024 annual report, the site has a massive daily refining capacity of 309,000 barrels. This extensive upstream refining capability ensures a stable and substantial supply of feedstocks (such as naphtha) for downstream aromatics production, solidifying Antwerp as a crucial node in ExxonMobil's European operations. While specific revenue figures for the Antwerp site are publicly limited, this immense material throughput serves as a strong proxy, indicating its significant economic contribution to the industrial cluster." However, as EU regulations continue to tighten on sustainability, there is growing concern at Exxon Mobil about continuing operations in Antwerp. According to the recent report, Exxon Mobil halted investment of 100 million euros in plastic recycling infrastructure in Antwerp due to financial unviability. On top of that, the company's vice president also said the company was considering the potential sale or closure of plants in Belgium. This is a warning sign for the petrochemical cluster, as a lack of effort and awareness in transitioning to sustainability could hinder the overall development of industrial symbiosis. Furthermore, misaligned investments in traditional infrastructure could create a technological lock-in effect, ultimately limiting the cluster's ability to improve material efficiency and waste treatment in the long run.

Challenges and problems

In chapters 4.1 and 4.2, we conducted multiple analyses of material use across various processes in the Antwerp petrochemical industrial cluster. In the series of analyses, material intensity, E factor, and greenhouse gas emissions intensity were examined. However, the environmental analyses indicated ample room for improvement for the aromatic companies, especially in waste treatment and material intensity. The aromatic companies have relatively low E-factor values, indicating that waste generation needs to be reduced. In addition, the material intensity of the aromatics process is high, indicating a strong dependence on virgin material. Furthermore, aromatics are among the most crucial base chemicals in the petrochemical industry, forming the foundation for the production and applications of various commodities. Therefore, the production quantity of the aromatics units is considerable, and the waste generated is substantial. Based on the current data we obtained, the waste stream flows from the aromatics units amount to 2900 kilotons per year. Additionally, existing studies indicate that waste from aromatics negatively impacts human health and the environment, either directly or indirectly. Aromatic chemicals can have multiple toxic effects on humans, depending on age and health status. For example, a 2016 study by Abdel-Shafy and Mansour reports that potential health effects of aromatic chemicals in humans include eye irritation, vomiting, diarrhea, disorientation, skin irritation, and inflammation. Furthermore, according to Garcia-Sanchez et al. (2018), aroma compounds can have varying toxic effects on aquatic organisms. As a result, waste treatment and material efficiency are key topics for aromatics units transitioning from a linear economy to a more circular production process. In the following section, the proposed intervention will center on these critical aspects.



4.5 Proposed Interventions

In the intervention selection process, the CRIO (context-interventions-mechanisms-outcomes) principle is the cornerstone of decision-making. It enables policymakers to improve existing systems and gain a comprehensive understanding of them. The context indicates that obtaining information about the project should be thorough, covering different aspects, including economic activities, material and energy analysis, and spatial characteristics. In addition, the second step involves interventions that use case-specific information, including the demand and supply potentials of relevant materials, to understand the current system and propose solutions.

Intervention A: increase the use of the waste stream

To enhance material efficiency and the E-factor of the aromatic units in the Antwerp industrial cluster, the proposed strategy is to increase the utilization of materials from waste streams. This presents an ideal opportunity to transition to a circular economy and to explore new product lines and revenue streams. In the existing database, the waste stream from aromatics production includes several chemicals with residual value, such as remaining heavy components (WS-2C9+), heavy components C9+ (WS-C9+), C1-C5 Hydrocarbons (WS-OFFGA), Non-aromatics (WS-RAFF), and other streams with lower flow rates. For the heavy components waste stream, the compounds contain Ortho-xylene, Para-xylene, Trimethylbenzene, Dicyclopentadiene, and Naphthalene. Naphthalene and Trimethylbenzene make up most of the flow, making it the ideal material for by-product production. The proposed interventions are precision fractionation and purification of the waste stream and co-carbonization of heavy aromatics for advanced carbon materials.

Since a certain amount of high-value chemicals, such as Trimethylbenzene and Naphthalene, remain in the heavy components, the proposed interventions are to purify and fractionate high-value pure chemicals from the heavy stream. These materials are the ingredients of Polyimide (PI) film precursors, specialty solvents, and intermediates for pharmaceuticals and dyes. This intervention significantly improves the economic value of the byproduct while lowering the total amount of waste.

After extracting the high-value materials from the waste stream, the second intervention proposal is to utilize the naphthalene-rich bottom residue. The materials can achieve mesophase development and form high-quality carbon structures through thermal treatment. These materials could serve as ingredients for anode materials for EV lithium-ion batteries (Needle Coke), ultra-high-power graphite electrodes for steelmaking, and specialty carbon black, playing a significant role in the sustainable energy transition. On top of that, the intervention successfully turns the most polluting bottom oils into critical materials, lowering the E-factor performance of aromatics production.

Intervention B: AOP wastewater treatment

The wastewater generated from aromatics plants contains highly toxic compounds, such as benzene, naphthalene, and xylene (BTX), which can easily inhibit or destroy traditional biological treatment systems. To tackle this issue, the process starts with an Advanced Oxidation Process (AOP) as an essential initial step. In this phase, ozone is activated by specific catalysts to generate highly reactive hydroxyl radicals. These radicals act as chemical scissors, targeting and cleaving the stable aromatic ring. This vital step effectively breaks down complex, toxic hydrocarbons into smaller, non-toxic, and highly biodegradable organic acids, without generating secondary solid waste. After detoxification and structural breakdown during oxidation, the effluent safely proceeds to deep biological treatment. The system integrates biological degradation with advanced physical membrane filtration, enabling extended sludge retention times. This environment fosters the growth of specialized, slow-growing microorganisms that consume residual trace organics. Consequently, the process produces high-quality, suspended-solids-free effluent, essential for promoting industrial water recycling. The treated water can be transferred to Reverse Osmosis (RO) units for desalination, then reused as cooling tower makeup or boiler feedwater. This significantly lowers the plant's overall material consumption and freshwater usage.

4.6 Selection of companies for further assessment and intervention design

TotalEnergies is selected as the representative company for the olefins process, which acts as one of the key upstream actors in the PolyCycle industrial park in the Port of Antwerp. Its role is particularly important due to its involvement in olefins production and the generation of key intermediates such as ethylene and pygas. From a



material perspective, the process involves a significant throughput of fossil-based feedstock, primarily naphtha, and produces strategic intermediates including ethylene and propylene, which serve as essential inputs for downstream processes. Based on the results presented in Chapter 4.2, the olefins process exhibits a high material intensity, indicating a strong dependence on virgin feedstock. In addition, the E-factor is significantly higher than the typical range for bulk chemicals, and higher compared to other units within the system. This highlights substantial material losses and indicates an urgent need for improvement. Overall, these results suggest that the olefins process operates with relatively low material efficiency and high waste generation, making it a critical bottleneck for achieving circularity within the industrial system. Several material-related challenges can be identified. First, a large proportion of by-products, such as propylene, butene mixtures, and heavy hydrocarbon streams (e.g., C9+ and non-aromatic fractions), are not fully utilized within the system and are often exported or treated as low-value outputs, resulting in material losses. Second, the process heavily relies on virgin fossil feedstock, particularly naphtha, which increases resource dependency and reduces system resilience to feedstock availability and price fluctuations. Third, the process generates a substantial amount of waste and emissions, including flue gas, effluent, and other by-product streams, indicating inefficient resource utilization and environmental pressure. Finally, there is a lack of internal material recirculation, as many streams are not reintegrated into upstream or downstream processes, limiting the overall circularity of the industrial park. It is important to note that the GHG emission intensity of the olefins process appears relatively low. This is because a large portion of carbon from fossil feedstock, such as naphtha, is not directly emitted as CO₂ within the olefins unit but instead remains embedded in intermediate products such as ethylene and other hydrocarbons. As a result, the direct emissions of the olefins process are underestimated when considering only this indicator. The environmental impact of fossil feedstock use is partially shifted to downstream processes. This can be observed in the EO/EG unit, where the GHG emission intensity is significantly higher (≈ 2.27), reflecting the further processing and conversion of carbon-containing intermediates such as ethylene. Therefore, while the olefins process appears to have low direct emissions, its overall carbon impact remains substantial when considering the entire system.

4.7 Proposed Interventions

Intervention A: plastic-to-naphtha pathway

The current olefins process relies heavily on virgin fossil-based naphtha, resulting in a predominantly linear material flow where carbon is extracted, converted into products, and eventually released as waste or emissions. At the same time, a significant amount of plastic waste, particularly post-consumer PET, is generated after the use phase but is not currently reintegrated into the production system. As a result, a plastic-to-naphtha pathway can be introduced, in which post-consumer plastic waste is converted into pyrolysis oil and further refined into alternative naphtha. The required plastic feedstock can be sourced from post-consumer waste streams collected through municipal and commercial waste management systems, as well as plastic residual generated such as packaging or off-spec materials within the industrial park. This recycled naphtha can then be reintroduced into the olefins process as a feedstock, partially replacing virgin fossil inputs. A key advantage of this approach is that alternative naphtha can be used in existing steam cracking infrastructure without requiring major modifications, making it technically feasible for industrial implementation. By recycling plastic waste produced in the use phase back into feedback, the carbon is redirected to the manufacturing phase instead of being lost after the use phase. This creates a closed-loop carbon system and strengthens the linkage between plastic consumption and upstream petrochemical production. Although this intervention may not significantly reduce the material intensity of the olefins process itself, it contributes to system-level circularity by replacing a portion of virgin fossil-based naphtha with recycled feedstock. As a result, intervention reduces dependence on primary resources, improves carbon retention within the industrial system, and supports the transition from a linear to a more circular material flow.

Intervention B: Cryo-GasSM process technology

Waste generation in olefins units is relatively high, with flue gas accounting for a significant share of the E-factor, while its direct utilization as a material remains limited. In contrast, hydrocarbon-rich waste streams such as propane-rich fuel gas can be more effectively utilized through established technical pathways. For example, Cryo-GasSM process technology can be applied to recover valuable hydrocarbons from waste gas streams. The recovered fuel gas can then be supplied to the CHP plant as secondary fuel, enabling internal energy recovery within the industrial park. This implementation enhances the utilization of waste streams, thereby reducing, improving overall resource efficiency, while strengthening industrial symbiosis between the olefins unit and the CHP plant.



CHAPTER 5. ENERGY ANALYSIS

5. Energy balance analysis:

5.1 Method and interpretation logic

The energy analysis was conducted for the five selected companies within the industrial park: the Natural Gas CHP plant, the Olefins Production plant, the Aromatics plant, the Ethylene Oxide–Ethylene Glycol (EO/EG) plant, and the PET production plant. The objective of this analysis is to understand both the overall utility demand at plant level and the internal process drivers behind this demand. Two complementary datasets were used. First, the energy balance tables provide information on net utility imports and exports for each plant. These tables quantify the overall energy demand or surplus at the system level. Second, the breakdown of requirements tables identifies the specific process units responsible for heating and cooling duties. This allows a more detailed understanding of where energy is consumed or generated within each process. Combining these two datasets is essential. While the energy balances indicate how much utility is required or available, the breakdown tables explain the origin of these demands within the process structure. This distinction is particularly important for the subsequent stages of the project, where the identification of heat sources, heat sinks, and opportunities for heat integration plays a central role. A consistent sign convention is applied throughout the analysis. Positive net utility values represent net imports, indicating that the plant requires external energy input. Negative values represent net exports, indicating that the plant generates a surplus of utility that can potentially be redistributed within the industrial park. In this context, a positive steam value reflects an external steam demand, whereas a negative steam value indicates excess steam available for integration with other processes.

Assumptions: To ensure consistency across the energy balance and KPI calculations, the same assumptions were applied throughout the analysis. The system boundary includes the five selected plants: Natural Gas CHP, Olefins Production, Aromatics, EO/EG, and PET Production. All hourly utility flows were converted to annual values using 8000 operating hours per year. The analysis includes electricity, fired heat, steam at different pressure levels, cooling water, and chilled water where relevant. A consistent sign convention was used. positive values represent net utility imports or demand, while negative values represent net exports or surplus flows that could potentially be redistributed within the industrial park. Cooling water demand is interpreted as an indicator of heat rejection and possible heat recovery potential, rather than as useful energy output. Finally, the efficiency and loss indicators are used as screening proxies to compare the functional roles of the plants, not as exact thermodynamic efficiencies or real energy losses.

Park-level energy balance

At the level of the industrial park, the system can be characterized as heat- and utility-dominated rather than electricity-dominated. This indicates that the primary energy flows within the park are related to thermal utilities, particularly steam and process heat, rather than electrical consumption. The aggregated utility balance therefore reflects a system in which heating and cooling demands play a central role in determining overall energy performance and integration potential. (see Energy Balance table in Appendix) The results indicate that the industrial park is not constrained by electricity availability. Instead, the dominant challenge lies in the provision and management of thermal utilities, particularly low- and medium-pressure steam, together with a significant cooling-water demand. This finding is critical for defining the system's optimization problem. The energy imbalance is primarily related to heat flows, meaning that the key opportunity is not additional electricity generation but improved heat integration and utility matching across processes. In other words, the system should be analyzed as a thermal network rather than an electricity-driven system. This interpretation is consistent with the course framework, which emphasizes that industrial systems are largely governed by thermal energy demand. Within this context, identifying heat sources, heat sinks, and opportunities for heat recovery becomes the central task for improving overall system efficiency.

Natural Gas CHP

Natural Gas CHP functions as the utility backbone of the park. It is a net exporter of electricity and all major steam levels, while importing fired heat and cooling water. Its net utility profile is:



Electricity: -1449.92 MJ/h	HP steam: -524.80 MJ/h	Cooling water: +1111.52 MJ/h
LP steam: -722.24 MJ/h	MP steam: -604.80 MJ/h	Fired heat: +708.04 MJ/h

This result confirms that the CHP plant does not function as a conventional process-energy consumer, unlike the chemical production units. Instead, its role is to convert primary energy from fuel combustion into usable utilities for the rest of the system. From a system perspective, the CHP acts as a centralized utility hub, supplying both electricity and steam to the industrial park. It establishes the baseline energy provision on which the other plants depend for their operation. This makes the CHP strategically essential for the current configuration of the park. At the same time, this centralization also concentrates fossil fuel dependency in a single node. As a result, while the CHP ensures stability and reliability of utility supply, it also represents a critical leverage point for future system transformation. Within this configuration, the CHP should therefore be interpreted as the main utility provider rather than as a final production facility.

Olefins Production Plant

The Olefins Production Plant has the most interesting mixed utility profile in the park. It is a large importer of LP steam and cooling water, but it is simultaneously a net exporter of MP, HP and VHP steam. Its net profile is:

Electricity: +448.23 MJ/h	HP steam: -5972.25 MJ/h	VHP steam: -1962.00 MJ/h
LP steam: +5121.24 MJ/h	MP steam: -3506.56 MJ/h	Cooling water: +13370.42 MJ/h

This result indicates that the Olefins plant cannot be interpreted as a simple utility consumer. Instead, it operates as a hybrid thermal node within the industrial system. While it requires low-pressure steam and significant cooling

Electricity: +530.64 MJ/h	MP steam: +14760.13 MJ/h	Chilled water: +332.27 MJ/h	LP steam: +3912.31 MJ/h
VLP steam: +222.77 MJ/h	HP steam: +5636.93 MJ/h	Cooling water: +26185.03 MJ/h	

capacity, it simultaneously generates and exports higher-pressure steam to the wider utility network. This dual role suggests that the process already includes substantial internal energy conversion and recovery mechanisms. The high cooling-water demand reflects large amounts of heat rejection, whereas the export of steam indicates that part of this thermal energy is already being recovered and upgraded into a usable form. The breakdown of requirements further supports this interpretation. The dominant process units exhibit large medium- and high-pressure thermal duties, with both positive and negative values. This pattern reflects a highly energy-intensive process with simultaneous heat generation and heat demand occurring within the system. From a system perspective, the Olefins plant should therefore be considered one of the key energy nodes in the industrial park, with significant potential for further heat integration and utility optimization.

Aromatics Production Plant

The Aromatics Production Plant is the dominant thermal utility sink in the park. It has the highest total imported energy and the highest cooling-water demand. Its net profile is: This result shows that the Aromatics plant is the most utility-intensive unit within the current industrial park configuration. The process exhibits a strong dependence on steam across multiple pressure levels, combined with a substantial rejection of heat through cooling water. This pattern is characteristic of processes with intensive reaction and separation stages. The breakdown of requirements further indicates that the plant's energy profile is dominated by a limited number of key units. In particular, units associated with high cooling-water demand and a major highpressure steam duty define the overall energy behavior



of the system. From a system perspective, the Aromatics plant represents the largest heat sink within the industrial park. As such, it should be considered a priority target for heat integration, with significant potential for recovering and redistributing thermal energy within the network.

EO/EG Plant

The Ethylene Oxide–Ethylene Glycol plant is another major thermal process. It is a large steam consumer and a large cooling-water user. Its net profile is:

Electricity:	HP steam:	MP steam:	LP steam:	Cooling water:
+248.52 MJ/h	+667.69 MJ/h	+3937.77 MJ/h	+1801.49 MJ/h	+12297.49 MJ/h

The energy profile of the EO/EG plant shows a clear coexistence of significant heat consumption and heat rejection within the same process. This indicates a thermally complex system in which multiple units operate at different energy levels. The breakdown of requirements provides further detail. The dominant contributors include the ethylene oxide reactor and desorber, the feed heater, the recycle gas heater, and the incinerator radiative section. These units exhibit both strongly positive and strongly negative thermal duties, reflecting simultaneous heat demand and heat generation across the process. This configuration confirms that EO/EG is a thermally dynamic system with substantial internal potential for heat recovery, heat upgrading, and utility optimization. From a system perspective, it represents one of the most promising candidates within the industrial park for advanced heat integration and for the implementation of heat-pump-based interventions in later design stages.

PET Production Plant

The PET Production Plant is much smaller than the other four plants. Its net profile is:

Electricity:	HP steam:	MP steam:	Cooling water:	LP steam:
+0.95 MJ/h	+4.93 MJ/h net combined	+1.20 MJ/h	+22.55 MJ/h	+9.09 MJ/h

Compared with the other plants, the PET unit functions as a relatively small downstream utility consumer. Its overall energy demand is modest at park level, especially when compared with the more utility-intensive upstream processes. However, the breakdown of requirements shows that the plant still has a distinct internal thermal structure. The main energy-related units include the azeotropic solvent recovery column, the esterification reactor, the overhead cooler, the heater, and the MEG recovery column. These units indicate that, despite its smaller scale, the plant contains identifiable heating and cooling demands that shape its utility profile. From a system perspective, PET is not a major energy bottleneck within the industrial park. Nevertheless, it remains relevant as a smaller heat sink that could benefit from symbiotic utility matching and from the recovery of thermal energy generated elsewhere in the park.

Main conclusions from the energy balance: The energy-balance analysis reveals a clear functional hierarchy within the industrial park. The CHP plant operates as the central supplier of electricity and steam, providing the baseline utilities required by the system. The Olefins plant functions as a hybrid node, combining significant utility demand with the generation and export of surplus steam. In contrast, the Aromatics plant represents the largest thermal utility sink, with a strong dependence on steam and extensive heat rejection. The EO/EG plant is characterized by high steam and cooling-water demand but also shows considerable internal potential for heat integration. The PET plant, while smaller in scale, acts as a downstream utility receiver within the network. At the system level, the dominant challenge is not electricity supply but the management of thermal utilities. In particular, the balance and distribution of low- and medium-pressure steam, together with the very high cooling-water demand, define the core energy problem of the park. As a result, future optimization efforts should prioritize steam-level matching, heat recovery, and the reduction of final cooling loads, rather than focusing primarily on electricity generation.



5.2 Performance evaluation:

5.2 KPI framework and interpretation The performance evaluation is based on a set of screening KPIs derived from the Python analysis. These indicators are used to compare the energy behavior of the selected plants and to identify priority cases for further system optimization. The main KPIs include total energy import, annual total energy consumption, electricity import, steam import, fired heat import, cooling-water demand, by-product energy recovery potential, a proxy for process energy efficiency, and a proxy for energy losses. These indicators should be interpreted as screening metrics rather than exact thermodynamic measures. In particular, the efficiency and loss proxies derived from the breakdown of duties do not represent precise efficiencies, but relative performance indicators across plants. Their purpose is to highlight differences in energy intensity, identify major sources of inefficiency, and support the selection of processes with the highest potential for heat integration and targeted intervention design. (see KPI results table in Appendix)

Analysis of KPI results, Total energy consumption

The results show a highly uneven distribution of energy demand across the industrial park. The Aromatics plant is clearly the dominant energy consumer, accounting for approximately 65.9% of the total annual imported energy among the five selected companies. This makes it the primary energy hotspot within the system. The EO/EG and Olefins plants follow, contributing around 17.5% and 14.7%, respectively. These plants also represent significant energy users, although their impact is considerably lower than that of the Aromatics unit. In contrast, the CHP plant accounts for only about 1.9% of annual imported energy. This is because, within the current dataset, it is modeled as a utility conversion unit rather than a final production process. The PET plant has a negligible contribution at park scale due to its relatively small energy demand. Overall, this distribution confirms that the energy burden within the park is not evenly shared. Instead, it is strongly concentrated in the Aromatics plant, with EO/EG and Olefins forming a secondary tier of energy-intensive processes.

Analysis of KPI results, Cooling-water demand

Cooling-water demand is especially high in Aromatics, Olefins, and EO/EG:

Aromatics: 26185.03 MJ/h	EO/EG: 12297.49 MJ/h	Olefins: 13370.42 MJ/h
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This is one of the most important findings of the chapter. The park's energy problem is not only steam demand; it is also the fact that very large amounts of heat are currently being rejected to cooling water. This strongly supports the case for future heat recovery and heat integration. It also explains why the later pinch-analysis and exchanger network logic of the course is relevant for this industrial park.

Analysis of KPI results, By-product energy recovery potential

The byproduct energy recovery KPI highlights which plants already provide useful utility surplus:

Olefins: 13370.42 MJ/h	PET: 3.69 MWh/y	EO/EG: 0	CHP: 7337.24 MWh/y	Aromatics: 0
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This shows that the strongest utility suppliers in the park are Olefines and CHP.

Systems Interpretation and KPI-Based Insights

The results highlight a key system-level insight: the industrial park already contains internal energy-recovery and utility-production mechanisms, but these are unevenly distributed across plants. From a symbiosis perspective, this implies a clear functional distinction between providers and receivers. CHP and Olefins should be treated as primary utility providers, while Aromatics and EO/EG act mainly as utility receivers.

Process Energy Efficiency Proxy

The efficiency proxy requires careful interpretation. Plants such as Aromatics and EO/EG show values close to 100



percent because they primarily import utilities without exporting usable energy. In contrast, CHP and Olefins show lower values due to their role in exporting steam and other utilities to the system. These indicators should therefore not be interpreted as thermodynamic efficiencies. Instead, they reflect the functional role of each plant within the energy network. Under this interpretation, Aromatics and EO/EG behave as strong utility sinks, while CHP and Olefins operate as conversion and supply nodes. This classification is consistent with the earlier company-level analysis.

Energy Losses Proxy

The breakdown-based losses proxy indicates the highest internal duty intensity in Olefins and EO/EG. These values do not represent actual energy losses to the environment. Rather, they indicate the presence of highly intensive internal heating and cooling loops. This suggests significant potential for internal heat recovery and improved integration. These indicators should therefore be used comparatively as screening tools for identifying priority plants for intervention.

Process-Level Interpretation

The breakdown tables show that, in each plant, a limited number of process units dominate overall energy behavior. In Aromatics, the energy profile is driven mainly by cooling-water duties and a single major high-pressure steam demand. This concentration suggests strong potential for targeted process integration. In EO/EG, the dominant units include the reactor, desorber, feed heater, recycle gas heater, and incinerator section. The presence of both high positive and negative duties indicates strong internal interactions between hot and cold streams, making this plant highly suitable for heat integration. In Olefins, large medium- and high-pressure steam duties with both positive and negative values confirm a thermally intensive process with internal redistribution and steam export capability. In PET, although the scale is smaller, identifiable units such as the solvent recovery column and esterification reactor still define a structured thermal profile, allowing for local optimization.

CO₂ Emissions Analysis by Plant

The emissions analysis shows a clear pattern across the industrial park, with CO₂ emissions largely driven by thermal utility demand rather than electricity consumption. **Aromatics:** Aromatics is the dominant CO₂ emitter in the system. This is directly linked to its very high imported steam demand, which represents the main source of emissions. Electricity-related emissions are comparatively minor. From a system perspective, the emissions profile of Aromatics is therefore primarily a function of thermal utility consumption rather than electrical use. **EO/EG:** EO/EG represents the second-largest emitter. Like Aromatics, its emissions are mainly associated with steam demand. This reinforces its role as a major thermal utility sink and confirms its relevance as a priority candidate for heat integration and thermal optimization. **Olefins:** Olefins show a moderate but still significant level of emissions. Its emissions are driven by both imported low-pressure steam and electricity. However, unlike Aromatics and EO/EG, Olefins also exports higher-pressure steam. It should therefore be interpreted as a hybrid node that both consumes and supplies utilities, rather than as a pure emissions sink. **CHP:** CHP differs fundamentally from the other plants. Its emissions are associated with fired heat input, reflecting direct fuel combustion rather than imported utilities. As a result, its CO₂ value represents a production-based emission rather than a consumption-based one. For system-level comparison, CHP emissions would need to be allocated to the plants that consume its steam and electricity. **PET:** PET has a negligible contribution in terms of both energy use and emissions. While it remains relevant as a downstream receiver within the system, it does not represent a significant emissions hotspot.

Main Conclusions from KPI Evaluation

The KPI analysis leads to several key conclusions. First, the park is dominated by thermal utility demand, particularly in Aromatics and EO/EG. Second, CHP and Olefins function as the main utility providers within the system. Third, the largest cooling-water demands occur in Aromatics, Olefins, and EO/EG, indicating substantial heat-recovery potential. Fourth, energy use within each plant is highly concentrated in a small number of process units. Fifth, Aromatics, EO/EG, and Olefins emerge as the most relevant targets for future intervention, due to their high energy intensity and strong integration potential.



Park-Level Utility Matching

Based on the combined analysis, a clear utility-matching structure can be defined for the industrial park. CHP remains the central utility hub, supplying electricity as well as low-, medium- and high-pressure steam to all plants. Olefins functions as a secondary supplier, providing surplus medium-, high-, and very high-pressure steam to the system. Aromatics and EO/EG should be treated as the primary receivers, as they account for the largest steam and cooling demands. PET operates as a smaller downstream receiver. This leads to a system logic in which electricity is distributed from CHP to all plants, while steam is supplied at different pressure levels depending on process requirements. Surplus steam from Olefins should be redirected to the largest receivers, and all plants remain connected to a shared cooling-water network. This configuration provides a consistent basis for the next phase of the project and for system representation.

Direction for Future Interventions: Although detailed interventions are addressed in the next stage, the current analysis already indicates several priority directions. For CHP, opportunities include flue-gas heat recovery and improved boiler-feedwater preheating. For Olefins, optimization should focus on steam-pressure cascading and reuse of high-temperature residual heat before rejection to cooling water. For Aromatics, the priority is process-level heat integration to reduce both steam demand and cooling load. For EO/EG, strong potential exists for internal heat recovery and for upgrading low-temperature heat using industrial heat pumps.

Translation of KPI Outcomes into Intervention Priorities: The KPI results show that the main intervention priority should be thermal utility optimization rather than electricity supply. Aromatics should be prioritized because it has the highest imported energy demand, cooling water demand, and CO₂ emissions, making it the main heat integration target. EO/EG is also important because its high steam and cooling water demand indicates potential for internal heat recovery or heat upgrading. Olefins should be treated as a hybrid utility node because it imports LP steam and cooling water, while also exporting MP, HP, and VHP steam; therefore steam cascading and surplus steam redistribution are relevant priorities. CHP remains the central utility hub, so flue gas heat recovery, feedwater preheating, renewable electricity integration and power-to-steam conversion are key system level interventions. PET has a much smaller energy demand, so only local optimization is relevant. Overall, the KPI outcomes support the focus on heat recovery, steam level matching, cooling load reduction, and improvement of the CHP utility hub.

5.3 Identification of Industrial Symbiosis Opportunities

Pinch Analysis in the Industrial Park

Company	Pinch role	Evidence
CHP	Central utility source	Exports electricity, LP, MP, and HP steam. Imports fired heat
Olefins	Hybrid source and sink	Imports HP steam but exports MP, HP, and VHP steam
Aromatics	Dominant heat sink	Largest MP, HP, LP steam demand and largest cooling water load
EO/EG	Main heat sink/integration	Large LP/MP/HP steam demand and high cooling water
PET	Small downstream sink	Small but stable steam and cooling demand

Pinch analysis is a standard heat-integration method used to determine the minimum external heating and cooling requirements of a system by matching internal heat sources and sinks. The method involves classifying streams as hot or cold, defining supply and target temperatures, calculating heat duties, and applying a minimum temperature difference (typically $\Delta T_{min} \approx 10^{\circ}\text{C}$) to identify feasible heat exchange. In this industrial park, pinch analysis is particularly relevant because the system is dominated by thermal utilities rather than electricity. The energy balance



shows that CHP supplies electricity and steam, Olefins exports higher-pressure steam, while Aromatics and EO/EG represent the main heat and cooling demand centers. This confirms that the core system challenge is thermal integration, consistent with the course framework where industrial energy systems are primarily governed by heat flows. At this stage, pinch analysis is applied at a park level as a screening tool. The objective is not to design detailed heat exchanger networks, but to identify potential energy exchanges between companies and define integration opportunities. A full process-level pinch analysis should be carried out in later stages using detailed stream temperature and duty data.

Current Utility and Temperature Basis

The utility condition table provides the physical basis for integration by defining the quality of available utilities. In particular, the different steam levels represent distinct temperature and pressure ranges, which determine how energy can be transferred and matched across the system.

Utility	Temperature	Pressure	Energy content
LP steam	155°C	5.5 bar	2097 kJ/kg
MP steam	214°C	21 bar	1880 kJ/kg
HP steam	265°C	51 bar	1632 kJ/kg
Fired heat	1000°C to 750°C	2 bar	—
Cooling water	25°C to 40°C	1.02 bar	4.18 kJ/kg

This table supports three important checks. First, it enables conversion from energy flow to mass flow. For example, the Aromatics plant requires 14,760.13 MJ/h of MP steam. With an MP steam energy content of 1.88 MJ/kg, the required mass flow is: MP steam demand = $14,760.13 / 1.88 = 7,851$ kg/h. Second, it confirms the technical feasibility of steam cascading. Since HP steam is available at approximately 265°C, MP steam at 214°C, and LP steam at 155°C, steam can theoretically be cascaded from higher to lower pressure levels while still respecting useful temperature differences. Third, it shows that cooling-water heat is low-grade. Cooling water leaves the system at around 40°C, which is not suitable for direct steam generation. However, it can still be useful for feed preheating or can be upgraded through a heat pump. This is consistent with the course principle that heat integration must respect temperature levels and the minimum temperature difference, ΔT_{min} .



Utility	Supply from internal sources	Demand from plants	Net result
Electricity	1449.92 MJ/h	1228.34 MJ/h	surplus 221.58 MJ/h
HP steam	6497.05 MJ/h	6309.55 MJ/h	surplus 187.49 MJ/h
MP steam	4111.36 MJ/h	18698.62 MJ/h	deficit 14587.26 MJ/h
LP steam	722.24 MJ/h	10842.95 MJ/h	deficit 10120.71 MJ/h
Cooling water	—	52987.02 MJ/h	very high cooling burden

The strongest conclusion from the park screening is that the system has sufficient electricity availability and almost enough HP steam, but still faces large MP and LP steam deficits. Therefore, the main energy problem is not electricity generation. It is steam-level matching, heat recovery, and reduction of external thermal utility demand. The cooling-water burden is also substantial. The combined cooling-water duty is 52,987 MJ/h. Using the cooling-water conditions from the utility table, the equivalent cooling-water mass flow can be estimated as: $Q = \dot{m} c_p \Delta T$, $\dot{m}_{cw} = (52,987.02 \times 1000) / (4.18 \times 15)$, $\dot{m}_{cw} = 845,088 \text{ kg/h}$. This means that the park requires approximately 845 t/h of cooling-water equivalent. This is a major heat rejection pathway and provides strong evidence that significant heat-integration opportunities exist within the system.

Pinch screening illustrations

Company	Pinch role	Evidence
CHP	Central utility source	Exports electricity, LP, MP, and HP steam. Imports fired heat
Olefins	Hybrid source and sink	Imports HP steam but exports MP, HP, and VHP steam
Aromatics	Dominant heat sink	Largest MP, HP, LP steam demand and largest cooling water load
EO/EG	Main heat sink / integration	Large LP/MP/HP steam demand and high cooling water
PET	Small downstream sink	Small but stable steam and cooling demand

The lecture on heat integration describes industrial systems as networks where processes needing cooling act as heat sources and processes needing heating act as heat sinks; the result is a heat exchanger network with lower utility demand and lower greenhouse gas emissions. In this park, Aromatics and EO/EG are the clearest heat sinks, while CHP and Olefins are the main supply-side nodes.

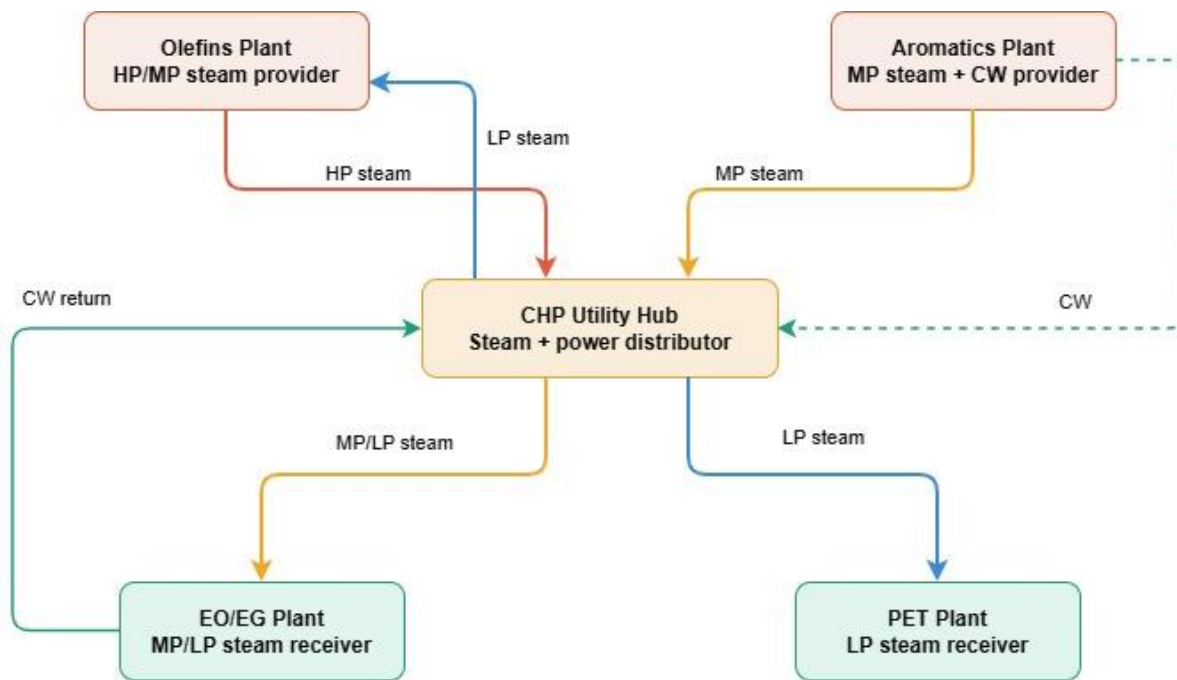


Figure 5.3. Overview of utility exchanges between process units in the PolyCycle park, grouped by provider and receiver role.

Current pinch result

Indicator	Current system result	Interpretation
Net electricity balance	+221.58 MJ/h surplus	electricity not the main bottleneck
Net HP steam balance	+187.49 MJ/h surplus	HP steam can be internally balanced
Net MP steam deficit	14587.26 MJ/h	major external thermal utility need
Net steam deficit	10120.71 MJ/h	major external thermal utility need
Total cooling	52987.02 MJ/h	large rejection/recovery opportunity
Cooling water equivalent	845 t/h	very large, shared cooling burden

This shows that the first design priority should be steam cascading and heat recovery. The second priority should be reducing cooling-water demand by using hot streams for preheating or upgrading low-grade heat with heat pumps.



Exchange	Supplier	Receiver	Utility / heat level	Current calculation basis	Feasibility	Expected benefit
1	CHP	All plants	Electricity	CHP surplus after demand 221.58 MJ/h	High	Internal electricity balance, less grid dependence
2	Olefin CHP	Aromatics, EO/EG, PET	HP steam	HP surplus = 187.49 MJ/h	High	Direct steam matching at high temperature
3	Olefins CHP	Aromatics, EO/EG	MP steam	MP deficit remains 14587.26 MJ/h	Medium and High	Internal supply reduces but still external MP steam need
4	CHP	Aromatics,EO/EG, Olefins, PET	LP steam	LP deficit remains 10120.71MJ/h	Medium	CHP supply useful but insufficient, reduction needed
5	Olefins VHP steam	Steam cascade / turbine / MP pool	VHP to HP/MP/P	VHP surplus = 1962 MJ/h	High, if pressure let down/turbine added	Converts high grade surplus into lower pressure or power
6	Aromatics, EO/EG, Olefins	Feed preheating / heat pump loop	Low grade heat	Cooling duty = 52987 MJ/h	Medium	Reduces cooling burden and preheating demand
7	CHP exhaust gas	CHP feedwater / steam cycle	High temperature heat	Fired heat stream 1000–750°C	High	Reduces natural gas use at CHP

Supply/Demand Matrix



5.4 Selection of Companies for Intervention Design

Based on energy balance, KPI evaluation, and identified symbiosis potential, three plants are selected for detailed intervention analysis: CHP, EO/EG, and PET. The selection reflects both system-level importance and potential for targeted improvement.

CHP Plant

Natural Gas CHP has been selected due to its role as the central utility hub of the industrial park. It supplies electricity and multiple steam levels to all other plants, meaning that any improvement at CHP propagates across the entire system. Although its direct energy import is relatively low compared to Aromatics and EO/EG, its strategic importance lies in determining the carbon intensity of shared utilities. The main limitation is its dependence on natural gas. Therefore, CHP is selected for interventions focused on flue-gas heat recovery and renewable hybridization.

EO/EG Plant

The EO/EG plant is selected because it combines high utility demand, high cooling-water rejection, and strong symbiosis potential. It imports approximately 6,655.5 MJ/h of utilities, primarily steam, and rejects around 12,297.5 MJ/h through cooling water. In addition, the process generates several environmental and operational challenges, including unrecovered low-grade heat, a CO₂ rich stream from the K₂CO₃ desorbed, untreated wastewater, and dependence on CHP supplied steam. From a system perspective, EO/EG is important because it contains both heat sources and heat sinks. This makes it a strong candidate for heat recovery, heat-pump upgrading, and potential CO₂ capture and reuse within the industrial network.

PET Plant

The PET plant is selected due to its position as a downstream receiver in the material and energy chain. It consumes monoethylene glycol (MEG) from EO/EG and represents the final polymer production stage. Although its overall energy demand is relatively small, it functions as a useful heat sink that can absorb low-pressure steam, recover heat, or hot-water streams. While PET is not a major energy hotspot, it is strategically relevant because it links upstream chemical production with final product manufacturing, making it an important node for integrating recovered energy flows within the park.



5.5 Proposed Interventions

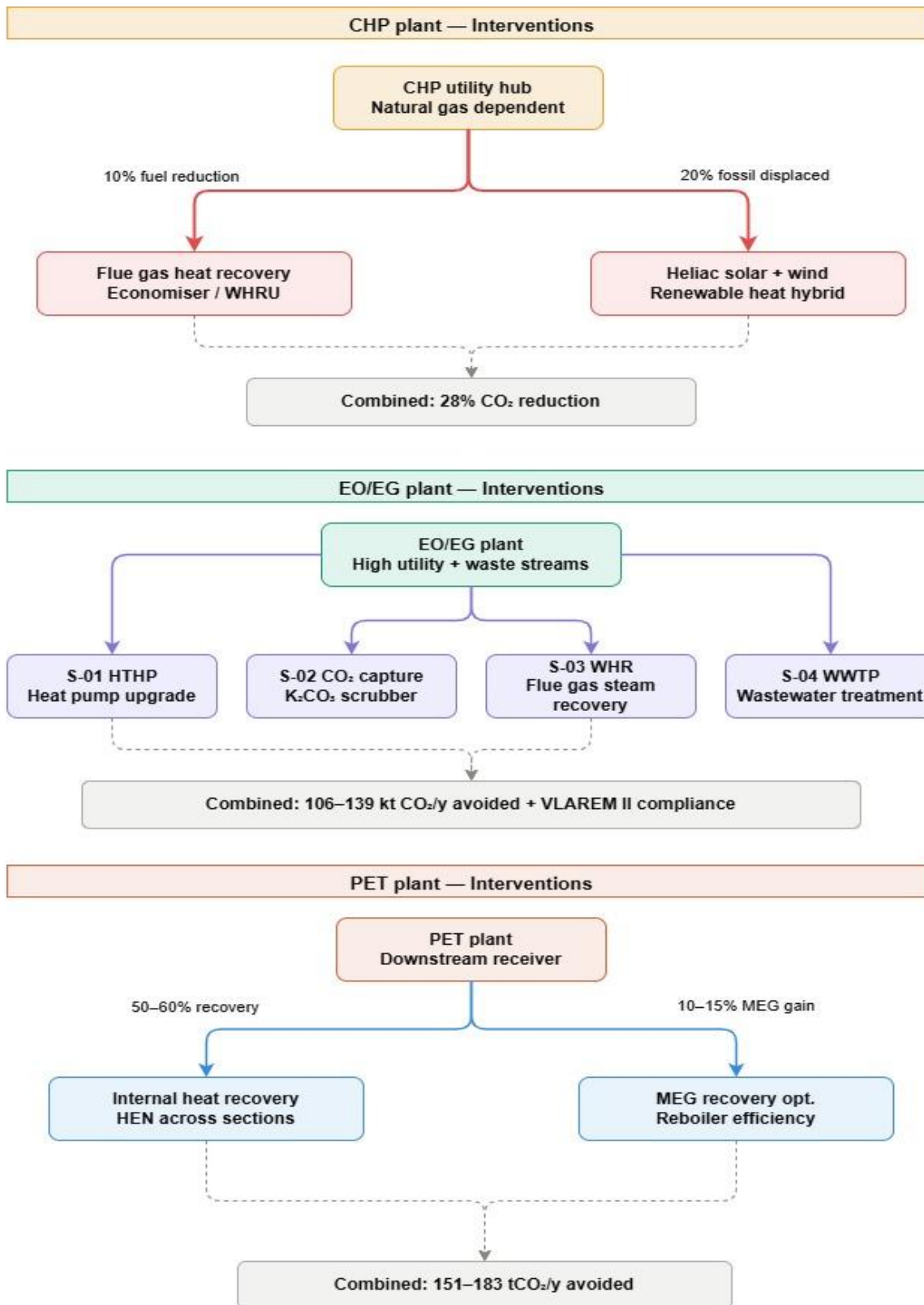


Fig 5.5. Overview of proposed interventions for CHP, EO/EG, and PET plants in the PolyCycle park, grouped by plant. Dashed arrows indicate combined outcomes.



5.5. Intervention introduction

The CHP plant is the central supplier of electricity and steam within the industrial park and therefore plays a critical role in the overall energy system. Because multiple companies depend on CHP utilities, improvements at this node create benefits across the wider network rather than only within a single process plant. However, the current CHP system relies heavily on natural-gas combustion, making it a significant source of direct CO₂ emissions and increasing the carbon intensity of the electricity and steam supplied to the other companies. To address this challenge, the proposed intervention combines two complementary strategies. The first strategy focuses on improving energy efficiency through WHRU/HRSG heat recovery. By recovering useful thermal energy from CHP exhaust gases, the system reduces heat losses and lowers natural-gas consumption. The second strategy focuses on reducing fossil-based utility demand through the integration of renewable electricity from a 1.5 MW wind turbine. Part of this renewable electricity is supplied directly to industrial users, while the remaining electricity is converted into low-carbon steam through a high-voltage electrode boiler. This intervention follows a logical energy-transition pathway. The first step is to reduce energy losses through heat recovery. The second step is to replace part of the fossil-based utility supply with renewable energy. The third step is to increase system flexibility by linking the electricity and steam networks through power-to-steam conversion. Together, these measures improve energy efficiency, reduce emissions, and strengthen the resilience of the utility system. The CHP plant is not replaced by the intervention. Instead, its role evolves from a predominantly fossil-based utility supplier into a hybrid utility hub that combines conventional CHP generation, recovered heat, renewable electricity, and power-to-steam integration. The CHP continues to provide reliable backup electricity and steam when renewable generation is insufficient or when utility demand increases. This approach maintains system reliability while significantly reducing the carbon intensity of the utilities supplied throughout the industrial park.

As a result, the CHP Utility-Hub Retrofit represents both an energy-efficiency measure and a decarbonization strategy. It improves the environmental performance of the utility system while creating the foundation for future integration of material-recovery and water-reuse interventions within the broader industrial symbiosis network.

Baseline CHP Emissions

Based on the Python energy analysis, the selected CHP retrofit boundary has a fired heat input of 708.04 MJ/h. Assuming 8,000 operating hours per year, this corresponds to an annual fuel input of:

$708.04 \times 8000 / 3600 = 1573.43 \text{ MWh/y}$ Using a natural-gas emission factor of 0.181 tCO₂/MWh:

CO₂ baseline = $1573.43 \times 0.181 = 284.79 \text{ tCO}_2/\text{y}$

Intervention 1: WHRU/HRSG Heat Recovery

The first intervention is the installation of a WHRU/HRSG (Waste Heat Recovery Unit / Heat Recovery Steam Generator) to recover thermal energy from CHP exhaust gases that would otherwise be lost to the environment. The recovered heat can be integrated into the existing utility system by preheating boiler feedwater, supporting steam generation, or reducing the thermal load on conventional fuel-fired equipment. As a result, less natural gas is required to produce the same amount of useful energy. The intervention is selected because CHP exhaust gases contain a significant amount of recoverable thermal energy and the CHP system is directly connected to the steam and feedwater network serving the industrial park. This makes waste-heat recovery one of the most straightforward and technically compatible efficiency improvements available within the utility system. For the screening-level assessment, a conservative fuel-saving assumption of 10% is applied. This value is not intended to represent a detailed engineering design but rather a realistic preliminary estimate of the potential energy benefit from exhaust-gas heat recovery. The assumption is consistent with typical waste-heat recovery screening studies and provides a reasonable basis for comparing intervention options at the conceptual stage. By reducing fuel consumption, the



WHRU/HRSG directly lowers operating costs and decreases CO₂ emissions associated with natural-gas combustion. In addition, the recovered heat improves overall CHP efficiency by increasing the amount of useful energy obtained from the same fuel input. Therefore, the intervention contributes simultaneously to energy efficiency, emissions reduction, and improved resource utilization within the industrial park.

The WHRU/HRSG is also an important component of the broader CHP Utility-Hub Retrofit. Although its direct emissions-reduction benefit is smaller than the combined contribution of renewable electricity and power-to-steam integration, it forms the first step of the energy-transition strategy by reducing energy losses before renewable-energy integration is introduced. This follows the principle of improving efficiency first and then replacing the remaining fossil-based energy demand with lower-carbon alternatives. Fuel demand after heat recovery:

$$1573.43 \times 0.90 = 1416.09 \text{ MWh/y} \quad \text{CO}_2 \text{ emissions after heat recovery: } 1416.09 \times 0.181 = 256.31 \text{ tCO}_2/\text{y}$$

$$\text{CO}_2 \text{ saving: } 284.79 - 256.31 = 28.48 \text{ tCO}_2/\text{y}$$

Intervention 2: Wind Electricity and High-Voltage Electrode Boiler for Steam

The second intervention is based on the integration of renewable electricity through a 1.5 MW wind turbine and a high-voltage electrode boiler. The wind turbine supplies renewable electricity to the CHP Utility Hub, where the generated electricity is distributed between direct electricity use and steam production. Part of the renewable electricity is supplied directly to industrial users, while the remaining electricity is converted into steam through the high-voltage electrode boiler. This creates a direct link between the electricity and steam networks and allows renewable electricity to contribute to the thermal needs of the industrial park. This intervention is particularly relevant because the energy analysis showed that the industrial park is primarily constrained by heat and steam demand rather than electricity demand. Significant MP and LP steam deficits were identified, while electricity demand was comparatively less critical. Therefore, the purpose of the wind turbine is not simply to increase renewable-electricity generation or address an electricity shortage. Instead, the renewable electricity is used to decarbonize the utility system and provide a power-to-steam pathway through the electrode boiler. The high-voltage electrode boiler converts renewable electricity into low-carbon steam that can support the LP and MP steam networks. This improves steam flexibility and reduces dependence on fossil-fuel-based steam generation. As a result, the intervention addresses both the decarbonization objective and the thermal challenges identified within the industrial park. The CHP remains an essential part of the system after implementation. Rather than replacing CHP operation, the intervention transforms CHP into a hybrid utility hub that combines conventional CHP generation, waste-heat recovery, renewable electricity, and power-to-steam integration. The CHP continues to provide backup electricity and steam when renewable-electricity generation is low or when steam demand exceeds the capacity of the electrode boiler. This ensures operational reliability while increasing the share of low-carbon utilities within the industrial park.

Consequently, the wind turbine and electrode boiler should not be viewed as independent technologies. Together they form a single integrated intervention that supports utility decarbonization, improves steam-network flexibility, and contributes directly to the reduction of fossil-fuel consumption and CO₂ emissions across the industrial park.

$$\text{The annual wind electricity generation is calculated as: } 1.5 \text{ MW} \times 8760 \text{ h/y} \times 0.291 = 3823.74 \text{ MWh/y}$$

$$\text{Of this amount, } 2294.24 \text{ MWh/y} \text{ is used directly as renewable electricity by the industrial park, while the remaining electricity is sent to the high-voltage electrode boiler: } 3823.74 - 2294.24 = 1529.50 \text{ MWh/y}$$

$$\text{Assuming 98\% electricity-to-steam conversion efficiency, the renewable steam production is: } 1529.50 \times 0.98 = 1498.91 \text{ MWh/y}$$

$$\text{Using 8000 operating hours per year, this equals: } 1498.91 \times 3600 / 8000 = 674.51 \text{ MJ/h steam support.}$$



Final CHP Utility-Hub Retrofit

The final CHP intervention combines three complementary technologies: WHRU/HRSG heat recovery, renewable-electricity generation from a 1.5 MW wind turbine, and power-to-steam conversion through a high-voltage electric boiler. These technologies are not evaluated as separate interventions but as components of a single CHP Utility-Hub Retrofit. Together, they transform the CHP from a predominantly fossil-fuel-based utility supplier into a hybrid utility hub that integrates heat recovery, renewable electricity, and low-carbon steam production. Each component contributes a different function within the system. The WHRU/HRSG recovers useful thermal energy from CHP exhaust gases and directly reduces natural-gas consumption. The wind turbine provides renewable electricity to the utility hub, reducing dependence on conventional electricity sources. The high-voltage electric boiler then converts part of this renewable electricity into steam, creating a direct connection between the electricity and steam networks and supporting the MP and LP steam deficits identified in the energy analysis. The direct emissions benefit from the WHRU/HRSG component alone is relatively modest, reducing emissions by approximately 28.48 tCO₂/year through lower natural-gas consumption. However, the largest contribution to overall system performance comes from the combined effect of renewable-electricity generation and renewable steam production. By displacing grid electricity and fossil-fuel-based steam generation, these components significantly increase the overall emissions-reduction potential of the utility system. The total estimated CO₂ reduction of the CHP Utility-Hub Retrofit is 651.22 tCO₂/year. This value includes direct fuel savings from heat recovery, avoided grid-electricity emissions, and avoided emissions associated with fossil-based steam production. Therefore, the reported CO₂ reduction should be interpreted as the total benefit of the integrated utility-hub retrofit rather than as a direct stack-emission reduction from the CHP unit alone. This value is used consistently throughout Chapters 6 and 7 as the final performance result of the selected energy intervention.

Scenario / Component	Energy value	CO ₂ reduction	Interpretation
Baseline CHP retrofit boundary	1573.43 MWh/y fuel input	—	Screening boundary for CHP retrofit calculation
WHRU/HRSG heat recovery	157.34 MWh/y natural-gas saving	28.48 tCO ₂ /y	Direct Scope 1 fuel-saving effect
1.5 MW wind turbine	3823.74 MWh/y renewable electricity	Included in total	Renewable electricity source for direct use and power-to-steam
Direct wind electricity use	2294.24 MWh/y	Included in total	Avoided grid electricity
Electricity to high-voltage electrode boiler	1529.50 MWh/y	Included in total	Power-to-steam input
Renewable steam from electrode boiler	1498.91 MWh/y	Included in total	Low-carbon steam support
Renewable steam support	674.51 MJ/h	Included in total	LP/MP steam-system support
Final CHP Utility-Hub Retrofit	WHRU/HRSG + wind turbine + electrode boiler	651.22 tCO ₂ /y	Final energy intervention used in Chapters 6-8

The Heliac solar-thermal and thermal-storage concept was evaluated during the initial screening phase as a potential decarbonisation option for the CHP system. The purpose of this assessment was to explore alternative pathways for reducing fossil-fuel dependence and supporting steam generation through renewable energy. Although the concept demonstrated potential benefits, it was not selected as the final intervention because the project later focused on a



more integrated utility-hub approach with stronger direct links to the identified steam deficits and utility-network requirements. Therefore, the Heliac-based solar-thermal hybrid should be considered an earlier screening option only and not part of the final proposed system. The final intervention carried forward into Chapters 6, 7, and 8 is the CHP Utility-Hub Retrofit consisting of three integrated components: WHRU/HRSG heat recovery, a 1.5 MW wind turbine, and a high-voltage electrode boiler. These technologies were selected because they directly address the key energy challenges identified in the industrial park, including fossil-fuel dependence, MP and LP steam deficits, and the need for greater utility-system flexibility. The WHRU/HRSG improves energy efficiency by recovering useful heat from CHP exhaust gases, the wind turbine supplies renewable electricity to the utility hub, and the high-voltage electrode boiler converts part of this renewable electricity into low-carbon steam. Together, these technologies form the final energy intervention assessed throughout the remainder of the project and provide the basis for the quantified energy, emissions, economic, and implementation analyses presented in Chapters 6–8.

Drivers and Barriers

The main drivers of the CHP Utility-Hub Retrofit are the commercial maturity of WHRU/HRSG heat-recovery systems, the availability of renewable wind electricity, the technical feasibility of high-voltage electrode boilers, and the direct reduction of natural-gas consumption. All three technologies are commercially available and have been successfully applied in industrial energy systems. In addition, the intervention is highly compatible with existing CHP operation because it builds upon the current utility infrastructure rather than replacing it entirely. This reduces implementation risk and allows the retrofit to be integrated into the existing energy network. A further driver is the strategic position of CHP within the industrial park. As the central supplier of electricity and steam, CHP influences the environmental performance of multiple companies simultaneously. Any reduction in fuel consumption or improvement in utility carbon intensity therefore generates benefits beyond the CHP itself. By lowering the carbon intensity of shared electricity and steam, the retrofit contributes to park-wide decarbonisation and supports future industrial-symbiosis initiatives. However, several barriers must also be considered. The most important challenges include capital investment requirements, grid-connection capacity, wind-resource variability, permitting procedures, electric-boiler sizing, steam-pressure compatibility, and control-system integration. Additional engineering work is required to ensure that recovered heat, renewable electricity, and steam production can be integrated safely and efficiently into the existing utility network. The intervention also faces organisational challenges. Although the retrofit is implemented at the CHP Utility Hub, its benefits extend to other companies through the shared electricity and steam systems. This requires coordination between the CHP operator and industrial users, particularly regarding utility tariffs, operational responsibilities, and benefit allocation. Furthermore, while the retrofit significantly improves the carbon intensity and flexibility of the utility system, it does not completely eliminate the MP and LP steam deficits identified in the energy analysis. For this reason, the CHP Utility-Hub Retrofit should be viewed as the first phase of a broader industrial-symbiosis strategy rather than as a complete stand-alone solution. It establishes the low-carbon utility backbone required for future material-recovery and water-reuse initiatives while providing substantial energy and emissions benefits in the short term. Subsequent integration of Olefins hydrocarbon recovery and Aromatics wastewater treatment and reuse can further improve overall system performance and support the development of a fully integrated energy-material-water symbiosis network.

Ethylene Oxide/Ethylene Glycol (EO/EG) Plant Profile and Symbiosis Potential

The EO/EG plant links the PolyCycle Park by converting ethylene into mono ethylene glycol (MEG), the key PET precursor. It is material-efficient (1.99 kg/kg MEG) but entirely fossil-dependent. Its E-factor (9.61 kg waste/kg MEG) exceeds bulk chemical norms due to CO₂ incineration and untreated wastewater, and it has the park's highest Scope 1 GHG emissions (2.27 tCO₂e/t MEG; 305,088 tCO₂e/y). The plant imports 6,655.5 MJ/h of utilities while wasting 12,297.5 MJ/h of low-grade heat at 40°C, the largest unrecovered stream in the park. Network analysis ranks it as the top symbiosis candidate: participation coefficient 0.5 and interlayer connectivity 1.0 are the highest in the five plants.



Challenges and Problems

The key material-related challenges identified for EO/EG are:

- Unrecovered 12,297.5 MJ/h of low-grade heat at 40°C, the largest thermal waste stream in the park.
- Incineration of 58.5 kt/y of high-purity CO₂ from the K₂CO₃ desorber (E1-T2), driving the highest Scope 1 GHG intensity in the park.
- Two untreated wastewater streams (biodegradable effluent and MEG-contaminated purge water) breaching VLAREM II discharge standards.
- Total reliance on CHP for 6,406.95 MJ/h of steam (of 6,655.5 MJ/h total utility imports) with no backup, creating a single point of failure.

Material Analysis

System Boundary

The plant receives ethylene feedstock from the Olefins plant, MP steam (3,506.56 MJ/h), stripping steam and make-up water from CHP and municipal supply, and electricity (248.55 MJ/h) from CHP. The sole product export is MEG to the PET plant. CO₂ off-gas from the K₂CO₃ desorber (E1-T2) is not exported but incinerated on-site, generating the plant's primary Scope 1 emissions. Cooling water and 12,297.5 MJ/h of low-grade heat at 40°C are internal and not exported under the base case.

Waste Stream Classification

The plant generates four waste streams. Effluent water (from make-up water and stripping steam) is organic and biodegradable. Cooling water purge contains both organic and inorganic matter, is poorly biodegradable, with MEG as the primary impurity. Both are currently untreated and breach VLAREM II. Flue gas from E1-FU1B is gaseous, contributing 0.20 tCO₂e per tonne processed. The K₂CO₃ scrubber purge is inorganic and currently incinerated. The latter two together drive the E-factor to 9.61 kg waste/kg MEG, above the bulk chemical threshold of 5.

Carbon Balance and Consistency

The carbon balance is consistent with the park-level Sankey. Ethylene carbon is either retained in MEG (material efficiency 1.99 kg/kg) or exits as CO₂ via EO reactor oxidation and scrubber purge incineration. The Scope 1 GHG intensity of 2.27 tCO₂e/t MEG (305,088 tCO₂e/y), the highest in the park, and the E-factor of 9.61 share the same root cause, a waste route centred on incineration. The main uncertainty is the scrubber purge flow rate, to which both indicators are sensitive.

Proposed Interventions

Four interventions are identified for the EO/EG plant using the CIMO-logic framework, with a phased sequence if pursued: S-03 first, S-04 second, then S-01 and S-02 in parallel. This package is diagnosed here but not carried forward as this project's primary recommendation (see Chapter 6 for the prioritization rationale). S-04 remains independently relevant regardless of that prioritization, since it addresses licence-to-operate precondition; VLAREM II wastewater compliance is a regulatory requirement at EO/EG irrespective of which energy intervention first.



S-01: High-Temperature Heat Pump (HTHP)

Phase	Content
Context	12,297.5 MJ/h of heat is wasted at 40°C and nearby greenhouses need 70–85°C, but no direct link exists. The greenhouse is treated as a potential external symbiosis partner, not a confirmed offtake, as noted in Chapter 3.
Intervention	Install an HTHP (R-1233zd(E), COP ≥ 3.0) on the CW exchangers to upgrade heat and supply 108–180 MJ/h to the S-01 greenhouse network without process changes. In Linny-R: CW exchangers → CW Heat → HTHP. No S-01 product exists in the model. The path ends at HTHP. The 108–180 MJ/h greenhouse-supply figure is a separate manual estimate, consistent with the note in the combined-results section below.
Mechanism	Cuts cooling load and replaces fossil heating, creating a heat exchanger link (EO/EG → greenhouse). Drivers: EU ETS, Flemish VEKP, €20–30M CAPEX, seasonal demand. The HTHP can be powered by renewable electricity, making S-01 compatible with full decarbonization of the heating supply.
Outcome	108–180 MJ/h supplied to greenhouse; 48–80 kt CO ₂ /y avoided (assuming displacement of natural gas boilers at ~0.20 tCO ₂ /MWh)

S-02: CO₂ Capture from the K₂CO₃ Scrubber

Phase	Content
Context	The EO desorber (E1-T2) produces 58.5 kt/y of 98% CO ₂ , currently incinerated, driving high Scope 1 emissions.
Intervention	Add a CO ₂ capture unit, compress and supply the stream to a potential greenhouse or urea plant recipient (both identified as unconfirmed external symbiosis partners in Chapter 3), replacing incineration. In Linny-R: E1-T2 → S-02 external sink at 58.5 kt/y.
Mechanism	Avoid on-site emissions and displaces fossil CO ₂ at the user, enabled by EU ETS incentives. Barriers include certification, reliability, and transport infrastructure.
Outcome	58.5 kt CO ₂ /y Scope 1 emissions cut

S-03: Waste Heat Recovery from E1-FU1B

Phase	Content
Context	Flue gas from E1-FU1B is wasted, while EO/EG relies on 6,406.95 MJ/h of CHP steam (steam-only; total utility import is 6,655.5 MJ/h) with no backup
Intervention	Install WHR boiler to recover 19 MJ/h of HP steam from flue gas. In Linny-R: E1-FU1B → S-03 at 19 MJ/h.
Mechanism	Convert waste heat into steam, reducing CHP dependence. Mature technology with short paybacks depends on incinerator operation.
Outcome	19 MJ/h HP steam recovered; ~11 tCO ₂ /y avoided



Combined Effect

Intervention	Benefit	TRL	Phase
S-01 HTHP	48–80 kt CO ₂ /y avoided	TRL 8–9	Phase 3
S-02 CO ₂ Capture	58.5 kt CO ₂ /y Scope 1 cut	TRL 7–8	Phase 3
S-03 WHR	~11 tCO ₂ /y avoided	TRL 9	Phase 1
S-04 WWTP	E-factor reduction + water reuse	BAT/IED	Phase 2

S-04: Centralized Wastewater Treatment (WWTP)

Phase	Content
Context	Two untreated streams (biodegradable effluent and MEG-contaminated purge water) violate VLAREM II and raise the E-factor to 9.61.
Intervention	Send wastewater to a central WWTP with staged treatment: (1) equalisation and pH control, (2) aerobic biological treatment, (3) MBR for MEG removal, (4) optional UV/H ₂ O ₂ polishing, and (5) reuse as cooling make-up water. Consistent with Best Available Techniques (BAT) under the IED.
Mechanism	Closes the water loop, ensures compliance, and lowers the E-factor via shared CAPEX. Driven by Flemish Blue Deal 2.0 and VLAREM II (40% water reduction by 2030). Barriers include inter-company coordination, MEG load variability, and land availability. Note: S-04 is not modelled in the Linny-R energy network as it addresses material flows and regulatory compliance rather than energy symbiosis.
Outcome	Full VLAREM II compliance. MEG recovery from purge water reduces the E-factor from 9.61 toward the bulk chemical threshold of 5. Treated effluent reused as cooling make-up water contributes to the Flemish Blue Deal 2.0 target of 40% water use reduction by 2030.

Together, the four interventions avoid 106–139 kt CO₂/y, reduce the E-factor toward bulk chemical thresholds, ensure full regulatory compliance, and position EO/EG as the park's main industrial symbiosis hub. Of this total, 58.5 kt/y (S-02) depends on an unconfirmed external CO₂ offtaker and should be read as future decarbonisation potential rather than a committed saving; see the Linny-R combined-energy result below, which accordingly excludes S-02.

In Linny-R, the EO/EG park network models two of the four proposed interventions directly: S-02 (external CO₂ capture, modelled as the EO desorber's CO₂ stream) and S-03 (steam recovery, modelled as the incinerator's waste-heat-recovery flow to the boiler). S-01 (greenhouse heat use) is not represented in the Linny-R network; its 48–80 kt CO₂/y figure is a separate estimate and is not part of the modelled energy results below. S-03's 19 MJ/h recovered steam corresponds to approximately 11 tCO₂/year avoided through displaced natural gas combustion, consistent with the intervention table above. S-02 is excluded from these modelled totals because it is a large-scale direct carbon capture measure (58.5 kt CO₂/year) without a confirmed stakeholder, and is therefore treated separately as a future decarbonization potential

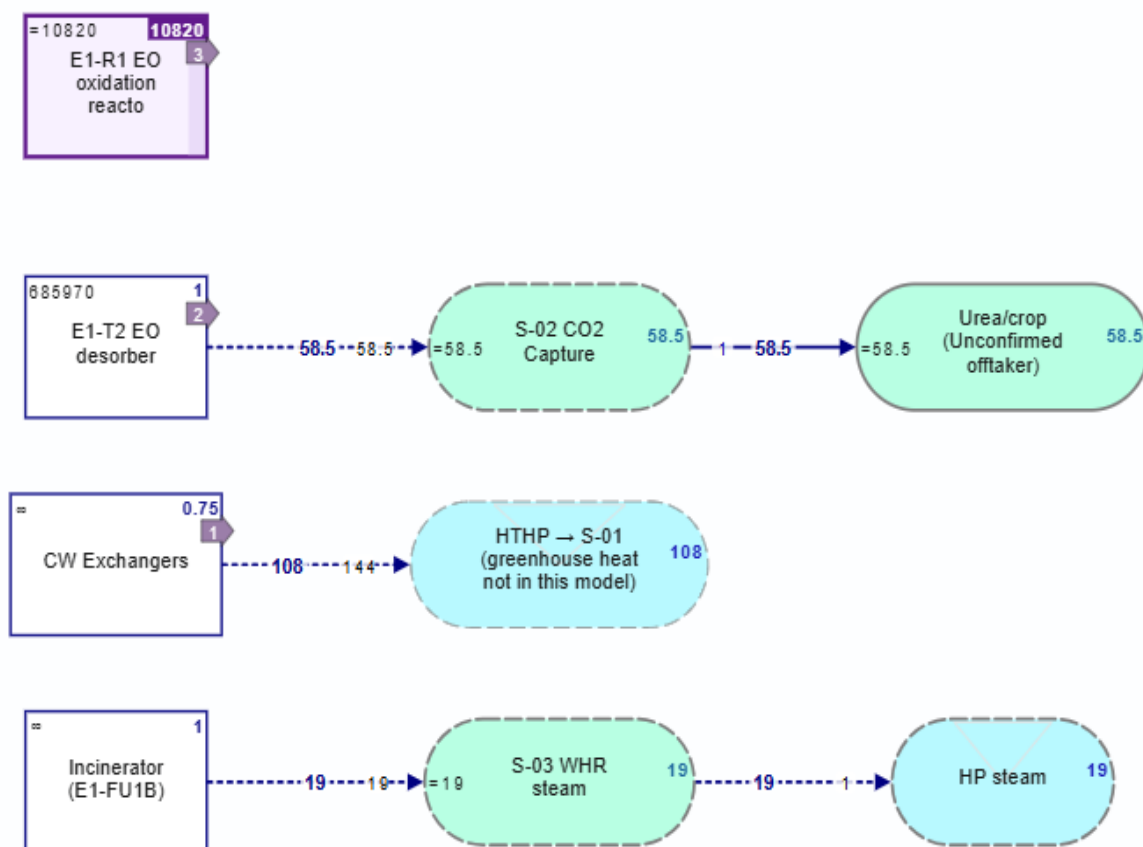


Figure: EO/EG plant Linny-R energy network, showing the modelled S-02 (CO₂ capture) and S-03 (waste heat recovery) flows. S-01 (greenhouse heat use) is not represented in the Linny-R network.

PET Plant

Why PET: The PET plant is selected because it functions as the downstream receiver in the industrial park. It uses MEG from the EO/EG plant and converts it into PET, effectively closing the main material chain. As noted in Chapter 3, the reference data sheet lists a base capacity of approximately 1 kt/year; all energy figures and intervention calculations below correspond to the scaled 100 kt/year configuration required to balance upstream material flows. Although its total energy demand is lower than Aromatics, EO/EG, or Olefins, it still contains both heat-demanding and heat-rejecting units. This makes it suitable for targeted internal heat integration. The analysis shows that electricity is not the main issue. Electricity demand is relatively low, while the process still requires high-pressure steam and rejects heat through multiple overhead coolers. This indicates that the main optimization potential is thermal rather than electrical.

Intervention 1: Internal Heat Recovery Network Description

The first intervention focuses on installing an internal heat recovery network across the esterification, polymerisation, and MEG recovery sections. The objective is to recover heat from hot process streams that are currently cooled and reuse this heat to supply cold streams that require heating. The main heat sources are overhead coolers and product cooling streams, including the esterification reactor overhead coolers, MEG recovery overhead cooler, low-polymerisation overhead coolers, and the PET product cooler. These streams provide approximately 5.684 TJ/year of recoverable heat. The main heat sinks are the esterification reactors, the MEG recovery column, and related process



units, with a combined heat demand of about 6.049 TJ/year. This creates a strong internal match between heat supply and demand. The theoretical maximum recovery is limited by the smaller of these values: $Q_{\max} = \min(5.684, 6.049) = 5.684$ TJ/year. In practice, full recovery is not achievable due to temperature constraints, heat exchanger limitations, fouling, and process integration issues.

Therefore, a realistic assumption is that 50–60% of the available heat can be recovered.

For a conservative 50% recovery:

Recovered heat = $5.684 \times 0.50 = 2.842$ TJ/year = 789.45 MWh/year,

CO₂ avoided = $789.45 \times 0.181 = 142.9$ tCO₂/year

For a higher 60% recovery:

Recovered heat = $5.684 \times 0.60 = 3.410$ TJ/year = 947.3 MWh/year,

CO₂ avoided : $947.3 \times 0.181 = 171.5$ tCO₂/year

KPI Effects

KPI	Expected change	Explanation
Total energy consumption	Decreases slightly	lower MP steam use
Energy intensity	Decreases	less energy per tonne PET
Material efficiency	Increases	more MEG recovered and recycled
Wastewater load	Decreases	less MEG lost in effluent
CO ₂ emissions	Decrease by 8–12 tCO ₂ /y	lower steam generation
Process stability	Improves	better control of recovery column

Intervention Result

Approach	Steam saved (TJ/y)	Steam saved (MWh/y)	Avoided CO ₂	Main effect
MEGrecovery optimisation, 10%	0.155	43.1	7.8	reduced reboiler duty
MEGrecovery optimization, 15%	0.233	64.7	11.7	energy/material benefit



MEG recovery optimization is a realistic intervention because PET production inherently relies on the use and recycling of glycols in esterification and polymerisation stages. In industrial practice, recovery and reuse of MEG is already integrated into process design. For example, Technip Energies (Zimmer PET technology) describes multi-stage PET production systems where process control and glycol recovery are central to efficient operation. Similarly, studies by PBL Netherlands Environmental Assessment Agency and TNO identify key PET process steps such as esterification, polymerisation, solvent recovery, cooling, and drying. These studies highlight energy efficiency and process optimization as primary decarbonization pathways, confirming that both heat recovery and MEG recovery are technically grounded interventions.

Combined PET Intervention Effect

The total impact of the PET intervention combines two components: the internal heat recovery network and MEG recovery optimization. A conservative scenario assumes 50% heat recovery combined with a 10% improvement in MEG recovery efficiency. A higher-performance scenario assumes 60% heat recovery combined with a 15% MEG recovery improvement. This combined approach reflects a realistic strategy where thermal integration and material recovery are implemented together, leading to cumulative reductions in energy demand and emissions while maintaining process stability.

Combined PET interventions	Energy saved (TJ/y)	Energy saved (MWh/y)	Avoided CO2 (tCO2/y)
Conservative: 50% heat recovery + 10% MEG optimisation	2.997	832.6	150.7
Higher: 60%heat recovery + 15% MEG optimisation	3.643	1012.0	183.2

Explanation

The combined PET intervention results in a reduction of steam-equivalent energy demand of approximately 3.0–3.6 TJ/year and an avoided emission range of about 151–183 tCO2/year. The majority of this impact is driven by the internal heat recovery network, while MEG recovery optimization contributes a smaller but still relevant additional benefit by improving both energy and material efficiency. Although PET is not the largest energy consumer in the park, this intervention remains credible because it targets well-defined internal heat flows and achieves measurable improvements without requiring major system restructuring.

Drivers and Barriers

Energy cost reduction is a primary driver, as lower steam demand directly reduces operating expenses. CO2 reduction is achieved through avoiding natural-gas-based steam generation, which improves the plant's environmental performance. Cooling-water demand is also reduced, since less heat is rejected to the cooling system. From a process perspective, PET is well suited for heat integration because it contains both hot and cold streams, enabling internal matching. In addition, continuous operation improves the feasibility of stable heat recovery. The intervention is also aligned with Best Available Techniques (BAT), particularly in terms of energy efficiency and process optimization. Temperature mismatch can limit recovery potential, as some cooling streams may not be hot enough to replace high-pressure steam. Fouling is another concern, since PET and MEG streams can affect heat exchanger performance over time. Retrofit complexity is also relevant, as additional heat exchangers and piping require space and integration into existing layouts.



Chapter 6. Systems Integration

6.1 Integrated System Definition

Integration of Synergies and Selected Interventions

The selected option is the Full Integrated System. This system was selected after comparing the baseline system, the energy retrofit, the material circularity option, the water reuse option, and the fully integrated system. The comparison shows that single domain interventions improve only one part of the industrial park. The energy retrofit performs strongly in terms of CO₂ reduction, renewable electricity integration, and steam flexibility. However, it does not improve material circularity or water circularity. The material circularity option reduces material losses and improves E factor performance, but it does not address fossil based utility dependence. The water reuse option improves water circularity, but its direct impact on CO₂ reduction and steam system performance is limited. For this reason, the full integrated system is selected because it is the only option that improves all major system dimensions simultaneously. The selected package combines one intervention from each domain: energy, material, and water. The energy intervention is the CHP Utility Hub Retrofit. CHP means Combined Heat and Power. In this project, the CHP plant is the central utility provider of the industrial park because it supplies electricity and steam to the process plants. In the baseline system, the CHP depends mainly on natural gas. After the intervention, the CHP becomes a hybrid utility hub that combines recovered heat, wind electricity, and power to steam conversion. The energy intervention consists of three connected components. The first component is WHRU/HRSG heat recovery. This unit recovers useful heat from CHP exhaust gas and transfers it to the feedwater or steam system. It therefore reduces natural gas consumption because part of the heat demand is supplied by recovered heat that would otherwise be rejected. The second component is a 1.5 MW wind turbine. The wind turbine supplies renewable electricity to the CHP electricity hub. The third component is a high voltage electrode boiler. A high voltage electrode boiler is an industrial electric boiler that uses electricity to produce steam or hot water by passing current through boiler water. It does not burn fuel at the point of use, and therefore it does not produce direct combustion emissions during operation. Industrial high voltage electrode boilers are suitable for power to steam applications because commercial systems are available at industrial scale, can operate at high steam pressures, and can respond quickly to variable electricity supply. Official supplier and technical sources report electrode boiler efficiencies close to 100%, while Energy.nl reports a wider electric/electrode boiler efficiency range of 95–99.9%. Therefore, a 98% electricity to steam conversion efficiency is used in this report as a conservative screening assumption. The 1.5 MW value used in the calculation represents the amount of wind electricity available for the electrode boiler pathway. It should not be interpreted as the final supplier nameplate capacity. In detailed design, the boiler could be sized according to supplier availability and operated at part load when less renewable electricity is available (PARAT, n.d.; VAPEC, n.d.; Energy.nl, n.d.). In this project, the high voltage electrode boiler converts part of the renewable wind electricity into low carbon steam. The electrode boiler is integrated inside the CHP Utility Hub and connected to the LP/MP steam network. This is technically realistic because industrial electrode boilers can be used as flexible power to steam units in industrial steam systems and can operate together with existing fossil fuel or CHP steam generation. Real industrial examples include electrode steam boilers installed in chemical or energy intensive industrial sites, which supports the feasibility of applying this type of technology in the PolyCycle Port utility network (PARAT, n.d.). The three energy components work together as one integrated energy intervention. Wind electricity is first used to supply the electricity demand of the industrial park. When wind electricity is available above the direct electricity demand, the remaining electricity is sent to the high voltage electrode boiler. The boiler converts this electricity into steam, which supports the LP/MP steam system. At the same time, the WHRU/HRSG recovers exhaust heat from the CHP and reduces natural gas input. The CHP remains available as backup and balancing capacity during periods with low wind generation or high steam demand. Therefore, the intervention does not replace the CHP. It changes the CHP from a fossil based utility supplier into a hybrid utility hub. The material intervention is Olefins hydrocarbon by product recovery and circular feedstock recirculation. The intervention focuses on reducing material losses in the Olefins plant by recovering underutilized hydrocarbon by products and redirecting suitable streams back into productive use. The earlier material flow analysis showed that Olefins has one of the highest material intensity values in the industrial park, together with a very high E factor. This indicates substantial generation of low value by products and waste streams. Recovering these hydrocarbon fractions and reintroducing them as feedstocks or process inputs can reduce virgin material demand, improve resource efficiency, and strengthen material circularity within the petrochemical network. A second possible route is plastic to naphtha, where post consumer plastic waste is converted into pyrolysis oil and upgraded into cracker compatible feedstock. In this report, the main material



intervention is therefore defined as Olefins hydrocarbon by product recovery and circular feedstock recirculation. Aromatics can still act as a linked receiver or processor of selected hydrocarbon fractions, but Olefins is the main material hotspot. The water intervention is Aromatics BTX wastewater treatment and reuse through AOP, biological or membrane treatment, and reverse osmosis. This intervention targets wastewater streams generated in the Aromatics plant that contain BTX related compounds and other contaminants. The treatment train combines advanced oxidation processes for contaminant degradation, biological or membrane treatment for further pollutant removal, and reverse osmosis to produce high quality reusable water. The treated water can then be reused in the industrial park as cooling water make up or boiler feedwater support. This reduces freshwater demand, lowers wastewater discharge, and improves water circularity. The intervention also creates a direct link between the water system and the CHP Utility Hub because treated water can support the utility system. Further integration opportunities are also identified. The S 01 to S 07 intervention set can be treated as additional future symbiosis opportunities. These measures show that the park has further integration potential beyond the selected package, especially through steam redistribution, cooling water heat upgrading, CO₂ utilization, and external greenhouse or urea connections. However, the main Chapter 6 and Chapter 7 analysis focuses on the selected integrated package to keep the system boundary consistent.

Domain	Final selected intervention	Justification
Energy	CHP Utility Hub Retrofit: WHRU/HRSG heat recovery + 1.5 MW wind turbine + high voltage electrode boiler	Provides the largest quantified CO ₂ reduction and significantly improves electricity and steam flexibility across the industrial park
Material	Olefins hydrocarbon by product recovery and circular feedstock recirculation	Olefins has the highest E factor and one of the highest material intensity values, indicating substantial material loss reduction potential
Water	Aromatics BTX wastewater treatment and reuse through AOP, biological/membrane treatment, and RO	Treats contaminated aromatic wastewater and produces reusable water for cooling water make up or boiler feedwater support

Table 6.1. Final selected integrated package

6.1 Why CHP Is Selected as the Priority Energy Node

The CHP Utility Hub is selected as the priority energy node because it occupies a central position within the industrial park energy network. CHP supplies electricity and multiple steam levels to the surrounding process plants, making it the main utility provider for the system. As a result, improvements implemented at the CHP level influence several companies simultaneously, whereas interventions at individual process plants primarily affect local operations. This makes CHP the most effective location for achieving system-wide energy and emissions improvements. The selection is further supported by the energy-balance results. The analysis showed that the main challenge within the industrial park is not electricity demand, but large MP and LP steam deficits together with significant cooling-water requirements. Therefore, the proposed intervention is designed to address the thermal imbalance of the system rather than focusing solely on renewable-electricity generation. The CHP Utility-Hub Retrofit combines three complementary technologies. First, the WHRU/HRSG recovers useful thermal energy from CHP exhaust gases and reduces natural-gas consumption. Second, a 1.5 MW wind turbine supplies renewable electricity to the utility hub. Third, a high-voltage electric boiler converts part of this renewable electricity into steam, creating a direct connection between the electricity and steam networks. Together, these technologies improve energy efficiency, reduce fossil-fuel dependence, and increase steam flexibility within the industrial park. The CHP intervention is also closely linked to the key findings of the pinch and utility analyses. The park experiences substantial steam deficits, particularly at the MP and LP levels, while excess heat is rejected through cooling systems. By recovering waste heat and converting renewable electricity into steam, the CHP retrofit directly addresses these system-wide energy challenges. This makes the intervention more relevant than a stand-alone renewable-electricity project because it contributes simultaneously to electricity supply, steam generation, and emissions reduction.



Finally, CHP represents the most important enabling node for the full integrated symbiosis system. Lower-carbon electricity and steam supplied by the utility hub can support future material and water interventions, including Olefins hydrocarbon recovery and Aromatics wastewater treatment and reuse. For this reason, CHP is not selected as an isolated technology but as the backbone of the integrated energy-material-water strategy. The intervention transforms CHP from a predominantly fossil-based utility supplier into a hybrid utility hub that combines heat recovery, renewable-electricity generation, and power-to-steam integration. This central role justifies its selection as the priority energy node for detailed assessment and implementation.

6.1.2 System Representation

The integrated system is represented as a multilayer industrial symbiosis network consisting of three resource layers: energy, material, and water. The CHP Utility Hub functions as the central node of the system because it manages the shared electricity and steam infrastructure of the industrial park. Within the CHP Utility Hub, the electricity hub and steam hub are represented as internal utility pools that distribute utilities to the other companies.

The energy layer includes the CHP plant, WHRU/HRSG heat recovery system, 1.5 MW wind turbine, high voltage electrode boiler, and the electricity and steam connections supplying Olefins, Aromatics, EO/EG, and PET. The purpose of this layer is to reduce fossil fuel dependence, increase renewable energy integration, and improve steam system flexibility.

The material layer focuses on Olefins hydrocarbon by product recovery and circular feedstock recirculation. Valuable hydrocarbon streams that are currently underutilized are recovered and redirected back into productive use. This reduces material losses and improves resource efficiency. The water layer consists of Aromatics BTX wastewater treatment using advanced oxidation processes, biological or membrane treatment, and reverse osmosis. The treated water is reused within the industrial park as cooling water make up or boiler feedwater support. Together, these three layers create an integrated industrial symbiosis system in which energy, material, and water flows are connected through shared infrastructure and resource exchanges rather than operating as separate systems. The system representation also reflects the wider park level utility balance. In the current park, CHP acts as the main utility provider for electricity and steam, while the process plants depend on this central utility system. The park still faces MP and LP steam deficits, Aromatics has the highest heat demand, and EO/EG releases a large cooling water heat stream. These findings confirm that the integrated system should not only add new technology, but also improve the way existing energy, material, and water streams are redistributed across the park.

Table 6.2. Linny R nodes and main flows

Linny R node	Role in the system	Main flows
CHP Utility Hub	Central utility provider and system coordinator	Natural gas input; electricity and steam supplied to industrial park users
WHRU/HRSG heat recovery	Heat recovery unit	CHP exhaust heat input; recovered heat to feedwater or steam system
Wind turbine	Renewable electricity source	Wind electricity supplied to the CHP electricity hub
High voltage electrode boiler	Power to steam conversion unit	Renewable electricity input; low carbon steam output
Olefins hydrocarbon recovery	Material circularity intervention	Hydrocarbon by products input; recovered hydrocarbon feedstock output
Aromaticswastewater treatment plant	Water circularity intervention	BTX containing wastewater input; treated reusable water output
Cooling and boiler water loop	Water reuse receiver	Treated water input; cooling water make up and boiler feedwater support



6.1.3 Physical Infrastructure Requirements

The selected integrated system requires modifications to the existing utility and process infrastructure. For the energy layer, the CHP Utility Hub Retrofit requires ducting from the CHP exhaust outlet to the WHRU/HRSG unit, heat exchanger or boiler equipment, feedwater and condensate connections, steam tie ins to the LP/MP steam headers, blowdown and drainage connections, and control system integration. The wind turbine requires a foundation, electrical connection, transformer, switchgear, metering, protection equipment, and grid connection. If direct on site installation is not feasible, a power purchase agreement can be used as an alternative renewable electricity supply route. The high voltage electrode boiler requires a medium or high voltage electrical connection, transformer or switchgear capacity, feedwater supply, water treatment control, blowdown connection, steam outlet connection to the LP/MP steam network, and automatic load control. Commercial electrode boilers are suitable for this role because they are designed for industrial steam and hot water production and can be integrated with automation and control systems. Fast load response and wide modulation capability are important because the boiler must follow variable wind electricity availability while maintaining reliable steam support. For the material layer, the Olefins hydrocarbon recovery intervention requires collection lines for underutilized hydrocarbon by products, separation or purification equipment, intermediate storage or buffer tanks, quality control instrumentation, and a return line to the cracker feedstock system or another compatible process unit. For the water layer, the Aromatics wastewater treatment intervention requires a dedicated BTX wastewater collection line, AOP reactor, biological or membrane treatment unit, RO polishing stage, treated water storage, water quality monitoring, and reuse pipelines to cooling water make up or boiler feedwater pre treatment. These infrastructure requirements are essential because the integrated system is not only a set of technologies, but a modification of the shared industrial utility backbone.

Table 6.3. Physical infrastructure requirements

Infrastructure element	Needed for	Main function
Exhaust ducting and WHRU/HRSG tie in	CHP retrofit	Transfers exhaust heat to recovery unit
Feedwater and condensate connection	WHRU/HRSG	Converts recovered heat into useful steam or hot water
LP/MP steam header connection	WHRU/HRSG + electrode boiler	Supplies low carbon steam to park users
Medium or high voltage electrical connection	High voltage electrode boiler	Supplies electricity for power to steam conversion
Electrical switchgear and transformer	Wind turbine + electrode boiler	Enables renewable electricity use and safe power to steam operation
Boiler automation and SCADA connection	High voltage electrode boiler	Allows fast load control and integration with variable wind electricity
Wind turbine foundation and grid connection	Wind turbine	Provides renewable electricity to CHP hub
Hydrocarbon recovery lines and storage	Olefins material intervention	Collects and reuses valuable by products
AOP, membrane and RO treatment line	Aromatics water intervention	Treats BTX wastewater for reuse
Treated water storage and reuse pipeline	Water loop	Sends reusable water to cooling or boiler feedwater support
Control and metering system	Whole integrated system	Measures flows, verifies savings, and coordinates operation



6.1.4 TONF Structure

The TONF framework is used to evaluate the feasibility of the integrated system. From a technical perspective, the system combines mature technologies, including CHP exhaust gas heat recovery, wind electricity generation, high voltage electrode boiler steam production, hydrocarbon recovery, and advanced wastewater treatment. The main technical challenge is not whether the technologies exist, but whether the flow qualities, capacities, temperatures, pressures, timing and purity requirements can be matched across the park. The electrode boiler is technically feasible for this integration because industrial electrode boilers can produce steam at high pressures, respond quickly, and operate as flexible power to heat units alongside existing fossil fuel or CHP steam systems. From an organizational perspective, the CHP operator should manage the shared electricity and steam hub. Olefins is responsible for the material recovery route, while Aromatics is responsible for the wastewater treatment and reuse route. The park manager should coordinate shared investment, data exchange, infrastructure planning, cost allocation and monitoring. This creates a need for a park level coordination mechanism because costs and benefits are distributed across several companies. From a network perspective, the integrated system increases the number of useful connections between companies and resource layers. Instead of a mainly linear system, the park becomes a multilayer symbiosis network with energy, material and water loops. The CHP node becomes more central because it coordinates electricity, steam, recovered heat, wind electricity, and treated water support. From a financial perspective, the full integrated system has higher CAPEX and coordination complexity than isolated interventions. However, it creates multiple benefit streams: lower natural gas use, avoided grid electricity, lower CO₂ emissions, reduced virgin material demand, reduced waste generation and water reuse. Therefore, the integrated system is more complex than the baseline, but it provides stronger system wide value.

TONF Dimension	Main Requirement	Application in the Integrated System	Main Risk or Barrier	Required Action
Technical	Technologies must be technically feasible and compatible with existing process conditions	The integrated system combines commercially mature technologies, including WHRU/HRSG heat recovery, a 1.5 MW wind turbine, an electric boiler, Olefins hydrocarbon by-product recovery and circular-feedstock recirculation, and Aromatics wastewater treatment using AOP, biological/membrane treatment, and RO reuse	Differences in steam pressure, exhaust-gas conditions, wastewater composition, hydrocarbon quality, and renewable-energy availability may limit performance	Perform detailed engineering studies, verify process conditions, confirm recoverable hydrocarbon streams, evaluate wastewater quality, and optimize equipment sizing
Infrastructural	Physical infrastructure must support resource exchange between companies and process units	The system requires WHRU/HRSG integration, steam-header connections, electrical infrastructure for the electric boiler, wind-generation connection, hydrocarbon-recovery infrastructure, wastewater-treatment facilities, and treated-water reuse pipelines	Existing infrastructure may have insufficient capacity or unsuitable connection points, requiring modifications and temporary shutdowns	Conduct utility tie-in studies, grid-capacity assessments, steam-network analysis, pipeline-routing studies, and phased implementation planning
Organizational	Companies must coordinate operation, responsibilities, data sharing, and investment decisions	CHP manages the utility hub, Olefins manages hydrocarbon recovery, Aromatics manages wastewater treatment, and the industrial park manager coordinates shared infrastructure and resource exchanges	Costs, risks, and benefits are distributed across multiple companies, creating potential conflicts regarding ownership, operation, and investment responsibilities	Develop utility agreements, cost-sharing mechanisms, operational responsibilities, governance structures, and data-sharing procedures
Network	Resource exchanges should strengthen integration across energy, material, and water layers	The system increases connectivity between CHP, Olefins, Aromatics, EO/EG, and PET. CHP acts as the central coordination node linking electricity, steam, recovered heat, renewable energy, material recovery, and water reuse	Greater integration increases dependency on the CHP Utility Hub and shared infrastructure, making reliability more critical	Maintain backup capacity, implement preventive maintenance, establish monitoring systems, and develop emergency operating procedures
Financial	Benefits must justify additional CAPEX, operational costs, and coordination efforts	The system generates value through reduced natural-gas use, renewable-electricity generation, renewable steam production, lower CO ₂ emissions, improved material recovery, and increased water reuse	Economic performance depends on energy prices, carbon prices, wind performance, technology costs, and utilization rates	Use supplier quotations, perform sensitivity analysis, evaluate multiple economic scenarios, and implement the system through phased investment



6.1.5 Trade Off Matrix

The integration score is a qualitative screening score based on five criteria: energy and CO2 improvement, material circularity, water circularity, network integration and implementation feasibility. Each option receives one point when it substantially improves one of these criteria. The score is therefore not a financial optimization result, but a reproducible screening method for selecting the most balanced integrated option.

Table 6.4. Scoring criteria for the trade off matrix

Scoring criterion	Meaning
Energy and CO2 improvement	Reduces fossil utility use or CO2 emissions
Material circularity	Reduces virgin feedstock use or recovers by products
Water circularity	Reduces freshwater demand or enables reuse
Network integration	Creates useful cross company or cross layer links
Feasibility	Uses technically realistic and implementable measures

The trade off matrix shows that the full integrated system is the strongest option. The energy retrofit reduces CO2 emissions and improves steam flexibility, but it does not solve material or water inefficiencies. The material only option improves circularity, but it does not reduce fossil utility dependence. The water only option improves water circularity, but it gives limited direct CO2 reduction. The full integrated symbiosis system is therefore selected because it addresses all three problem domains at the same time: fossil based utility dependence, material losses and low water circularity.

Trade-Off Matrix

PolyCycle Port Integrated System Options

- Pass / Acceptable
- Fail / Not Acceptable
- Non-Negotiable = Yes

							Acceptable Ranges		Non-Negotiable	Reference
Criteria	Unit	Baseline	CHP Utility-Hub Retrofit	Material Circularity	Water Reuse	Full Integrated System	Minimum Pass	Maximum Pass		
CO2 Reduction	tCO2/y	0.000	651	80.0	20.0	740	300	1000	Yes	Main Decarbonisation Target
Renewable Electricity Share	% Annual Electricity Demand	0.000	140	0.000	0.000	140	30.0	100	Yes	Energy Transition Objective
Steam Flexibility	MJ/h Low-Carbon Steam Support	0.000	674	0.000	0.000	674	300	1500	Yes	Steam Deficit
Material Circularity	Normalized Score	0.250	0.250	0.800	0.300	0.850	0.600	1.00	Yes	Olefins Hydrocarbon Recovery
E-Factor Improvement	Normalized Score	0.100	0.100	0.700	0.200	0.750	0.500	1.00	No	Olefins Waste Reduction
Water Circularity	Normalized Score	0.200	0.200	0.250	0.800	0.850	0.600	1.00	Yes	Aromatics AOP / Membrane / RO Reuse
Network Integration	Normalized Score	0.350	0.700	0.550	0.500	0.900	0.600	1.00	Yes	Multilayer Network Density
Financial Feasibility	Normalized Score	0.800	0.650	0.550	0.500	0.550	0.500	1.00	No	CAPEX / OPEX
Implementation Simplicity	Normalized Score	0.900	0.600	0.550	0.550	0.400	0.400	1.00	No	Technical / Organisational Complexity
Policy Alignment	Normalized Score	0.400	0.900	0.850	0.850	0.950	0.700	1.00	Yes	EU Decarbonisation / Circularity

Option	Energy intervention	Material intervention	Water intervention	Main benefit	Main weakness	Score	Decision
A. Baseline system	None	None	None	No additional CAPEX	Fossil dependence, high waste and limited circularity	1/5	Rejected
B. Energy only retrofit	CHP Utility Hub Retrofit with WHRU/HRSG, wind turbine and high voltage electrode boiler	None	None	Strong CO2 and energy reduction	Does not address material or water circularity	3/5	Insufficient for Chapter 6
C. Material only circularity	None	Olefins hydrocarbon by product recovery and circular feedstock recirculation	None	Reduces virgin material demand and waste	Does not reduce fossil utility emissions from CHP	3/5	Insufficient alone
D. Water only reuse	None	None	Aromatics BTX wastewater treatment with AOP, biological/membrane treatment and RO reuse	Improves water circularity and reduces freshwater demand	Limited direct CO2 reduction	2/5	Insufficient alone
E. Energy + material	CHP Utility Hub Retrofit	Olefins hydrocarbon recovery and circular feedstock substitution	None	Strong energy, carbon and material circularity benefits	Does not close the water loop	4/5	Strong but incomplete
F. Energy + water	CHP Utility Hub Retrofit	None	Aromatics AOP and RO water reuse	Strong utility water integration	Does not address material circularity	4/5	Strong but incomplete
G. Full integrated symbiosis system	CHP Utility Hub Retrofit	Olefins hydrocarbon by product recovery and circular feedstock recirculation	Aromatics BTX wastewater treatment and RO reuse	Improves energy, carbon, material and water KPIs together	Highest coordination and implementation complexity	5/5	Selected

Table 6.5. Comparison of integration options



6.1.6 Flow Representation

Table 6.6. Main integrated flows

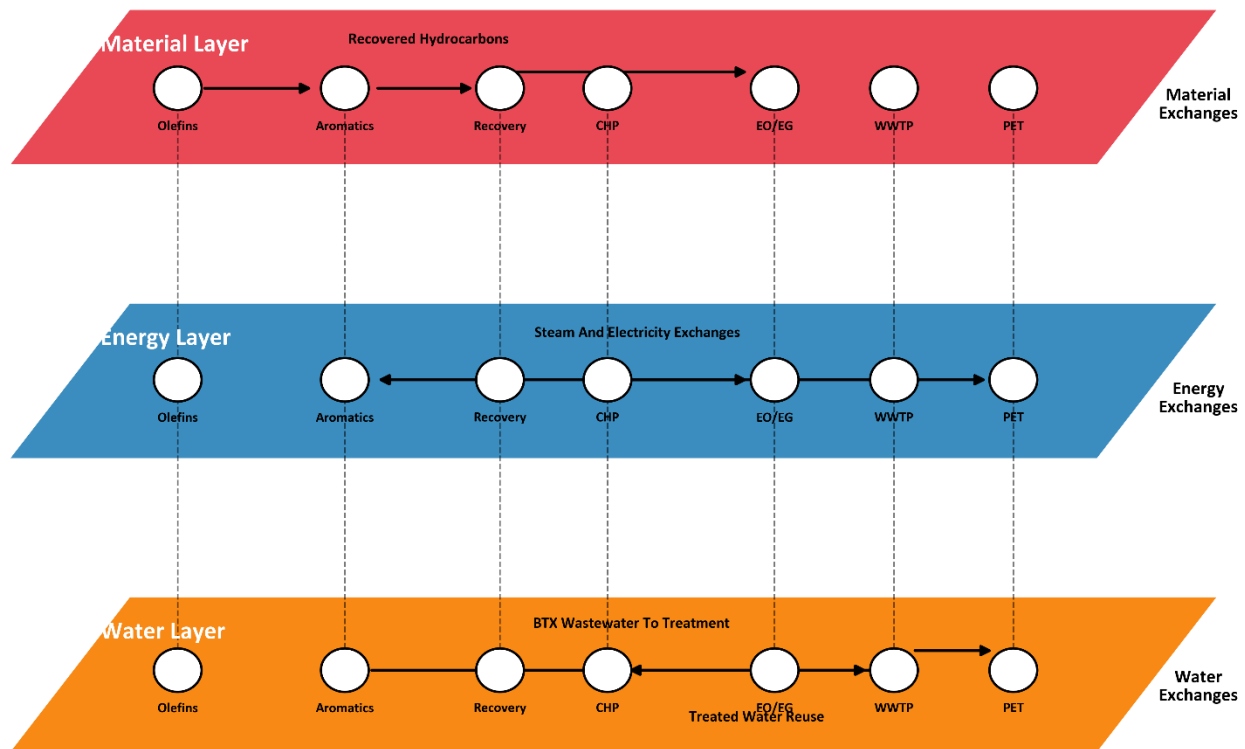
From	To	Flow
Natural gas	CHP	Fuel
CHP	CHP electricity hub	Electricity
CHP	Steam hubs	LP/MP/HP steam
CHP exhaust	WHRU/HRSG	Waste heat
WHRU/HRSG	Feedwater or steam hub	Recovered heat
Wind turbine	Electricity hub	Renewable electricity
Electricity hub	High voltage electrode boiler	Wind electricity
High voltage electrode boiler	LP/MP steam hub	Renewable steam
Electricity hub	All plants	Electricity
Steam hubs	All plants	Steam
Olefins	Olefins hydrocarbon recovery	Hydrocarbon by products
Olefins hydrocarbon recovery	Olefins or compatible receiver	Recovered feedstock
Aromatics	Aromatics WWTP	BTX wastewater
Aromatics WWTP	Cooling and boiler water loop	Treated water
Cooling and boiler water loop	CHP / plant users	Cooling water make up or boiler feedwater support

6.1.7 Network Analysis

The network analysis shows that the integrated system creates a more connected industrial symbiosis network than the baseline system. New infrastructure elements such as WHRU/HRSG heat recovery, wind turbine, high voltage electrode boiler, hydrocarbon recovery unit and wastewater treatment system increase the total number of nodes within the network. At the same time, additional exchanges of energy, material and water increase the number of links between companies and resource systems. As a result, network density increases because a larger share of potential connections is actively used. The integrated system also improves layer participation, since companies are no longer connected only through utility supply chains but also through material recovery and water reuse loops. The role of CHP becomes even more important in the integrated configuration. While CHP already functions as the main utility supplier in the baseline system, it becomes the central coordination hub for electricity, steam, renewable energy integration, recovered heat and supporting water flows.

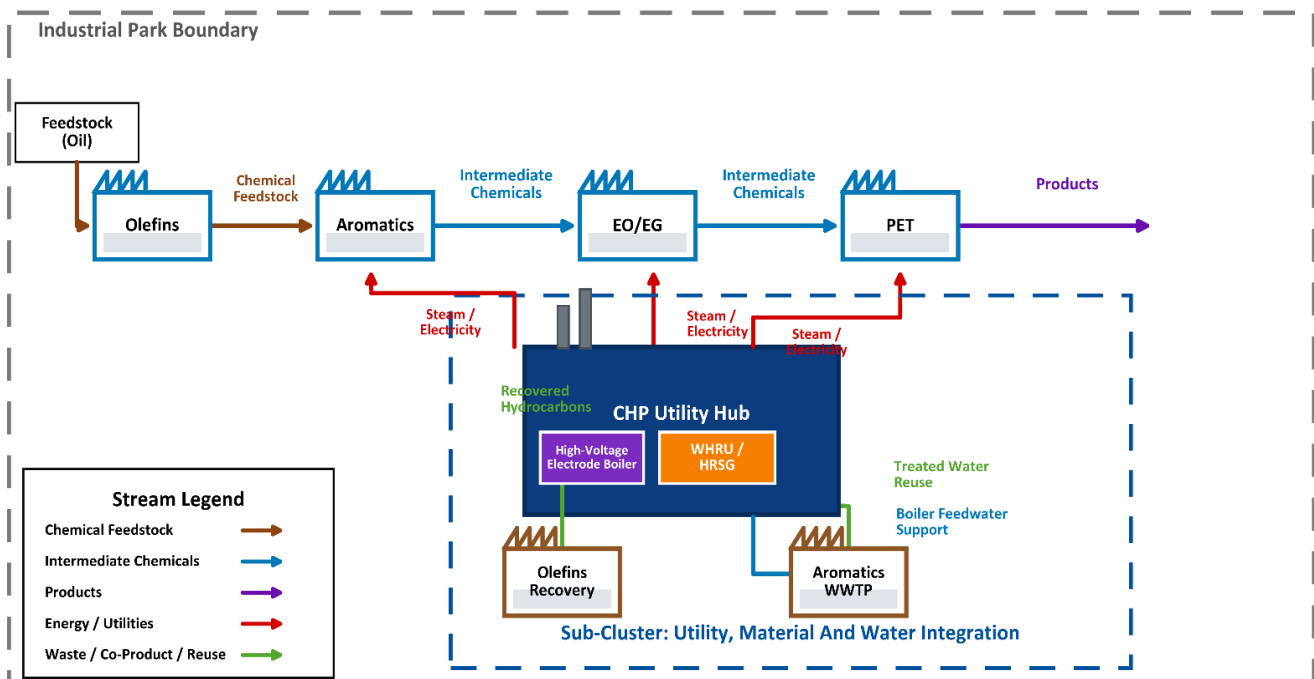


Multilayer Network Representation Of PolyCycle Port



Dashed Vertical Lines Connect The Same Company Across Material, Energy And Water Layers

PolyCycle Port Industrial Symbiosis Network



Selected Integrated Package: CHP Utility-Hub Retrofit + Olefins Hydrocarbon Recovery + Aromatics Wastewater Treatment And Reuse



Table 6.7. Network properties before and after intervention

Network property	Before intervention	After intervention
Energy layer nodes	CHP, Olefins, Aromatics, EO/EG and PET	CHP, Olefins, Aromatics, EO/EG, PET, WHRU/HRSG, wind turbine and high voltage electrode boiler
CHP centrality	High	Very high
Energy layer density	Lower	Higher
Network resilience	Strong dependence on natural gas	More diversified energy supply, but still dependent on CHP reliability
Renewable energy links	Absent	Strong integration through wind turbine
Steam electricity coupling	Limited	Strong through power to steam conversion
Cross layer integration	Limited interaction between energy, water and material systems	Stronger integration through CHP linked energy, water and material interventions

Table 6.8. Multiplex network interpretation

Metric	Baseline system	Full integrated system	Interpretation
Number of nodes	Lower	Higher	Additional nodes are introduced, including WHRU/HRSG, wind turbine, high voltage electrode boiler, wastewater treatment system and hydrocarbon recovery unit
Number of links	Lower	Higher	Additional exchanges are created across the energy, material and water layers
Network density	Lower	Higher	The number of actual connections increases relative to the total number of possible connections
CHP centrality	High	Very high	CHP evolves from a utility supplier into the central coordination hub for electricity, steam and supporting resource flows
Layer participation	Low	Higher	More companies participate simultaneously in energy, material and water exchanges
System resilience	Lower	Higher, but still dependent on CHP reliability	Resource supply becomes more diversified, but overall system performance still depends strongly on CHP reliability

The intervention increases the functional diversity of the CHP node. In the baseline system, CHP mainly supplies fossil based electricity and steam to the industrial park. After the intervention, CHP becomes a hybrid utility hub that coordinates multiple energy functions, including recovered heat, wind electricity and power to steam conversion. This changes the network structure without replacing the existing utility backbone. The intervention increases network density, adds new energy pathways and improves the resilience of the energy layer. However, because the CHP Utility Hub becomes more central after integration, redundancy, backup steam capacity, preventive maintenance and clear utility management rules are required.

6.1.8 Wider System Context

Additional park level performance information from the wider intervention set shows that the industrial park still has major steam network integration potential. The external steam demand is reduced by 2642 MJ/h when S 04, S 05 and S 06 are considered together, mainly because S 06 contributes 1962 MJ/h, or about 74% of the total recovered



energy. The MP steam deficit decreases from 14587 MJ/h to approximately 12625 MJ/h, a reduction of about 13%, while the LP steam deficit decreases from 10121 MJ/h to approximately 9512 MJ/h. These values show that the selected CHP retrofit should be understood as part of a broader steam network optimization problem rather than as a complete closure of all steam deficits. These values represent wider park level integration opportunities and should not be interpreted as the isolated effect of the CHP Utility Hub Retrofit alone.

6.2 Performance Evaluation

6.2.1 Key Assumptions

The performance evaluation compares the baseline system, the energy retrofit and the full integrated system. The results are based on a screening level assessment because detailed engineering data, exact flow qualities and final supplier quotations are not yet available. Therefore, the values should be interpreted as conceptual design results rather than final implementation guarantees.

Table 6.9. Key assumptions

Assumption	Value / treatment	Reason
System boundary	PolyCycle Port industrial park	Matches Chapter 6 integration scope
Selected energy intervention	CHP Utility Hub Retrofit	Priority energy intervention from Chapter 5
Selected material intervention	Olefins hydrocarbon recovery and circular feedstock recirculation	Olefins has high E factor and material intensity
Selected water intervention	Aromatics BTX wastewater treatment and reuse	Targets contaminated aromatic wastewater and enables reuse
Wind turbine capacity	1.5 MW	Selected energy scenario
Wind capacity factor	29.1%	Used to estimate annual wind generation
Wind electricity generation	3823.74 MWh/y	Used for renewable electricity and power to steam calculation
Wind electricity to direct use	2294.24 MWh/y	Direct replacement of grid electricity
Wind electricity to electrode boiler	1529.50 MWh/y	Power to steam route
Renewable steam generation	1498.91 MWh/y	Produced through electrode boiler pathway
Electrode boiler efficiency	98%	Conservative value within official/technical 95–99.9% efficiency range
WHRU/HRSG fuel saving	157.34 MWh/y	Direct CHP natural gas saving
CO ₂ saving	651.22 tCO ₂ /y	Includes avoided grid electricity, renewable steam and direct fuel saving
Water reuse	Qualitative / normalized	Exact water flow data remain uncertain
Material circularity improvement	Qualitative / normalized	Exact recovery yield depends on by product composition and treatment route
Wider steam deficit values	Supporting context only	Not part of isolated CHP economic boundary



6.2.2 Energy and CO₂ Assessment

The energy retrofit delivers substantial improvements in both energy performance and carbon reduction. The WHRU/HRSG heat recovery system reduces natural gas consumption by approximately 157.34 MWh/y through recovery of exhaust heat that would otherwise be lost. This alone avoids around 28.48 tCO₂/y. The largest contribution comes from renewable electricity supplied by the 1.5 MW wind turbine. Annual wind generation is estimated at approximately 3823.74 MWh/y. Of this, 2294.24 MWh/y is supplied directly to industrial electricity users, while 1529.50 MWh/y is sent to the high voltage electrode boiler. Assuming 98% conversion efficiency, this produces approximately 1498.91 MWh/y of renewable steam. This efficiency assumption is conservative because official and technical sources report electrode boiler efficiency values close to 100% and a wider electric/electrode boiler range of 95–99.9% (VAPEC, n.d.; Energy.nl, n.d.). This power to steam configuration is important because the industrial park has significant steam demand. The electrode boiler therefore increases renewable energy utilization and supports the steam system at the same time. Its technical role is similar to real industrial applications where electrode boilers are used to produce steam in chemical or energy intensive industrial sites and to operate with existing steam systems. The total CO₂ reduction of the CHP Utility Hub Retrofit is estimated at 651.22 tCO₂/y. This value includes avoided grid electricity, avoided fossil based steam production and direct natural gas savings from heat recovery. It should not be interpreted as direct CHP stack emission reduction only. The direct heat recovery component is 28.48 tCO₂/y, while the larger system benefit comes from renewable electricity and renewable steam integration.

Table 6.10. Energy and CO₂ results

Indicator	Value
Natural gas saved through WHRU/HRSG heat recovery	157.34 MWh/y
Direct CO ₂ reduction from WHRU/HRSG	28.48 tCO ₂ /y
Total wind electricity generation	3823.74 MWh/y
Wind electricity supplied directly to industrial users	2294.24 MWh/y
Wind electricity supplied to high voltage electrode boiler	1529.50 MWh/y
Electrode boiler efficiency	98%
Renewable steam produced by electrode boiler	1498.91 MWh/y
Renewable steam support	674.51 MJ/h
Total estimated CO ₂ reduction	651.22 tCO ₂ /y

The 1.5 MW wind turbine produces approximately 140% of the annual electricity demand of the selected industrial park on an annual energy basis. However, this does not mean that the park becomes fully self sufficient at every moment. Wind generation is variable and depends on weather conditions. Therefore, the park still requires grid connectivity, operational flexibility and support from the CHP Utility Hub. During periods of high wind generation, renewable electricity can be supplied directly to industrial users or converted into steam through the high voltage electrode boiler. During periods of low wind generation, CHP and the grid continue to provide reliable utility support.



6.2.3 System Wide KPI Assessment

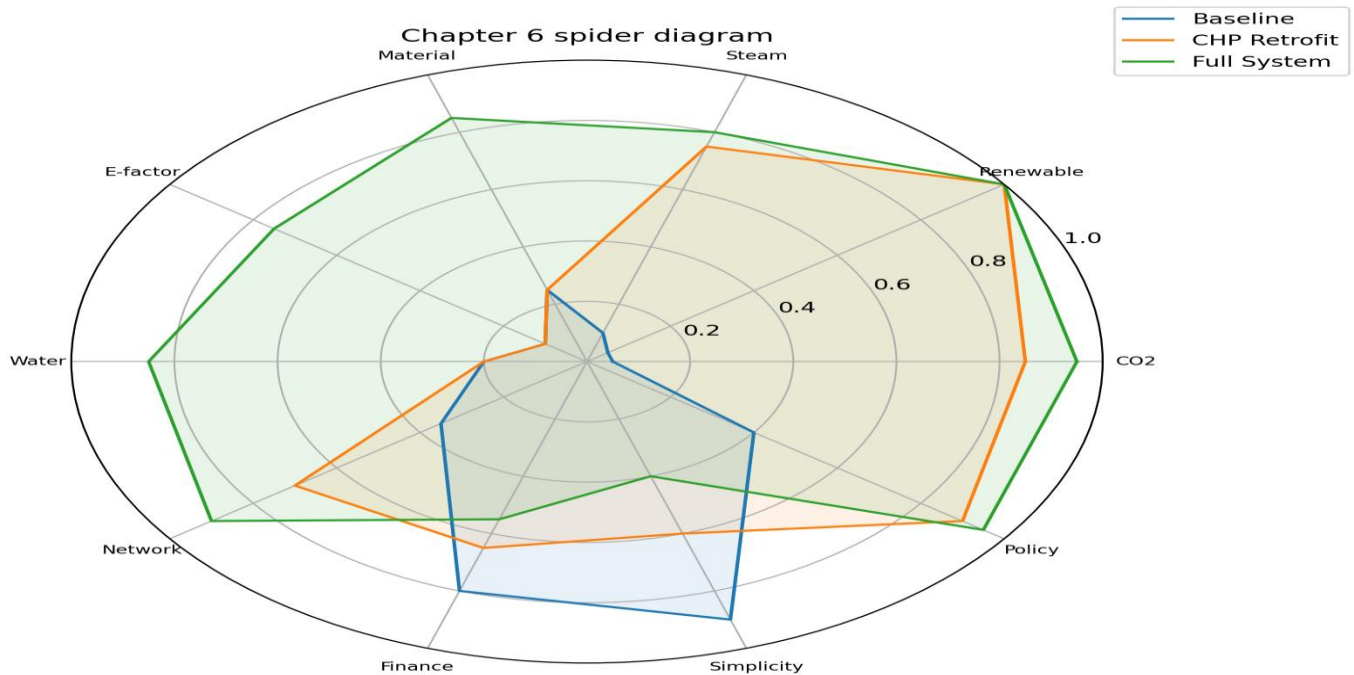
Table 6.11. System wide KPI assessment

KPI	Baseline	Energy intervention	Material intervention	Water intervention	Full integrated effect
CHP fuel input	1573.43 MWh/y	-157.34 MWh/y	No direct effect	No direct effect	Lower natural gas demand
CO2 reduction	0	651.22 tCO2/y	Indirect reduction possible	Small indirect effect	Strong decarbonization
Renewable electricity	0	3823.74 MWh/y generated	Supports recovery operations	Supports treatment	Lower carbon utility supply
Renewable steam	0	1498.91 MWh/y	Supports material processing	Supports water treatment if needed	Stronger steam flexibility
Material intensity	High in Olefins	No direct effect	Expected reduction through recovery	No direct effect	Improved material circularity
E factor	High in Olefins	No direct effect	Expected reduction through by product recovery	No direct effect	Reduced waste burden
Water circularity	Low reuse of contaminated wastewater	No direct effect	No direct effect	Higher through AOP/RO reuse	Reduced freshwater and wastewater burden
Network integration	Limited	Higher energy integration	Higher material integration	Higher water integration	Multilayer symbiosis

The full integrated system performs better than the individual options because it combines benefits across several dimensions. Energy performance improves through heat recovery, wind electricity and electrode boiler steam production. Material performance improves through reduced upstream material losses and improved recovery of hydrocarbon by products. Water performance improves through wastewater treatment and reuse. Network performance improves because more companies become connected through useful exchanges rather than only through linear supply chains.

6.2.4 Spider Diagram

The spider diagram compares the baseline system, the energy retrofit and the full integrated system using normalized indicators. The energy retrofit strongly improves CO2 reduction, renewable energy share, steam flexibility and policy alignment. However, it leaves material circularity and water circularity close to baseline values. The full integrated system performs better as a park level solution because it combines the CHP Utility Hub Retrofit, Olefins hydrocarbon recovery and Aromatics wastewater treatment and reuse. As a result, the full integrated system improves not only energy and emissions performance, but also material circularity, E factor improvement, water circularity and network integration. The main trade off is that the full integrated system has lower financial feasibility and lower implementation simplicity than the baseline because it requires higher investment, more coordination and more complex operation.



6.2.5 Social Impact Assessment

The social impact assessment evaluates both intended and unintended consequences of the integrated system. The intervention is mainly technical, but it affects people, organizations and working routines within the industrial park. Therefore, the assessment considers workers, plant operators, utility managers, maintenance teams, local communities, the port authority, the grid operator and company managers. The intended social impacts include improved energy security, reduced exposure to fossil fuel price volatility, reduced emissions, improved resource efficiency and strengthened technical capability within the industrial park. The intervention may also create new skill requirements related to WHRU/HRSG operation, wind energy monitoring, high voltage electrode boiler operation, wastewater treatment and material recovery. These changes can support workforce development and industrial competitiveness. However, there may also be unintended consequences. The system increases interdependence between companies because more operations depend on the CHP Utility Hub and shared infrastructure. If the CHP Utility Hub, steam network or water reuse system fails, several companies may be affected. Wind turbine implementation may also create concerns about visual impact, noise, permitting or land use. Therefore, safety, reliability, backup capacity, fair cost allocation and transparent communication are important requirements.

Table 6.12. Social indicators

Social indicator	Reason for inclusion
Worker skill development	New technologies require new operational and maintenance skills
Operational reliability	Shared utility systems increase interdependence between companies
Safety perception	WHRU/HRSG, high voltage electrode boiler and wastewater treatment require safety management
Stakeholder acceptance	Wind turbine and shared infrastructure may raise acceptance concerns
Cost benefit fairness	Costs and benefits are distributed across different companies
Local environmental perception	Lower emissions, waste reduction and water reuse may improve public perception



6.2.6 Interview Questions

For workers and operators:

1. What new skills would be needed to operate or maintain the WHRU/HRSG, high voltage electrode boiler, wastewater treatment and material recovery systems?
2. Do you expect the integrated system to improve or complicate daily operations?
3. What safety or reliability concerns do you see in shared steam, electricity and water infrastructure?

For company managers:

1. How would your company evaluate the benefits of joining a shared utility and circularity network?
2. What contractual or financial conditions would be needed before investing in shared infrastructure?
3. Which risks are most important: technical reliability, cost allocation, data sharing or operational dependence?

For local and institutional stakeholders:

1. Would lower CO2 emissions, reduced waste and water reuse improve your perception of the industrial park?
2. What environmental impacts are most important to monitor after implementation?
3. How should the industrial park communicate progress and risks to the local community?

6.2.7 Follow Up After Social Impact Evaluation

After collecting interview responses, the results should be discussed in a stakeholder workshop with company representatives, CHP operators, workers, the port authority, grid operator and local stakeholders. The workshop should present the expected CO2 reduction, infrastructure changes, wind permitting implications, safety measures and cost benefit allocation principles. Stakeholder feedback should be used to revise the implementation roadmap before the final investment decision. This step helps translate the social impact assessment into practical design changes and improves acceptance of the integrated system.

Table 6.13. Follow up actions after social assessment

Social assessment output	Follow up action
Concern about wind turbine noise or visual impact	Include in permitting and stakeholder communication plan
Concern about operational reliability	Add backup capacity and maintenance protocols
Concern about cost benefit fairness	Develop shared utility tariff and cost allocation model
Concern about worker skills	Add training for WHRU/HRSG, electrode boiler, water treatment and recovery systems
Concern from local community	Organize public information workshop



Chapter 7. Economic Assessment

7.1 Identification of Priority Technology

Although Chapter 6 identifies the Full Integrated Symbiosis System as the preferred park level solution, the detailed economic assessment in Chapter 7 focuses on the CHP Utility Hub Retrofit. This approach is justified because the CHP retrofit is the central enabling intervention within the integrated system. By reducing fossil fuel consumption and supplying lower carbon electricity and steam, the CHP Utility Hub supports not only the energy transition of the industrial park but also the future implementation of material recovery and water reuse initiatives. As a result, improvements at the CHP level generate benefits across multiple parts of the industrial symbiosis network.

The selected priority technology is the CHP Utility Hub Retrofit, which combines a WHRU/HRSG heat recovery system, a 1.5 MW wind turbine, and a high voltage electrode boiler for power to steam conversion. This option was selected because it delivers the largest system wide impact among the evaluated energy interventions. The earlier CHP heat recovery only intervention reduced emissions by approximately 28.48 tCO₂/y. In contrast, the integrated utility hub retrofit achieves a total reduction of approximately 651.22 tCO₂/y by combining direct fuel savings with renewable electricity integration and renewable steam production.

The high voltage electrode boiler is technically suitable for this priority technology because industrial electrode boilers are commercially available, high efficiency, fast reacting power to steam units. Official and technical sources report industrial electrode boiler capacities in the MW scale, steam pressure capability above 70 bar in some commercial systems, and efficiency values close to 100%. These data show that the proposed power to steam role is technically realistic. However, final boiler sizing should be checked by supplier quotation because the 1.5 MW value in this assessment represents the renewable electricity input available for the boiler pathway, not necessarily the final supplier nameplate capacity (PARAT, n.d.; Energy.nl, n.d.; VAPEC, n.d.).

Table 7.1. Formal scoring of energy technology options

Criterion	Weight	Heat recovery only	Wind only	Electrode boiler only	CHP Utility Hub Retrofit
CO ₂ reduction potential	30%	2	4	3	5
Steam system benefit	20%	3	1	4	5
Renewable energy integration	15%	1	5	3	5
Compatibility with existing CHP hub	15%	5	3	4	5
Implementation feasibility	10%	4	3	4	3
Strategic value for full integrated system	10%	3	3	4	5
Weighted result	100%	3.0	3.35	3.55	4.8

The CHP Utility Hub Retrofit receives the highest score because it combines the strengths of the individual options. Heat recovery alone provides direct fuel savings but limited total CO₂ reduction. Wind alone provides renewable electricity but does not solve steam flexibility. The electrode boiler alone enables power to steam but depends on low carbon electricity. The combined retrofit is therefore selected because it improves electricity supply, steam flexibility, fuel use, CO₂ emissions and the utility backbone required for the full integrated system.



7.2 Definition of Assessment Scope

The techno economic assessment focuses on the integrated CHP Utility Hub Retrofit within the PolyCycle industrial park. The system boundary includes CHP exhaust gas recovery through the WHRU/HRSG, wind electricity generation, and steam production through the high voltage electrode boiler. The assessment is performed over a 25 year project lifetime using a discount rate of 8% and 8000 operating hours per year. All costs are expressed in 2024 euros. Unlike Life Cycle Assessment, which focuses on environmental impacts over the full life cycle, Techno Economic Assessment evaluates the economic feasibility of the intervention. Therefore, the objective of this chapter is to determine whether the proposed retrofit can achieve meaningful CO₂ reductions while remaining economically attractive.

Table 7.2. Assessment scope

Item	Description
Geographic boundary	PolyCycle industrial park in the Port of Antwerp context
Process boundary	CHP exhaust gas recovery through WHRU/HRSG, wind electricity generation, and power to steam production through the electrode boiler
Technology	WHRU/HRSG heat recovery system, 1.5 MW wind turbine, and high voltage electrode boiler
Time horizon	25 years
Discount rate	8%
Operating hours	8000 h/y
Cost basis	2024 EUR
Main outputs	Lower carbon electricity and steam supplied to the industrial park
Main avoided emissions	Direct natural gas combustion emissions (Scope 1) and avoided grid electricity emissions (Scope 2)

Table 7.3. Included and excluded items

Item	Included?	Reason
WHRU/HRSG equipment	Yes	Direct component of CHP retrofit
1.5 MW wind turbine or PPA equivalent renewable supply	Yes	Provides renewable electricity to CHP hub
High voltage electrode boiler	Yes	Converts renewable electricity into steam
Electrical, steam and control integration	Yes	Required to connect technologies to utility hub
Avoided natural gas use	Yes	Direct fuel saving benefit
Avoided grid electricity	Yes	Scope 2 reduction and cost saving
Avoided fossil steam	Yes	Steam decarbonization benefit
Olefins material recovery CAPEX	No	Belongs to material intervention and requires different process data
Aromatics WWTP CAPEX	No	Belongs to water intervention and requires wastewater flow and treatment data
External S 01 to S 07 opportunities	No	Future opportunities, not part of priority technology boundary
Full upstream LCA emissions	No	Chapter 7 is TEA screening, not full LCA
Biomass or feedstock upstream emissions	Not applicable	No biomass is used in the selected energy intervention



The excluded material and water interventions are not ignored in the integrated system; they are included qualitatively and in the Chapter 6 system comparison. However, they are excluded from the Chapter 7 economic boundary because they require different equipment, different ownership structures and different flow data. Including them without detailed cost and design information would bias the economic result. The Chapter 7 boundary is therefore limited to the CHP Utility Hub Retrofit, which has the strongest quantified energy, CO₂ and cost data.

7.3 Energy and CO₂ Calculation Basis

The 1.5 MW wind turbine is estimated to generate approximately 3823.74 MWh of electricity per year. This corresponds to about 140% of the annual electricity demand of the selected industrial park on an annual energy basis. However, this does not mean that the park becomes fully self sufficient at every moment of operation. Wind generation is variable and depends on weather conditions. For this reason, the industrial park still requires grid connectivity, operational flexibility and support from the CHP Utility Hub. During periods of high wind generation, electricity can be supplied directly to industrial users or converted into steam through the high voltage electrode boiler. During periods of low wind availability, CHP and the electricity grid continue to provide reliable utility support. Therefore, the wind turbine should be viewed as a major contributor to annual renewable electricity supply rather than as a complete replacement for conventional electricity generation. The electrode boiler calculation uses 98% electricity to steam conversion efficiency. This is intentionally conservative. Official supplier data and technical databases report that industrial electrode boilers can reach nearly 100% efficiency, while Energy.nl gives a broader range of 95–99.9% for electric/electrode boilers. The 98% assumption is therefore realistic and avoids overstating the steam benefit.

Table 7.4. Energy and CO₂ calculation basis

Indicator	Value
Natural gas saved through WHRU/HRSG heat recovery	157.34 MWh/y
Direct CO ₂ reduction from WHRU/HRSG	28.48 tCO ₂ /y
Total wind electricity generation	3823.74 MWh/y
Wind electricity supplied directly to industrial users	2294.24 MWh/y
Wind electricity supplied to electrode boiler	1529.50 MWh/y
Electrode boiler efficiency	98%
Renewable steam produced by electrode boiler	1498.91 MWh/y
Renewable steam support	674.51 MJ/h
Total estimated CO ₂ reduction	651.22 tCO ₂ /y

The calculations are:

Wind electricity generation = 1.5 MW × 8760 h/y × 0.291 = 3823.74 MWh/y

Electricity to electrode boiler = 3823.74 – 2294.24 = 1529.50 MWh/y

Renewable steam production = 1529.50 × 0.98 = 1498.91 MWh/y

Renewable steam support = 1498.91 × 3600 / 8000 = 674.51 MJ/h

The total CO₂ reduction of 651.22 tCO₂/y includes several contributions. These consist of avoided grid electricity consumption through wind power generation, renewable steam production through the high voltage electrode boiler, and direct fuel savings from WHRU/HRSG heat recovery. It is important to distinguish between the total system benefit and the direct CHP fuel saving effect. The WHRU/HRSG system reduces natural gas consumption by approximately 157.34 MWh/y, corresponding to a direct emissions reduction of about 28.48 tCO₂/y. The remaining CO₂ reduction is associated with displacement of grid electricity and fossil based steam production through renewable electricity and power to steam integration.



7.4 Capital Cost Estimation

The capital cost estimate includes the wind turbine, the WHRU/HRSG package, the high voltage electrode boiler, and integration costs. Since the project is still at a conceptual design stage, the values should be interpreted as screening level estimates with an uncertainty of approximately $\pm 30\%$. The 1.5 MW wind turbine represents the largest investment, accounting for most of the total project cost. Additional investment is required for the WHRU/HRSG system, the electrode boiler, and the integration of electrical, steam and control systems. The total estimated capital cost is approximately €2.53 million. The electrode boiler cost is treated as a screening level installed cost for the power to steam package needed to absorb up to 1.53 MW of wind electricity in the modeled pathway. The capacity is smaller than many published large electrode boiler projects, but it remains technically defensible because the project uses a conceptual screening boundary rather than final vendor sizing. In detailed design, the supplier may recommend a larger standard module operating at part load or a different electric boiler configuration. Therefore, the cost should be interpreted as a conceptual installed allowance and must be refined using vendor quotations.

Table 7.5. CAPEX assumptions

Component	Assumption
Wind turbine	1.5 MW
Installed wind cost	€1200/kW
WHRU/HRSG package	€150000
High voltage electrode boiler	Sized for up to 1.53 MW renewable electricity input in screening calculation
Electrode boiler installed cost	€250000
Electrical, steam and control integration	Included as subcomponents
Cost accuracy	Conceptual estimate, $\pm 30\%$

Table 7.6. CAPEX breakdown

CAPEX component	Basis	Estimated cost (€2024)	Comment
1.5 MW wind turbine	€1200/kW $\times$ 1500 kW	1800000	Main renewable electricity asset
WHRU/HRSG package	Screening estimate	150000	Heat recovery from CHP exhaust
High voltage electrode boiler package	Screening estimate	250000	Power to steam conversion package; final nameplate to be checked by supplier quotation
Electrical integration	7% of equipment subtotal	154000	Switchgear, metering, cabling
Steam and feedwater tie ins	4% of equipment subtotal	88000	Steam piping, valves, condensate/feedwater connection
Control and monitoring	2% of equipment subtotal	44000	Sensors, control system and performance monitoring
Engineering and installation contingency	2% of equipment subtotal	44000	Conceptual stage allowance
Total CAPEX	—	2530000	Conceptual estimate, $\pm 30\%$



All costs are expressed in 2024 euros. When current 2024 screening values or supplier style assumptions are used, no CEPCI correction is applied. If older vendor or literature cost data are used, they should be updated using the following equation:

$$\text{Cost}_{2024} = \text{Cost}_{\text{base}} \times \text{CEPCI}_{2024} / \text{CEPCI}_{\text{base}}$$

Because the present estimate is conceptual and not based on final vendor quotations, an uncertainty range of $\pm 30\%$ is applied.

Table 7.7. CAPEX uncertainty range

Case	CAPEX (€M)	Interpretation
Low CAPEX	1.77	–30% optimistic case
Base CAPEX	2.53	Conceptual estimate
High CAPEX	3.29	+30% conservative case

7.5 Operating Cost and Revenue Estimation

The operating cost estimate separates variable effects, fixed operating costs and avoided cost streams. This is necessary because the intervention does not only create savings; it also creates new operation, maintenance, labour and monitoring requirements.

Table 7.8. Variable operating effects and savings

Variable item	Unit	Baseline	Intervention	Economic effect
Natural gas use	MWh/y	1573.43	1416.09	157.34 MWh/y saved
Grid electricity purchase	MWh/y	Current grid demand	–2294.24	Avoided grid electricity purchase
Fossil steam production	MWh/y	Fossil based steam	–1498.91	Replaced by renewable steam pathway
Electricity to electrode boiler	MWh/y	0	1529.50	Internal renewable electricity use
Water consumption	qualitative	No direct change	Minor indirect change	Excluded from CHP TEA
Waste disposal	qualitative	No direct change	No direct change	Excluded from CHP TEA
Feedstock	qualitative	No direct change	No direct change	Excluded from CHP TEA

Table 7.9. Fixed OPEX assumptions

Fixed OPEX item	Treatment	Estimated value
Wind turbine maintenance	2.5% of wind CAPEX	€45000/y
WHRU/HRSG maintenance	4% of WHRU/HRSG CAPEX	€6000/y
Electrode boiler maintenance	3% of electrode boiler CAPEX	€7500/y
Integration and monitoring maintenance	1% of integration CAPEX	€3300/y
Labour and supervision	Annual monitoring and operation allowance	€25000/y
Insurance and property tax	1% of total CAPEX	€25300/y
Plant overhead	20% of maintenance and labour	€17360/y
Total fixed OPEX	Sum of above items	€129460/y



Capital costs are represented through an annualized capital charge based on the project lifetime and selected discount rate.

$$\text{Annualized CAPEX} = \text{Total CAPEX} \times \text{CRF}$$

$$\text{CRF} = r(1+r)^n / [(1+r)^n - 1]$$

Where $r = 8\%$ and $n = 25$ years. This gives a capital recovery factor of approximately 0.0937. Therefore, the annualized CAPEX is approximately:

$$€2530000 \times 0.0937 = €237007/\text{y}$$

Table 7.10. Revenue and avoided cost streams

Avoided cost stream	Calculation	Annual value
Avoided grid electricity	$2294.24 \text{ MWh/y} \times €120/\text{MWh}$	€275309/y
Avoided natural gas from WHRU/HRSG	$157.34 \text{ MWh/y} \times €45/\text{MWh}$	€7080/y
Avoided fossil steam through electrode boiler	$1498.91 \text{ MWh/y} \times €45/\text{MWh}$	€67451/y
Potential avoided CO2 cost	$651.22 \text{ tCO}_2/\text{y} \times €75/\text{tCO}_2$	€48842/y
Total benefit excluding CO2 price	—	€349840/y
Total benefit including CO2 price	—	€398682/y

Table 7.11. Annual economic balance

Item	Annual value
Annualized CAPEX	€237007/y
Fixed OPEX	€129460/y
Total annual cost	€366467/y
Total benefit excluding CO2 price	€349840/y
Total benefit including CO2 price	€398682/y
Net annual balance excluding CO2 price	-€16627/y
Net annual balance including CO2 price	+€32215/y

This result shows that the project is close to economic break even without carbon price value and becomes economically attractive when avoided CO2 cost is included. This also shows why the result should not be presented as automatically profitable. The intervention has strong strategic and environmental value, but its financial performance depends on electricity price, carbon price, wind performance and CAPEX control.

7.6 CO2 Avoidance Cost Calculation

The CO2 avoidance cost is calculated by comparing the annualized net cost of the intervention with the annual CO2 reduction. The total CO2 reduction is 651.22 tCO2/y.

$$\text{CO2 avoidance cost} = \text{Net annual cost} / \text{Annual CO2 avoided}$$


Table 7.12. CO2 avoidance cost

Case	Net annual cost	CO2 saving	Avoidance cost
Without CO2 price benefit	€16627/y	651.22 tCO2/y	+€26/tCO2
Including CO2 price benefit	-€32215/y	651.22 tCO2/y	-€49/tCO2

The positive value without CO2 price means that, when only energy savings are considered, the project is close to break even but still has a small annual net cost. The negative value including CO2 price means that the project creates a net annual financial benefit when the avoided carbon cost is counted. For robustness, the main interpretation should focus on the first case, while the second case can be used to show policy and carbon market relevance.

7.7 Investment Indicators

The economic assessment should not only focus on annual cost savings. Therefore, simple payback, NPV, benefit cost ratio and CO2 avoidance cost are also included.

Simple Payback = CAPEX / Net Annual Cash Benefit

NPV = $\sum \text{CashFlow}_t / (1+r)^t - \text{CAPEX}$

BCR = Present Value of Benefits / Present Value of Costs

CO2 Avoidance Cost = Net Annual Cost / Annual CO2 Avoided

For the base case, the annual cash benefit before annualized CAPEX but after fixed OPEX is:

With CO2 price:

€398682 – €129460 = €269222/y

Without CO2 price:

€349840 – €129460 = €220380/y

Table 7.13. Investment indicators

Indicator	Without CO2 price	Including CO2 price
Simple payback	11.5 years	9.4 years
Annualized net balance	-€16627/y	+€32215/y
NPV over 25 years	-€177493	+€343885
Benefit cost ratio	0.93	1.14
CO2 avoidance cost	+€26/tCO2	-€49/tCO2

The base case result should be interpreted as a screening level feasibility result rather than a final investment decision. The project appears economically attractive only if a large share of the wind electricity is used internally and if avoided electricity, gas, steam and CO2 costs remain sufficiently high. If electricity prices fall, wind capacity factor is lower than expected, CAPEX increases, or grid connection costs rise, the avoidance cost and payback period become less favourable.

7.8 Sensitivity Analysis

The sensitivity analysis tests the influence of electricity price, CAPEX, wind capacity factor, natural gas price and carbon price. These parameters were selected because they directly affect the two main sides of the economic balance: annual benefits and investment cost.



Table 7.14. Sensitivity analysis

Parameter	Low case	Base case	High case	Expected effect
CAPEX	–30%	€2.53M	+30%	Higher CAPEX worsens NPV and payback
Electricity price	€80/MWh	€120/MWh	€160/MWh	Higher electricity price improves avoided cost value
Natural gas price	€30/MWh	€45/MWh	€70/MWh	Higher gas price improves fuel and steam savings
Wind capacity factor	24%	29.1%	34%	Higher wind output improves savings and CO2 reduction
Carbon price	€0/tCO2	€75/tCO2	€120/tCO2	Higher carbon price improves financial attractiveness

Table 7.15. Scenario results

Case	Assumption	CO2 saving	Payback	Avoidance cost
Low electricity price	€80/MWh	651.22 tCO2/y	around 13 years	Less attractive
Base case	€120/MWh	651.22 tCO2/y	9–12 years depending on CO2 price treatment	Close to break even or attractive with carbon value
High electricity price	€160/MWh	651.22 tCO2/y	around 7 years	Strongly attractive
CAPEX +30%	€3.29M	651.22 tCO2/y	around 12–15 years	Less attractive
CAPEX –30%	€1.77M	651.22 tCO2/y	around 6–8 years	Strongly attractive
Low wind capacity factor	24%	lower than base case	around 11–13 years	Less attractive
High wind capacity factor	34%	higher than base case	around 8 years	Stronger

The sensitivity analysis shows that electricity price and wind performance are the most important parameters because they determine the value of avoided grid electricity and the amount of renewable electricity available for power to steam conversion. CAPEX also remains important, especially because the wind turbine is the largest cost item. The project remains technically attractive in all scenarios because it reduces fossil dependence and improves steam flexibility. However, its economic attractiveness depends strongly on securing favourable electricity value, achieving realistic wind performance, controlling investment costs and using a large share of wind electricity internally.



7.9 Discussion of Economic Results

The CHP Utility Hub Retrofit has strong environmental and strategic value because it reduces fossil fuel use, increases renewable electricity integration and supports low carbon steam production. The intervention also strengthens the role of CHP as a flexible utility hub instead of a purely fossil based supplier. This is important for the wider integrated system because lower carbon electricity and steam can support future Olefins hydrocarbon recovery and Aromatics wastewater treatment.

The largest capital cost contributor is the 1.5 MW wind turbine. The WHRU/HRSG and high voltage electrode boiler have smaller individual CAPEX values, but they are important because they connect the renewable electricity and steam layers. The largest annual benefit is avoided grid electricity purchase, followed by avoided fossil steam production and avoided CO₂ cost. Direct natural gas saving from WHRU/HRSG is smaller, but it is still important because it represents a direct Scope 1 reduction at the CHP level.

The analysis also shows that the positive result relies on several assumptions. First, the wind electricity must be used internally as much as possible. If a large share of the wind electricity cannot be used by the park or the electrode boiler, the avoided cost value decreases. Second, the electricity price strongly affects the result. Lower electricity prices reduce the benefit of avoided grid electricity. Third, the CAPEX estimate is still conceptual and should be refined using supplier quotations. Fourth, the total CO₂ reduction includes both Scope 1 and Scope 2 effects, not only direct CHP stack emission reductions. Fifth, the electrode boiler calculation is technically realistic, but final design must verify the required steam pressure, feedwater quality, grid connection capacity, supplier module size and control logic.

Overall, the CHP Utility Hub Retrofit is technically feasible and strategically well aligned with the integrated system. Economically, it is close to break even without carbon price value and becomes attractive when carbon cost avoidance and policy support are included. Therefore, the intervention should proceed to a more detailed feasibility study, including supplier quotations, grid connection assessment, wind resource verification, electrode boiler sizing, steam pressure matching and updated financial modelling.



CHAPTER 8 STRATEGIC PLANNING/POLICY ALIGNMENT

8.1. Technology Action Plan

TAP organization matrix

The proposed CHP Utility-Hub Retrofit is based entirely on mature technologies with a Technology Readiness Level (TRL) of 9. Both Waste heat recovery systems and industrial electric boilers are widely deployed in industrial utility networks, while wind power is already well established in Belgium and the Port of Antwerp region. The main implementation challenges include securing investment capital, obtaining permits for wind generation, integrating new equipment into the existing steam network, and ensuring reliable control of the combined CHP High-voltage electrode boiler system. A phased implementation strategy is therefore recommended. The project should begin with a detailed feasibility study and resource assessment, followed by engineering design, safety studies, procurement, and staged installation. This approach reduces technical risk while allowing gradual integration into existing operations. If successfully implemented, the intervention would significantly reduce emissions while improving utility flexibility and renewable energy integration. In addition to the direct CO₂ reduction of approximately 651 tCO₂/year, the project would provide renewable electricity and renewable steam to the industrial park, supporting both decarbonization and long term energy resilience.

At the regional level, the selected intervention is considered technically within the Belgian industrial context and the Port of Antwerp Bruges region. WHRU/HRSG (Waste Heat Recovery Unit) systems are commercially established technologies for industrial flue gas heat recovery and can be integrated into existing CHP and steam generation systems. Industrial electric boilers are also commercially available and are increasingly being used in industrial power to heat and steam production applications. In addition, wind energy is a mature technology in Belgium, supported by existing industrial energy infrastructure and extensive experience with renewable energy deployment.

Additionally, the Port of Antwerp-Bruges hosts major petrochemical industries, CHP operators, utility providers, and engineering contractors capable of designing, installing, and operating waste heat recovery systems, wind power infrastructure and electric boilers. The region is also home to several renewable energy projects and mature industrial utility networks. Together, these provide strong local implementation capacity and a favourable environment for implementation in terms of technical expertise, infrastructure, and industrial ecosystem support, making deployment of the CHP Utility-Hub Retrofit technically feasible.

Awareness of industrial decarbonisation is generally high among regional stakeholders, particularly within the industrial sector, due to climate targets, carbon pricing mechanisms, and the port's transition strategy. However, implementation still depends on effective coordination among multiple stakeholders. Existing policies and regulations also provide a supportive framework for implementation. The intervention is consistent with the European Green Deal, the Fit for 55 package, the Renewable Energy Directive, the Energy Efficiency Directive, and the EU Emissions Trading System (EU ETS). At the regional level, the Port of Antwerp-Bruges transition strategy further supports industrial decarbonisation, electrification, and renewable energy integration, creating favourable regulatory conditions for deployment.

As a result, technology readiness and availability are not expected to be major barriers. Instead, the main challenges are related to implementation and integration. These include matching exhaust gas conditions with heat recovery requirements, integrating renewable steam into existing steam pressure levels, securing grid connections, obtaining permits, ensuring sufficient land availability, meeting safety and operational requirements, and establishing contractual arrangements between the CHP operator and industrial park users. Therefore, successful implementation depends primarily on engineering integration and stakeholder coordination rather than technological development.

Technology / Component	TRL	Current Status	Availability	Main Barriers	Measures and Activities	Primary Stakeholders	Secondary Stakeholders	Roles and Responsibilities	Timeline	Expected Outcome
WHRU/HRSG heat recovery for CHP exhaust gas	9	Commercially mature technology widely applied in industrial CHP and boiler systems	Available from industrial heat recovery suppliers and engineering contractors	Exhaust gas characterization, corrosion and fouling risks, integration with existing steam and feedwater systems	Measure exhaust gas flow and temperature, perform feasibility assessment, obtain supplier quotations, conduct HAZOP study, and develop detailed integration design	CHP operator, EPC contractor	Heat recovery supplier, park manager	CHP operator provides operational data; supplier specifies equipment; EPC contractor performs engineering, installation, and commissioning	Q3 2026–Q4 2028	Approximately 157.34 MWh/year natural gas saving and 28.48 tCO2/year direct emissions reduction
1.5 MW horizontal wind turbine or renewable PPA	9	Wind generation is a mature technology and well established within the Belgian energy system	Available through wind developers, energy providers, and PPA agreements	Permitting requirements, visual and noise impacts, grid connection constraints, and variable electricity generation	Conduct wind resource assessment, evaluate site suitability, perform permitting study, assess grid capacity, and compare on site installation	Park manager, wind developer	Port authority, grid operator, local stakeholders	Wind developer provides project development and installation; port authority manages site constraints; grid operator assesses connection requirements	Q4 2026–2029	Approximately 3823.74 MWh/year renewable electricity generation

					with PPA alternatives					
High-voltage electrode boiler for power to steam integration	9	Commercially available and increasingly used in industrial electrification projects	Available from industrial boiler manufacturers and engineering suppliers	Steam pressure matching, electricity supply integration, and operational control requirements	Size the boiler according to LP and MP steam demand, connect to the electricity hub, and develop operational control logic with CHP backup support	CHP operator, EPC contractor	Boiler supplier, industrial users	CHP operator manages operation; EPC contractor integrates electrical and steam systems; industrial users provide steam demand requirements	Q1 2027–Q4 2028	Approximately 1498.91 MWh/year renewable steam production and 674.51 MJ/h additional steam support

Technology Readiness and Implementation Assessment

Physical infrastructure requirements are an important part of the proposed intervention because the CHP Utility-Hub Retrofit not simply the installation of individual technologies. Instead, it represents a modification of the shared utility infrastructure of the industrial park. The WHRU/HRSG system requires physical integration with the CHP exhaust stream through dedicated ducting, a heat recovery boiler or heat exchanger package, condensate and feedwater connections, blowdown and drainage systems, and integration with the existing steam or feedwater network. These components allow recovered heat to be transferred safely into the utility system and reduce natural gas consumption. The High-voltage electrode boiler requires a high capacity electrical connection together with appropriate switchgear, transformer capacity where necessary, control system integration, and connection to the LP and MP steam headers. This infrastructure enables renewable electricity to be converted into useful process steam and supports flexibility within the steam network. The wind energy component requires either direct physical installation or an alternative power purchase arrangement. For an on site installation, infrastructure requirements include turbine foundations, site access, electrical cabling, grid connection, metering systems, and protection equipment. If an off site power purchase agreement is selected, the physical requirements are reduced, but contractual and grid integration arrangements remain necessary. Additional instrumentation and control upgrades are required throughout the utility hub. These systems must coordinate wind electricity generation, High-voltage electrode boiler operation, recovered heat from the WHRU/HRSG, CHP backup operation, and fluctuating steam demand from industrial users. Reliable monitoring and control are essential to ensure safe and efficient operation of the integrated utility system. The water reuse intervention also requires supporting infrastructure. Treated water from the Aromatics wastewater treatment system must be transported through dedicated piping, with storage or buffer capacity where necessary. Water quality monitoring systems and connections to cooling water make up or boiler feedwater pre treatment systems are also required.



8.2 Implementation Roadmap

The implementation strategy follows a phased approach that reduces technical and financial risk while allowing gradual integration into the existing utility system. The first stage focuses on establishing a reliable technical baseline by confirming CHP operating conditions, utility demand profiles, and renewable energy potential. The second stage develops the technical design of the intervention. This includes wind resource assessment, preliminary Waste heat recovery engineering, High-voltage electrode boiler sizing, and an updated techno economic evaluation using supplier quotations. At this point, the project moves from a conceptual design toward a realistic implementation plan. The third stage focuses on permitting, safety assessment, and stakeholder engagement. Because the intervention combines heat recovery, renewable electricity generation, and power to steam conversion, successful implementation requires coordination between the CHP operator, engineering contractors, the port authority, and the grid operator. The final stages involve procurement, installation, commissioning, and performance verification. By 2029, the system should be fully operational and capable of delivering the projected reductions in natural gas consumption and CO₂ emissions while increasing renewable electricity and renewable steam supply across the industrial park. The implementation plan is structured around a series of decision gates. Progression from one phase to the next should only occur when the required technical, financial, organisational, and regulatory conditions have been successfully verified. This approach reduces project risk and prevents premature commitment to large capital investments before key uncertainties have been resolved. The early decision gates focus on confirming the technical feasibility of the intervention. These include verification of CHP exhaust gas flow and temperature, assessment of recoverable heat potential, evaluation of wind resource availability, confirmation of grid connection capacity, and validation of steam pressure compatibility within the existing utility network. Without these conditions, the expected energy and emissions benefits cannot be guaranteed. Subsequent decision gates focus on economic and institutional feasibility. These include vendor quotations, updated economic assessments, permitting requirements, stakeholder acceptance, contractual arrangements, and agreement on investment responsibilities. Only after these conditions are satisfied should procurement and installation activities begin. This phased decision gate approach ensures that technical feasibility, economic viability, and stakeholder support are confirmed before major investments are made. As a result, the implementation strategy reduces uncertainty, improves project governance, and increases the likelihood of successful deployment of the integrated utility hub retrofit.

Decision-Gate Framework

Decision Gate	Required Evidence	Go / No-Go Criterion
Gate 1: Technical and Environmental Feasibility	CHP exhaust-gas temperature and flow measurements, steam-demand profile, wind-resource assessment, grid-capacity evaluation, and updated CO ₂ -reduction estimate	Proceed only if sufficient recoverable heat is available, wind electricity can be effectively utilized within the industrial park, and the expected CO ₂ reduction remains at least 500 tCO ₂ /year
Gate 2: Economic Feasibility	Vendor quotations, updated CAPEX and OPEX estimates, electricity and natural-gas price assumptions, carbon-price sensitivity analysis, and payback calculations	Proceed only if the project remains close to break-even under conservative assumptions and the CO ₂ avoidance cost remains acceptable
Gate 3: Permitting, Safety and Stakeholder Acceptance	Wind-permitting assessment, HAZOP study, safety review, environmental evaluation, and stakeholder-consultation results	Proceed only if permitting risks are manageable, safety risks can be controlled, and stakeholder concerns can be addressed before final design
Gate 4: Final Investment Decision	EPC proposal, financing strategy, contractual agreements, monitoring plan, and utility-tariff structure	Proceed only if investment costs, operational responsibilities, risks, benefits, and monitoring obligations are clearly allocated among participating parties



The first gate focuses on technical and environmental feasibility. At this stage, the project team must verify that sufficient waste heat is available from the CHP exhaust stream, that wind electricity can be integrated into the park's utility system, and that the expected environmental benefits remain substantial. The minimum target of 500 tCO₂/year ensures that the intervention continues to deliver meaningful decarbonisation benefits. The second gate evaluates economic viability. Updated supplier quotations, revised operating-cost estimates, and sensitivity analyses are required to determine whether the project remains economically attractive under realistic market conditions. Particular attention should be given to electricity prices, natural-gas prices, carbon costs, and wind-performance assumptions. The third gate addresses regulatory, safety, and social feasibility. Permitting requirements, operational safety, environmental impacts, and stakeholder acceptance must be assessed before detailed engineering begins. This stage helps ensure that technical feasibility is supported by institutional and social feasibility. The final gate represents the formal investment decision. By this stage, financing arrangements, contractual responsibilities, monitoring requirements, and benefit-allocation mechanisms should be clearly defined. Only after these conditions are satisfied should procurement and construction activities proceed. This phased decision-gate approach reduces implementation risk and improves the likelihood of successful deployment of the CHP Utility-Hub Retrofit. The expected CO₂ reduction from the CHP Utility-Hub Retrofit is approximately 651.22 tCO₂/year, including contributions from heat recovery, renewable electricity generation, and renewable steam production. To ensure that the project continues to provide substantial environmental value after more detailed engineering and updated calculations, a minimum threshold of 500 tCO₂/year is adopted as the go/no-go criterion. This threshold is intentionally conservative because it allows for uncertainty in assumptions related to wind generation, operational performance, energy demand, and system integration while still requiring a significant reduction in emissions.

Contingency Planning and Risk Management

Key Assumption Not Met	Risk	Contingency Measure
Wind resource is lower than expected	Lower renewable-electricity generation and lower CO ₂ reduction	Use a renewable PPA instead of relying solely on on-site wind generation, or resize the wind component following the wind-resource assessment
Grid capacity is limited	Wind electricity and electric-boiler operation cannot be fully integrated	Prioritize direct on-site electricity consumption, phase the electric-boiler connection, or negotiate grid reinforcement with the grid operator
Wind permit is delayed or rejected	Project implementation is delayed	Use an off-site renewable PPA as a temporary or permanent alternative to on-site wind generation
CAPEX is higher than expected	Longer payback period and reduced economic attractiveness	Implement the project in phases, beginning with WHRU/HRSG heat recovery and adding wind generation and the electric boiler later
Electric boiler cannot provide the required steam pressure	Renewable steam cannot be supplied effectively to the LP/MP steam network	Resize the electric boiler, maintain CHP backup operation, or restrict operation to compatible steam-pressure levels
CO ₂ reduction falls below 500 tCO ₂ /year	Environmental performance becomes lower than expected	Reassess electricity allocation between direct use and power-to-steam, increase internal renewable-electricity utilization, or identify additional heat-recovery opportunities
Stakeholder acceptance is weak	Delays or opposition related to wind, safety, visual impact, or cost allocation	Organize additional stakeholder workshops, revise design and location choices where possible, and improve benefit-sharing arrangements



Although the CHP Utility-Hub Retrofit is based on commercially mature technologies, several assumptions must be validated during implementation. If one or more key assumptions are not met, contingency measures should be available to maintain technical feasibility and project performance.

The purpose of these contingency measures is to ensure that the project remains viable even if initial assumptions prove inaccurate. Rather than treating technical, economic, or social uncertainties as project-ending risks, the implementation strategy provides alternative pathways that allow the intervention to adapt to changing conditions.

This approach increases the resilience of the implementation plan and supports informed decision-making at each project stage. It also aligns with the decision-gate framework, where uncertainties are progressively reduced before major investments are committed. As a result, project development can continue while maintaining acceptable technical performance, environmental benefits, and stakeholder support.

Phase	Timeline	Activity	Responsible Actor	Expected Output
Phase 1	Q3 2026	Confirm CHP exhaust gas temperature, exhaust flow rate, steam demand, and electricity demand profile	CHP operator	Technical baseline
Phase 2	Q4 2026	Conduct wind resource assessment and evaluate site suitability or power purchase agreement (PPA) options	Wind developer and park manager	Wind feasibility report
Phase 3	Q1 2027	Develop preliminary Waste heat recovery design and integration concept	NEM type supplier or engineering contractor	Preliminary design
Phase 4	Q1–Q2 2027	Size the High-voltage electrode boiler based on LP and MP steam requirements	Engineering contractor	High-voltage electrode boiler specification
Phase 5	Q2 2027	Update the techno economic assessment using vendor quotations and refined design data	Project team	Updated TEA
Phase 6	Q3 2027	Perform HAZOP study, safety assessment, permitting process, and stakeholder consultation	CHP operator and port authority	Permit package
Social	Q3 2027	Responsible Actor: Park manager and CHP operator	Outcome: stakeholder concerns and acceptance conditions integrated into project design	
Phase 7	Q4 2027 – Q2 2028	Procurement and detailed engineering	CHP operator and EPC contractor	Final engineering design
Phase 8	Q3–Q4 2028	Install Waste heat recovery and High-voltage electrode boiler during planned shutdown	EPC contractor	Operational Waste heat recovery and electric boiler
Phase 9	2028–2029	Install wind turbine or establish renewable electricity supply through PPA	Wind developer and grid operator	Renewable electricity supply
Phase 10	2029	Monitor performance and verify energy and CO2 reductions	CHP operator	Verified energy and CO2 savings



Stakeholder	Role in the Project	Main Responsibilities	Interest and Expected Benefits
CHP operator	Central utility owner and operator	Provide operational data, operate the WHRU/HRSG and High-voltage electrode boiler systems, maintain backup capacity, and verify energy and emissions savings	Reduce fuel consumption, lower CO2 cost exposure, improve utility reliability, and strengthen CHP performance
Industrial park manager	System coordinator	Facilitate cooperation between companies, coordinate investment planning, and oversee shared utility arrangements	Improve overall park sustainability, competitiveness, and resource efficiency
EPC contractor	Engineering and implementation partner	Perform detailed engineering, procurement, installation, commissioning, and integration of the retrofit components	Deliver a technically successful project within budget and schedule constraints
WHRU/HRSG supplier	Technology provider	Design and specify the heat recovery system based on CHP exhaust conditions and steam requirements	Demonstrate reliable industrial heat recovery performance and successful technology deployment
Wind developer or PPA provider	Renewable electricity supplier	Assess wind resources, manage permitting, coordinate grid connection, and deliver renewable electricity through direct installation or PPA arrangements	Expand renewable energy deployment and generate revenue from electricity supply
Grid operator	Electricity network operator	Assess grid capacity, connection requirements, metering systems, protection measures, and import/export constraints	Maintain grid stability, reliability, and operational safety
Port authority	Land use and permitting stakeholder	Support site selection, infrastructure planning, permitting procedures, and safety coordination	Promote sustainable industrial development and support regional decarbonization objectives
Olefins, Aromatics, EO/EG, and PET	Utility users and beneficiaries	Provide utility demand information, participate in operational coordination, and contribute to utility tariff agreements where required	Secure reliable, lower carbon electricity and steam supply while improving sustainability performance
Local and community stakeholders	External stakeholders affected by the project	Participate in consultation processes and provide feedback on visual, noise, environmental, and safety impacts	Ensure that environmental and social impacts remain acceptable and transparent



8.3 Monitoring and verification plan

A structured monitoring program is required to verify whether the expected benefits of the CHP Utility-Hub Retrofit are achieved in practice. Energy performance indicators focus on fuel savings, heat recovery, renewable electricity generation, and renewable steam production. These indicators provide direct evidence of whether the intervention performs according to the design assumptions. Environmental performance is monitored through annual CO₂ reduction, which combines savings from heat recovery, renewable electricity generation, and renewable steam production. This indicator is particularly important because emissions reduction is one of the main objectives of the intervention. Operational performance should be evaluated through reliability and downtime indicators. Since the CHP Utility Hub remains the central utility node within the industrial park, maintaining reliable operation is essential for all connected companies. Finally, stakeholder complaints and concerns should be monitored as a social performance indicator. Tracking stakeholder feedback helps identify potential acceptance issues and ensures that social and organisational aspects receive attention alongside technical and environmental performance. Together, these indicators provide a balanced monitoring framework covering technical, environmental, operational, and social dimensions of the intervention. This allows project stakeholders to verify performance, identify improvement opportunities, and support future decisions regarding expansion of the full integrated energy material water symbiosis system.

Monitoring Indicator	Unit	Purpose	Responsible Stakeholder
CHP natural gas input	MWh/year	Verify fuel savings achieved through WHRU/HRSG heat recovery	CHP operator
WHRU recovered heat	MWh/year	Measure and verify heat recovery performance	CHP operator and WHRU supplier
Wind electricity generation	MWh/year	Verify renewable electricity production and turbine performance	Wind developer and CHP operator
Electricity supplied to electric boiler	MWh/year	Confirm operation of the power to steam system	CHP operator
Renewable steam production	MWh/year or MJ/h	Verify contribution of the High-voltage electrode boiler to the steam network	CHP operator
CO ₂ reduction	tCO ₂ /year	Assess environmental performance and emissions savings	Park manager and CHP operator
System downtime and reliability	Hours/year	Monitor operational performance and identify reliability issues	CHP operator
Stakeholder complaints and concerns	Number/year	Evaluate stakeholder acceptance and social performance	Park manager

8.4 Barriers, Risks, and Mitigation Measures

The main challenges associated with the intervention are not technological, since all major technologies already have a high technology readiness level and are commercially available. Instead, the largest risks are related to economic, organizational, and implementation issues. Financially, the largest concern is the initial investment cost. Although the techno economic assessment indicates positive long term performance, high upfront costs can delay adoption. Institutional barriers, particularly wind permitting and grid connection requirements, may also influence project timing. From a technical perspective, the key risks involve integrating renewable electricity and heat recovery systems into the existing utility network. Wind variability, steam pressure requirements, and long term WHRU



reliability must be managed carefully to ensure stable operation. Finally, social and organizational barriers should not be overlooked. Acceptance of wind infrastructure, together with fair allocation of costs and benefits among industrial users, will be important for successful implementation. For this reason, stakeholder engagement and transparent utility management arrangements should be considered as important as the technical design itself.

Barrier/Type	Risk	Mitigation
High capital cost /Financial	Longer payback period and reduced economic attractiveness	Use available subsidies, power purchase agreements (PPAs), and phased investment strategies
Wind permitting /Institutional	Delays in project implementation or permit rejection	Consider port area wind development or renewable electricity supply through a PPA instead of direct on site installation
Wind variability /Technical	Mismatch between renewable generation and utility demand	Maintain CHP as backup generation and use the High-voltage electrode boiler to increase operational flexibility
Steam pressure matching/Technical	High-voltage electrode boiler may not deliver steam at the required pressure level	Size the High-voltage electrode boiler specifically for LP and MP steam requirements within the utility network
WHRU corrosion and fouling/Technical	Reduced performance and reliability of heat recovery equipment	Apply appropriate material selection, monitoring, and temperature control strategies
Grid constraints /Infrastructure	Limitations on electricity import or export capacity	Prioritize direct on site consumption and power to steam conversion through the electric boiler
Social acceptance	Concerns regarding visual impact, noise, or project acceptance	Conduct early stakeholder engagement, communication activities, and interviews
Infrastructure	Existing steam, electricity, and water infrastructure may have insufficient capacity or unsuitable connection points for the proposed retrofit	Conduct detailed utility tie in studies, electrical and pipeline route assessments, and phased shutdown planning before final engineering design
Data and Monitoring	Energy, emissions, and operational improvements cannot be demonstrated reliably if baseline and post implementation measurements are insufficient	Install dedicated monitoring systems for exhaust heat recovery, wind electricity generation, High-voltage electrode boiler consumption, steam production, natural gas use, and CO2 reduction verification
Organisational and Contractual	Investment costs may be concentrated at the CHP Utility Hub while benefits are distributed across multiple companies	Develop shared utility tariffs, cost sharing agreements, and transparent benefit allocation mechanisms before the final investment decision
Future Integration	Material recovery and water reuse interventions may not be implemented at the same pace as the CHP retrofit	Implement the CHP Utility-Hub Retrofit as the first phase and connect Olefins hydrocarbon recovery and Aromatics water reuse as subsequent integration phases



8.5 Policy/Strategy alignment

Although the detailed Technology Assessment Plan focuses on the CHP Utility Hub Retrofit, the policy relevance of the selected system extends beyond the energy sector. The CHP retrofit functions as the central enabling intervention within the integrated industrial symbiosis system because it provides lower carbon electricity and steam to the industrial park and creates the utility foundation required for future resource integration measures. The intervention therefore supports more than industrial decarbonization. Renewable electricity generation, heat recovery, and power to steam integration strengthen the energy transition by reducing fossil fuel dependence and lowering CO₂ emissions. At the same time, the improved utility infrastructure can support future implementation of Olefins hydrocarbon recovery by providing lower carbon process energy and can support Aromatics water reuse through reduced utility related emissions and stronger integration between water and energy systems.

As a result, the selected intervention contributes simultaneously to several policy objectives, including industrial decarbonization, circular economy development, water resilience, resource efficiency, and industrial symbiosis. The CHP Utility-Hub Retrofit should therefore be viewed not only as an energy project, but as the enabling platform for the broader energy material water integration strategy proposed for the industrial park. The intervention is strongly aligned with European and Belgian energy transition policy. By combining Waste heat recovery, wind electricity, and power to steam conversion, it directly supports decarbonization, energy efficiency, and renewable energy integration. This policy alignment strengthens the feasibility of the project because it may improve access to subsidies, reduce regulatory risk, and increase acceptance among public and industrial stakeholders.

Policy Area	Relevance to the CHP Utility Hub Retrofit	Relevance to the Full Integrated System
European Green Deal	Supports industrial decarbonization through reduced fossil fuel consumption and lower CO ₂ emissions	Combines climate objectives with circular use of energy, materials, and water
Fit for 55	Reduces greenhouse gas emissions through heat recovery, renewable electricity, and electrification	Supports lower carbon industrial production and resource efficient value chains
EU Emissions Trading System (EU ETS)	Reduces exposure to carbon costs associated with natural gas combustion and purchased electricity	Improves long term economic competitiveness through lower carbon intensity
Energy Efficiency Directive	WHRU/HRSG heat recovery improves energy efficiency and reduces primary energy demand	Encourages efficient use of shared utilities and industrial resource flows
Renewable Energy Directive	Wind generation increases the share of renewable electricity in the utility system	Renewable electricity supports steam production, material recovery, and water treatment processes
Circular Economy Action Plan	Provides lower carbon utility services that support circular industrial activities	Directly supports hydrocarbon recovery, resource efficiency, and material recirculation within the industrial park
Water Resilience and Flemish Water Policy	Treated water can support cooling water and boiler feedwater requirements	Reduces freshwater demand through wastewater treatment and water reuse
Port of Antwerp Bruges Transition Strategy	Supports industrial decarbonization, electrification, and modernization of utility infrastructure	Strengthens the role of the port as an integrated industrial symbiosis and circular economy cluster



Belgium already has a well developed wind sector, making wind integration a realistic option for the proposed intervention. According to the IEA Wind TCP, Belgium had approximately 5.594 GW of installed wind capacity in 2024. In addition, the IEA highlights offshore wind expansion and renewable energy deployment as important components of Belgium's energy transition strategy. This indicates that the policy, infrastructure, and market conditions required for wind development are already largely in place. At the European level, the transition toward cleaner electricity is also evident. The European Environment Agency reports that the greenhouse gas emission intensity of EU electricity generation declined in 2024 compared with 2023 and is substantially lower than a decade ago. This trend demonstrates that renewable electricity and electrification are becoming increasingly important drivers of industrial decarbonization. The proposed CHP Utility-Hub Retrofit strongly aligned with both EU and Belgian policy objectives. The Waste heat recovery system improves energy efficiency by recovering waste heat and reducing fuel consumption, which supports the objectives of the Energy Efficiency Directive. The 1.5 MW wind turbine contributes to renewable energy deployment and is consistent with the Renewable Energy Directive and the Fit for 55 package. The High-voltage electrode boiler further strengthens the intervention by supporting industrial electrification and sector coupling, allowing renewable electricity to be converted directly into useful steam. Together, these measures reduce dependence on natural gas, lower the carbon intensity of electricity and steam supplied within the industrial park, and decrease exposure to future EU ETS carbon costs. From both a technical and policy perspective, the intervention is therefore well aligned with current European decarbonization strategies and long term industrial transition goals. Following the social impact assessment and stakeholder interviews, a stakeholder engagement workshop should be organized to support the transition from technical feasibility to organisational and social feasibility. The workshop should bring together representatives from the CHP operator, industrial companies, workers, the port authority, the grid operator, technology suppliers, and local stakeholders. The purpose of the workshop is to present the proposed intervention, discuss expected benefits and risks, and collect feedback before the final investment decision. Key topics should include expected CO₂ reductions, energy and resource savings, required infrastructure modifications, wind permitting considerations, operational and safety requirements, and principles for allocating costs and benefits among participating companies. The workshop should also provide an opportunity to identify stakeholder concerns related to reliability, operational impacts, visual and noise effects, permitting requirements, and long term governance of the shared utility system. Feedback collected during the workshop should be incorporated into the implementation plan, risk management strategy, and stakeholder engagement process. This step is important because successful implementation depends not only on technical performance but also on stakeholder acceptance, institutional support, and effective coordination between actors. By incorporating stakeholder feedback before committing to major investments, the project can reduce implementation risks and strengthen the social and organisational feasibility of the integrated industrial symbiosis system.

8.6 Uncertainty Discussion

The most important policy uncertainty relates to electricity-market conditions because the intervention depends on renewable-electricity generation and power-to-steam operation. Changes in electricity-market rules, tariffs, or grid-access conditions could influence both operational savings and investment attractiveness. Similarly, future changes in subsidy programs may alter the economic feasibility of the project, particularly for renewable-energy and electrification components. Carbon pricing under the EU ETS also represents a significant uncertainty. Because part of the project's value is linked to avoided emissions, future carbon prices directly influence economic performance. For this reason, sensitivity analysis should include multiple carbon-price scenarios rather than relying on a single assumption. Regulatory uncertainty may also affect implementation schedules. Wind-energy permitting and grid-connection approvals can take considerable time and may delay deployment even when the technologies themselves are technically feasible. Maintaining flexibility through renewable PPAs and phased implementation reduces this risk. Overall, the project remains well aligned with current policy objectives, but a flexible implementation strategy is required to manage regulatory uncertainty. Regular updates of the techno-economic assessment, evaluation of alternative implementation pathways, and continued stakeholder engagement will improve project resilience under changing policy conditions.



Policy Uncertainty	Possible Impact	Response Strategy
Electricity-market regulation	Changes in grid tariffs, connection conditions, or electricity-market prices may affect project revenues and savings	Update the techno-economic assessment before the final investment decision and maximize direct internal use of renewable electricity
Subsidy schemes	Reduced support for renewable energy, electrification, or industrial decarbonisation may weaken project economics	Identify multiple funding sources and implement the project through phased investment
EU ETS carbon price	Lower carbon prices reduce the financial value of avoided CO ₂ emissions	Evaluate low, medium, and high carbon-price scenarios within the economic assessment
Wind-permitting regulations	Delays in permitting may postpone implementation and increase project uncertainty	Maintain renewable PPA options as an alternative to on-site wind generation
Grid-connection regulation	Limited grid capacity may restrict wind integration or electric-boiler operation	Engage early with the grid operator and consider staged connection and infrastructure upgrades



Chapter 9

9.1 Discussion of Results

The results of this project show that the selected industrial park in Antwerp cannot be improved by focusing on only one flow. The main problems are connected across energy, material, and water systems. The park depends on fossil-based utilities, has material losses in the petrochemical chain, and has limited water circularity. Therefore, the full integrated system is the most suitable long-term option because it combines energy, material, and water interventions. From the material analysis, Olefins is an important hotspot because it has high material intensity and a high E-factor. This means that valuable hydrocarbon byproducts are not fully used. The selected material intervention, Olefins hydrocarbon by product recovery and circular feedstock recirculation, can reduce material losses and lower dependence on virgin fossil feedstocks. However, the recovered streams must meet the quality requirements of the process. If not, extra separation, treatment, or blending may be needed. The water analysis showed that Aromatics wastewater contains BTX related contaminants and cannot be reused directly. Therefore, the proposed water intervention uses AOP, biological or membrane treatment, and reverse osmosis. This can produce reusable water for cooling water make-up or boiler feedwater support. This improves water circularity and reduces freshwater demand, but it also requires reliable treatment, monitoring, storage and reuse of pipelines. The energy analysis showed that the main issue is not only electricity supply, but steam and heat management. The CHP plant is the central utility provider, while the process plants depend on it for electricity and steam. The park still has MP and LP steam deficits, and some heat is rejected through cooling water. Therefore, the main energy opportunity is better heat recovery, renewable electricity use, and steam flexibility. Chapter 5 also identified EO/EG as the park's highest-emission, highest-waste-heat node, and diagnosed a four-intervention package for it using the CIMO-logic framework. This package is not carried forward as the project's primary recommendation, since it was not developed to the same investment-grade standard, CAPEX, NPV, payback, and sensitivity analysis, as the CHP Utility Hub Retrofit, and extending an equivalent techno-economic case to EO/EG was outside this project's scope. The CHP Utility Hub Retrofit is therefore the main enabling intervention. It includes WHRU/HRSG heat recovery, a 1.5 MW wind turbine, and a high voltage electrode boiler. The wind turbine generates about 3823.74 MWh/year of renewable electricity. Part of this electricity is used directly, while the rest is converted into renewable steam. With 98% boiler efficiency, this produces about 1498.91 MWh/year of renewable steam. This improves steam flexibility, but it does not fully solve the steam deficit. Therefore, it should be seen as a first step in a wider steam network improvement strategy. The environmental results are positive. The WHRU/HRSG saves 157.34 MWh/year of natural gas and directly avoids 28.48 tCO₂/year. The total estimated CO₂ reduction is 651.22 tCO₂/year when avoided grid electricity and renewable steam are included. This shows that the largest benefit comes from combining heat recovery, renewable electricity, and power to steam conversion, not only from direct fuel savings. The economic results are promising but should be interpreted carefully. The CHP Utility Hub Retrofit has an estimated CAPEX of about €2.53 million with ±30% uncertainty. Without avoided CO₂ price benefits, the project has a small negative annual balance and an avoidance cost of +€26/tCO₂. When avoided CO₂ costs are included, it becomes more attractive, with a positive annual balance and an avoidance cost of -€49/tCO₂. This means that the project is close to break even based on energy savings alone, but carbon pricing makes it more economically attractive. The main trade off is between sustainability performance and implementation complexity. The full integrated system gives the strongest environmental and circularity benefits, but it also requires higher investment, more infrastructure, and stronger coordination between companies. The network analysis also shows that the CHP Utility Hub becomes even more central. This improves coordination, but it also creates dependency. If the CHP hub or steam network fails, several plants could be affected. Therefore, backup capacity, monitoring, preventive maintenance, and clear utility management rules are needed. Social and organizational factors are also important. The system requires cooperation between the CHP operator, Olefins, Aromatics, EO/EG, PET, the park manager, the port authority, the grid operator, suppliers, workers, and local stakeholders. Benefits such as lower emissions, improved energy security, and new technical skills are expected. At the same time, concerns about cost sharing, safety, ownership, operational reliability, and wind turbine impacts must be addressed through stakeholder engagement and clear agreements.



Overall, the full integrated system is the best long-term direction because it addresses energy, material, and water challenges together. However, implementation should be phased. The CHP Utility Hub Retrofit should be implemented first because it creates the lower carbon electricity and steam foundation needed for later material recovery and water reuse.

The main trade-off of the selected solution is between overall system performance and implementation complexity. The full integrated system achieves the strongest environmental, circularity, and network-integration benefits because it simultaneously addresses energy, material, and water challenges within the industrial park. Compared with single-domain interventions, it provides greater CO₂ reduction, improved material recovery, higher water circularity, and stronger industrial symbiosis.

However, these benefits come at the cost of increased complexity. The integrated system requires coordination between multiple companies, shared infrastructure investments, data exchange, operational agreements, and more complex governance arrangements. In contrast, energy-only, material-only, or water-only interventions are generally easier to implement because they involve fewer stakeholders and more limited infrastructure modifications.

Therefore, the full integrated system offers the highest overall sustainability performance but also requires the greatest level of organisational coordination, infrastructure integration, and investment commitment. This trade-off is considered acceptable because the additional complexity is compensated by broader system-wide benefits and stronger long-term circularity performance.

9.2. Limitations and Uncertainties

In this report, the main limitation is that the analysis is based on literature values, engineering assumptions, and simplified mass and energy balances rather than real industrial data. In a real-world scenario, the outcomes could be affected by several factors. For instance, actual utility consumption, waste heat availability, wastewater composition, and production rates may vary over time and differ from the assumptions used in this study. Furthermore, the study relies on a number of technological and data-related assumptions, while the project scope remains limited. Therefore, the results cannot be directly extrapolated to a real industrial site without further validation. Additionally, the technologies proposed in this study, including the plastic-to-naphtha pathway, Aromatics wastewater treatment, and the CHP Utility-Hub Retrofit, may be influenced by site-specific conditions, operational constraints, and maintenance requirements, creating uncertainties regarding their implementation and both short-term and long-term performance. The implementation of these technologies is also subject to economic and policy uncertainties. Their feasibility may depend on future electricity prices, natural gas prices, carbon costs, and policy incentives. While these factors are recognized as important drivers and sources of uncertainty, they were not explicitly modelled within the scope of this project, and their future development remains difficult to predict and control. In addition, the availability and reliability of renewable electricity may influence the performance of the proposed interventions. Although the Antwerp industrial area generally has strong energy infrastructure, temporal mismatches between renewable electricity generation and industrial energy demand were not assessed in this study and may affect the practical implementation of the proposed system. Despite a number of limitations and uncertainties, the study shows the potential for a more integrated and resource-efficient industrial system. By combining energy, material, and water interventions, the proposed system can reduce fossil-fuel dependence, improve resource utilization, and strengthen industrial symbiosis within the industrial park.



9.3. Conclusions

The aim of this project was to assess the potential of industrial symbiosis interventions at the PolyCycle Port industrial park. The study evaluated how these interventions could improve environmental performance, economic outcomes, and circularity. This was achieved through the integrated management of energy, material, and water flows. Throughout the two quarters of this project, our analysis consistently showed that no single intervention is sufficient to achieve a significant transition toward an eco-industrial park. Better results can be achieved by treating energy, material, and water systems as interconnected components of the same industrial network. A key finding of this study was that the CHP Utility Hub is the most important industrial symbiosis hub in the park. It plays a central role due to its system-wide influence and integration potential, and it is the only intervention in this assessment supported by a full investment-grade economic case (Chapter 7). EO/EG was also identified as having the highest greenhouse gas emissions and the largest single waste-heat stream in the park, and a four-intervention package was diagnosed for it in Chapter 5. This package is not advanced as the project's primary recommendation here, since equivalent costing and investment analysis were not developed for it within this project's scope; it remains the most promising candidate for future assessment once the CHP retrofit's implementation is underway. The analysis of intervention options showed that industrial symbiosis is fundamentally a systems issue. Single interventions can help in one area. But they do not deal with the broader resource flows and connections across the park. The integrated system delivered the strongest overall performance. It enhanced environmental outcomes, circularity, network connectivity, and policy compliance simultaneously. However, achieving these benefits requires greater coordination, investment, and collaboration among stakeholders. This project showed that technical solutions alone are not enough. Successful implementation also requires strong organizational support and governance. It is essential for industrial partners, utility providers, regulators, and other stakeholders to work together effectively. The CHP utility hub plays a key role, highlighting the need for reliable operation, contingency planning, and clear agreements on shared infrastructure and responsibilities.

In summary, our results show that PolyCycle Port's transition to a more circular and low-carbon industrial park is technically possible and supported by available technologies. The project demonstrated that industrial ecology is about more than optimizing individual processes. It is about rethinking how they are connected. The strategy improves resource efficiency, reduces environmental impact, and strengthens industrial symbiosis by integrating energy, material, and water systems. The results of this project provide guidance for future implementation and a model for similar industrial clusters to speed up their sustainability transition.



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Appendix A – Detailed Stakeholder Roles and Operations

The following table provides a detailed description of stakeholder functions, responsibilities, and interactions with the EO/EG plant at PolyCycle Port

Stakeholder	Function & Responsibilities	Interaction with the Industrial Park	Motivation
Port of Antwerp-Bruges	Concession authority managing port land, shared infrastructure (pipelines, steam grids), and environmental permitting. Drives EIP strategy.	Approves cross-company symbiosis upgrades (CO ₂ pipeline, DH connection) and coordinates shared investments such as the WWTP.	Advance NextGen District's circular mandate and maintain Antwerp's competitive position
EO/EG Plant Operator (eg, INEOS / Borealis)	Asset owner responsible for daily operations, catalyst management, EU ETS reporting, and all CAPEX decisions on process upgrades.	Central park node: receives C ₂ H ₄ (269 kt/y) from Olefins Plant, supplies MEG (135 kt/y) to PET Plant, and consumes all CHP steam (6,407 TJ/y).	Reduce EU ETS liability and avoid stranded asset risk under Fit-for-55
Olefins Plant (eg, BASF / TotalEnergies)	Produces ethylene via steam cracking and supplies 269.4 kt/y to the EO/EG plant via a dedicated pipeline.	Primary upstream feedstock dependency — any supply disruption cascades directly to EO/EG production.	Secure long-term offtake and reduce shared fossil feedstock price exposure
PET Plant	Consumes 135 kt/y MEG as co-monomer for PET production; manages shared pipeline and tank infrastructure.	Downstream offtake partner; co-stakeholder in catalyst upgrade decisions that affect MEG purity and supply continuity.	Ensure MEG supply continuity and meet downstream sustainability requirements



CHP Plant	Supplies all HP/MP/LP steam to the park via the steam grid. Currently, gas-fired; no backup boiler exists in the park.	Single point of failure for the EO/EG plant. A transition to H ₂ /biomethane as a fuel is the primary lever for reducing Scope 2 CO ₂ .	Transition fuel mix toward hydrogen/biomethane ahead of rising carbon costs
Urea Plant	Produces urea fertilizer using CO ₂ as a reactant (demand: 740 kt/y); currently procures CO ₂ externally.	Target receiver for CCU symbiosis arc: 58.5 kt/y of 98% CO ₂ from the EO/EG K ₂ CO ₃ scrubber, reducing costs for both parties.	Reduce external CO ₂ procurement costs and improve emission profile
Centralised WWTP	Shared wastewater treatment facility applying BAT-LVOC stages (activated sludge, MF/UF, carbon polishing) for multiple plants.	Receives 35 kt/y of MEG-contaminated purge water from EO/EG; enables MEG recovery and ensures VLAREM II compliance.	Expand throughput and demonstrate circular water management under Flemish Blue Deal 2.0
Greenhouses / DH Operator	Agricultural greenhouses and a district heating network operator seeking low-cost thermal energy (<100 °C) and CO ₂ .	Target receiver for heat pump output (1,200–1,600 TJ/y at 70–85 °C) and purified flue-gas CO ₂ for crop fertilisation.	Reduce fossil heating costs and boost crop yields via elevated CO ₂
European Commission / ETS	Sets EU Green Deal, Fit-for-55, EU ETS carbon pricing (€65–85/t CO ₂), and funds innovation via Horizon Europe and the Innovation Fund.	Indirect high-power driver: ETS carbon price determines payback for key interventions; Innovation Fund can co-finance CAPEX.	Accelerate industrial decarbonisation toward Fit-for-55 targets
Flemish Govt Agency (OVAM/VMM)	Regional environmental regulator enforcing VLAREM II discharge limits, air quality	Issues and enforces environmental permits; COD discharge, EO emissions, and CO ₂ reporting are primary	Demonstrate measurable industrial transition progress under VEKP 2021-2030



	norms, and Flemish circular economy strategy.	compliance touchpoints.	
Local Communities	Residents affected by air quality (EO, VOC, NO _x) and industrial risk; social licence depends on visible environmental progress.	Benefit from reduced EO emissions, district heating supply, and CCU — constituting LCSP Principle 9 community co-benefits.	Achieve tangible air quality improvements and visible environmental progress
Uantwerp / VITO	Process optimisation and industrial symbiosis expertise	Applied research, pilots, and LCA assessments	Secure research funding and contribute to regional transition policy
Investors / Lenders	Green bond investors and EIB/IFC lenders are evaluating ESG criteria and alignment with the EU Taxonomy for capital deployment.	Gate capital for Interventions 2, 4 & 5 (~€50–80M combined); eligibility directly linked to documented CO ₂ KPI reductions.	Deploy capital into Taxonomy-aligned assets and meet institutional ESG mandate
Workers & Trade Unions	Operate and maintain the plant; unions represent labour interests in process changes, retraining, and EO safety procedures.	Directly affected by catalyst turnaround (Int. 1) and heat integration upgrades (Int. 4); safety is a key LCSP Level 3 indicator.	Secure employment continuity and retraining opportunities during upgrades

Table 3.1 Stakeholder role, interaction, and motivation



Appendix: Supplementary Linny-R Figures

The figure in Chapter 5 shows only the modelled S-01/S-02/S-03 interventions discussed in the text. The two figures below show additional detail from the same Linny-R model (Full-Park-Network-With-All-Intervention.lnr) for readers who want to verify the EO/EG plant's full utility connections, or see how the EO/EG plant fits within the park's wider steam and electricity network.

A.1 EO/EG plant, full utilities detail

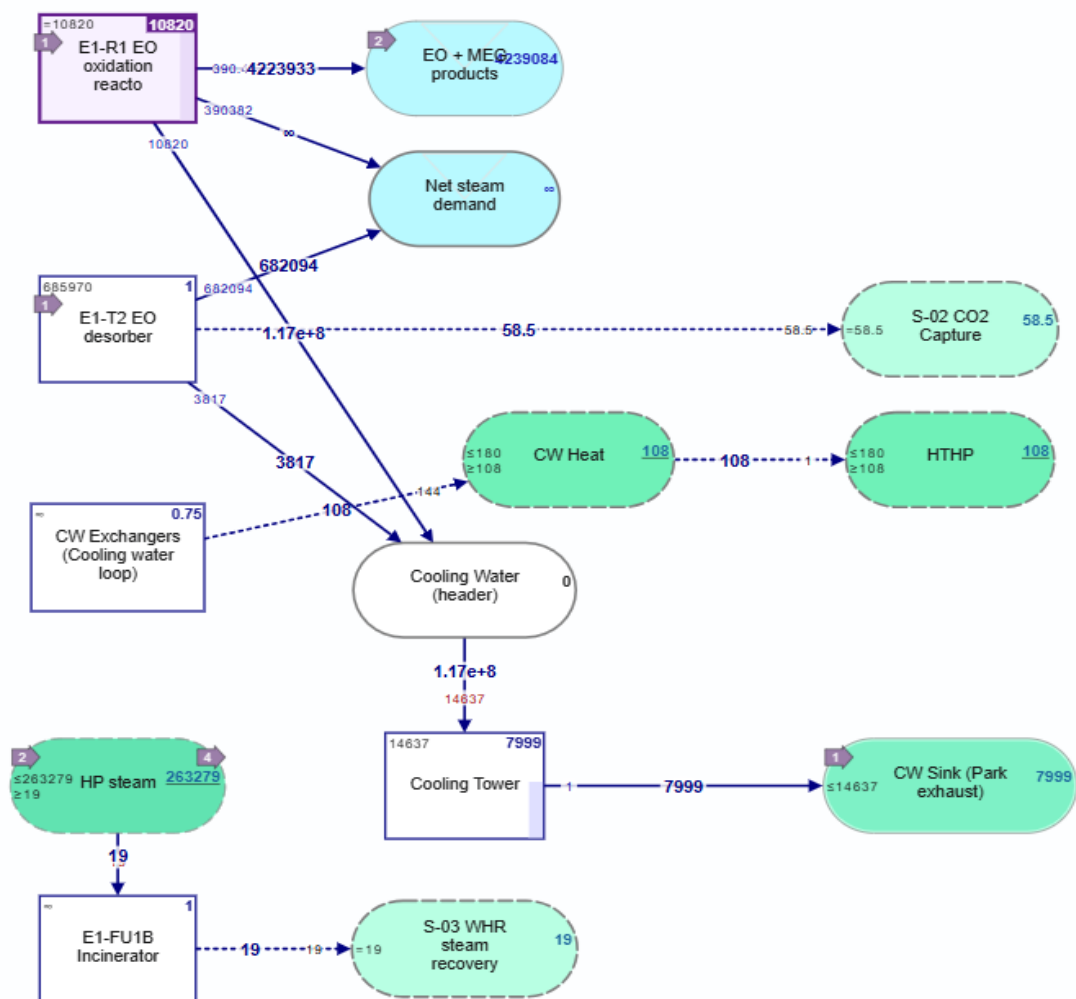
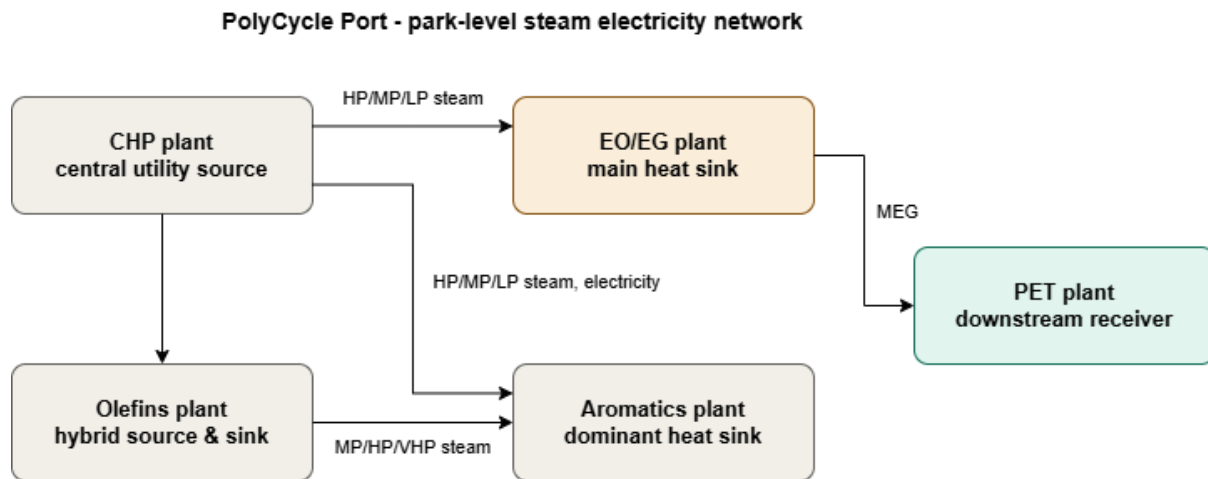


Figure A.1: EO/EG plant Linny-R network including utility connections (cooling tower, CW sink, HTHP, net steam demand) not shown in the main-text figure.



A.2 Full park network, plant-level overview



CHP Exports electricity, LP, MP and HP steam. Olefins imports LP steam, exports MP, HP and VHP steam. Aromatics and EO/EG are the park's largest steam and cooling-water demand centers. PET receives MEG from EO/EG

Figure A.2: PolyCycle Port park-level steam and electricity network, summarising the stakeholder roles described in Chapter 5 (Table: CHP as central utility source, Olefins as hybrid source and sink, Aromatics and EO/EG as the park's main heat sinks, PET as downstream receiver of MEG).

Appendix B – Power–Interest Mapping Methodology

Methodology: Each stakeholder was evaluated across two dimensions: Power and interest. Power includes influence over investment decisions, control of key inputs and infrastructure, regulatory authority, and control over critical off-take relationship with the value chain. Interest reflects the extent to which the plant's performance affects the stakeholder's operations, costs, or compliance. Scores were normalized to a 0–1 scale and cross-referenced with Table 3.1 and the intervention scenarios. Within each quadrant, the positioning reflects relative influence; for example, the EO/EG Plant Operator, CHP Plant and PET Plant are placed in the "Manage Closely" category due to their direct operational roles and strong economic interdependencies, while actors such as the European Commission exert high power but lower direct operational interest.



Appendix C – Water Flow Analysis

Producti on Process	Stre am Typ e	Name of the flow	Sources	Proc ess Uses	Water Types	Quality	Main Impuritie s
Natural Gas CHP	RM	FS-WATER	Surface Water	Boiler Feed Water	Demineralized pure water		N/A
Natural Gas CHP	WS	PS- LPS/MPS/HP S	Demineraliz ed pure water	Heat Carrier	High-purity gaseous water		N/A
Olefins Production Plant	RM	Steam	Natural Gas CHP	Heat Carrier	High-purity gaseous water		N/A
Olefins Production Plant	RM	NaOH	Surface Water	Washi ng	Inorgani c Wastew ater	Non biodegrada ble	Na ₂ S, Na ₂ CO ₃
Olefins Production Plant	RM	Dilution Water	Municipal Water / Steam Condensate Return	Dilutio n	Organic Wastew ater	Biodegrad able	Organic Waste
Olefins Production Plant	RM	Make up water	Surface Water (A)/Treated Wastewate r Effluent	Boiler Feed Water	Inorgani c Wastew ater	Poorly biodegrada ble	Industrial Dust
Olefins Production Plant	WS	Effluent	NaOH	Washi ng	Mixed Organic- Inorgani c Wastew	Toxic organics (incinerati on)	Cyclopenta diene, Na ₂ S, Sodium formate



ater							
Olefins Production Plant	WS	Dirty water (WS-DW)	Dilution Water	Quenching	Organic Wastewater	Poorly biodegradable	Benzene, Propyl mercaptan
Olefins Production Plant	WS	Water	Municipal water	Dilution	Inorganic Wastewater	Biodegradable	N/A
Olefins Production Plant	WS	Waste water	Municipal water	Washing	Organic Wastewater	Biodegradable	BDI-13, NMP
Aromatics Production Plant	RM	Water for wash	Surface Water (A)/Treated Wastewater Effluent	Washing	Demineralized pure water		N/A
Aromatics Production Plant	WS	Water and sulfonate	Surface Water (A)/Treated Wastewater Effluent	Washing	Organic Wastewater	Poorly biodegradable	SL
Ethylene Oxide-Ethylene Glycol	RM	Stripping steam	Natural Gas CHP	Stripping	High-purity gaseous water		N/A
Ethylene Oxide-Ethylene Glycol	RM	Make-up water	Municipal water	Reagent	Demineralized pure water		N/A
Ethylene Oxide-Ethylene	WS	Effluent water	Make-up water / Stripping	Evaporation	Organic Wastewater	Biodegradable	N/A



Glycol		stream			ater		
Ethylene Oxide-Ethylene Glycol	WS	Purge water	Cooling water loop	Impurity control	Mixed Organic-Inorganic Wastewater	Poorly biodegradable	MEG
PET Production Plant	RM	Solvent Water	Municipal water	Solvent	Demineralized pure water		N/A
PET Production Plant	WS	Excess Water	Municipal water	Solvent recovery	Organic Wastewater	Biodegradable	Acetic acid
PET Production Plant	WS	Waste water	Municipal water	Washing	Organic Wastewater	Poorly biodegradable	PTA

Energy Balance table, Overview

Utility	Net Park balance (MJ/h)	Interpretation
Electricity	-221.58	slight net surplus
HP steam	-187.49	almost balanced / slight surplus
MP steam	14587.26	main steam deficit
LP steam	10120.71	large steam deficit
Cooling water	52987.02	high cooling-water burden



Contribution Table

Participation	Rasoul	Andrew	Salih	Juting	Mona
Chapter 1	*	*	*	*	*
Chapter 2	*	*		*	*
Chapter 3	*	*	*	*	*
Chapter 4		*		*	
Chapter 5	*		*		*
Chapter 6	*		*	*	
Chapter 7	*		*		
Chapter 8	*			*	
Chapter 9	*		*	*	*
Presentation 1	*	*	*	*	*
Presentation 2	*	*	*	*	*
Presentation 3		*			
Improve Chapters after Feedback	*	*	*	*	*
Provide for Final Submission	*				
Review	*	*	*	*	*

Use Of AI

Artificial intelligence was used throughout the project as a support tool. It helped with improving writing quality, identifying mistakes and inconsistencies, searching for relevant literature, summarizing articles and checking Python code for possible errors. AI was not used to generate results or make research decisions. All calculations, analyses, interpretations, and conclusions were reviewed and completed by the group.